VU Medical Zone

Admin: Amaan Khan

BIOLOGICAL TECHNIQUE

ZOO101 FINAL TERM

Merge ppt

Lecture 41 to 104

LESSON 41
Separation Techniques

Dr. Aneela Ulfat

Separation Techniques

- Separation techniques are those techniques that can be used to separate two different states of matter such as liquids and solids.
- Separation processes or a separation method or simply a separation is methodology to attain any mass transfer phenomenon that convert a mixture of substances into two or more distinct product mixtures.
- Separation is an important asset to purify component of interest from a mixtures.

Need of Separation Techniques:

- To identify what to be separated from mixture.
- To obtain Important and pure substances.
- To remove unwanted particles





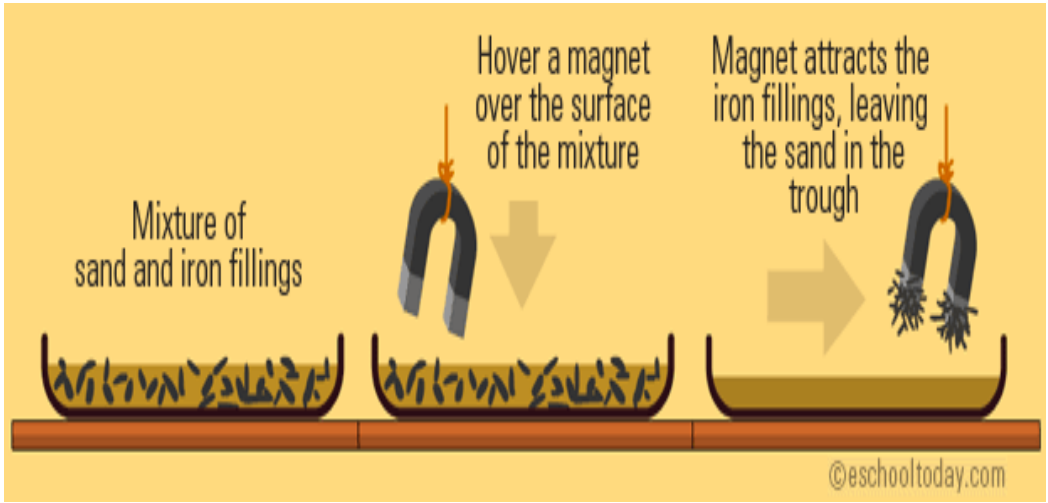


Classification of Separation Techniques

Basis Of Separation	Separation Technique(s)
Size	Filtration; Dialysis; Size-exclusion Chromatography
Mass Or Density	Centrifugation
Complex Formation	Masking
Change In Physical State	Distillation; Sublimation; Recrystallization
Change In Chemical State	Precipitation; Electrode position; Volatilization
Partitioning Between Phases	Extraction; Chromatography

Magnetic Separation

- This method involves the separation of magnetic substances from non-magnetic substances by means of magnet.
- Takes advantage of physical property of magnetism, so it useful only for certain substances such as ferromagnetic (materials strongly affected by magnetic fields) and paramagnetic (materials that are less affected, but the effect is still noticeable).
- This method involves the separation of magnetic substances from non-magnetic substances by means of magnet

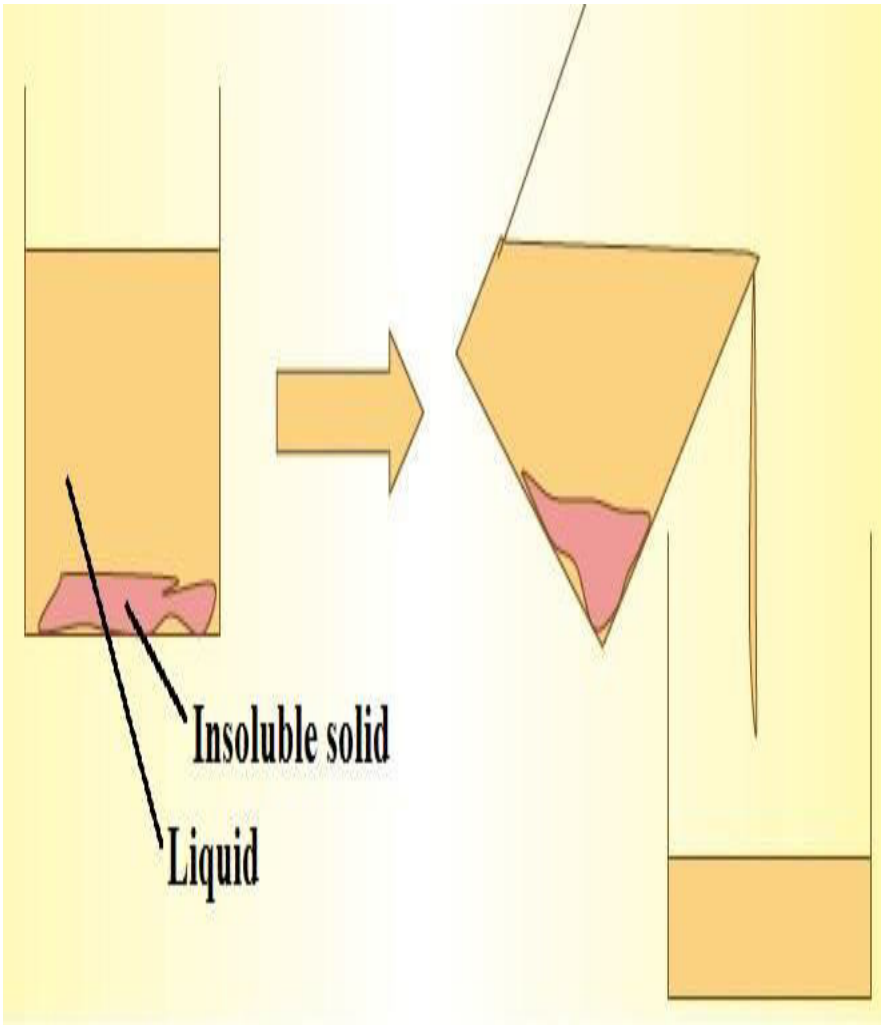


Applications

- Waste management, low-magnetic field separation in water purification and separation of complex mixtures.
- To remove metal contaminants from pharmaceutical product streams.
- Magnetic cell separation. It is currently being used in clinical therapies, more specifically in cancer and hereditary diseases researches.
- These techniques are combined with PCR (polymerase chain reaction), to increase sensitivity and specificity of results.

DECANTATION

- Decantation is a process for the separation of mixtures of immiscible liquids or of a liquid and a solid mixture such as a suspension. Immiscible liquid separation:
- Takes advantage of differences in density of the liquids.
- A separator funnel is an alternative apparatus for separating liquid layers. It has a valve at the bottom to allow draining off the bottom layer. It can give a better separation between the two liquids.
- Example: separation of mixture of oil and water. Takes advantage of gravity/sedimentation of solids in case solid-liquid separation.



- Sedimentation: The tendency of particles in suspension to settle down in the fluid due to certain forces like gravity, centrifugal acceleration, or electromagnetism is called as sedimentation.
- The solid that gets settled down is called as sediment.
- In laboratory it can be done in test tubes.
- To enhance productivity test tubes should be placed at 45° angle to allow the sediments to settle at the bottom of the apparatus.
- A decanter centrifuge may be used for continuous solid-liquid separation.

Applications

- Decantation is frequently used to purify a liquid by separating it from a suspension of insoluble particles.
- Decantation is also present in nanotechnology. In the synthesis of high-quality silver nanowire (AgNW) solutions and fabrication process of high-performance electrodes, decantation is also being applied which greatly simplifies the purification process.
- Fat is determined in butter by decantation.
- In sugar industry, processing of sugar beets into granular sugar many liquid - solid separations are encountered.

FILTRATION

- Takes advantage of physical property of state of matter, its size and solubility in liquid. The substance which is left behind in the filtration medium is called **residue**.
- The liquid which passes through the filtration medium is called **filtrate**. Mechanical, physical or biological operations that separates solids from fluids (liquids or gases) by adding a medium through which only the fluid can pass can be called as **filtration**.
- **Applications**

Most important techniques used by chemists to purify compounds. HEPA filters in air conditioning to remove particles from air. In the laboratory, a Buchner funnel is often used, with a filter paper serving as the porous barrier.

Centrifugation

- Centrifugation is one of the techniques that scientists can use to separate mixtures.
- In its simplest terms, a centrifuge is a mechanical device designed to separate particles from a solution by way of centrifugation. Centrifugation involves subjecting a sample to Relative Centrifugal Force (RCF). The success of the separation will depend on the size, shape, density, viscosity of the medium and the rotor speed.
- If a solution has some particles that are a higher density than others, then the lighter particles will separate from the heavier ones. The user will see the heavier substances forced to the bottom of the container (tube, bottle, bag etc.) and the lighter ones will remain suspended at the top.
- If there is no difference in the density of the particles (known as isopycnic conditions), then the sample will not separate.



Applications

- Micro centrifuges are used to process small volumes of biological molecules, cells, or nuclei.
- Used in diagnostic laboratories for blood and urine tests.
- Aids in separation of proteins using purification techniques such as salting out.
- Differential Centrifugation used to separate organelles and membranes found in cells.



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Lesson 42

Separation Techniques

Separation Techniques

- Separation is an important asset to purify component of interest from a mixtures

Evaporation

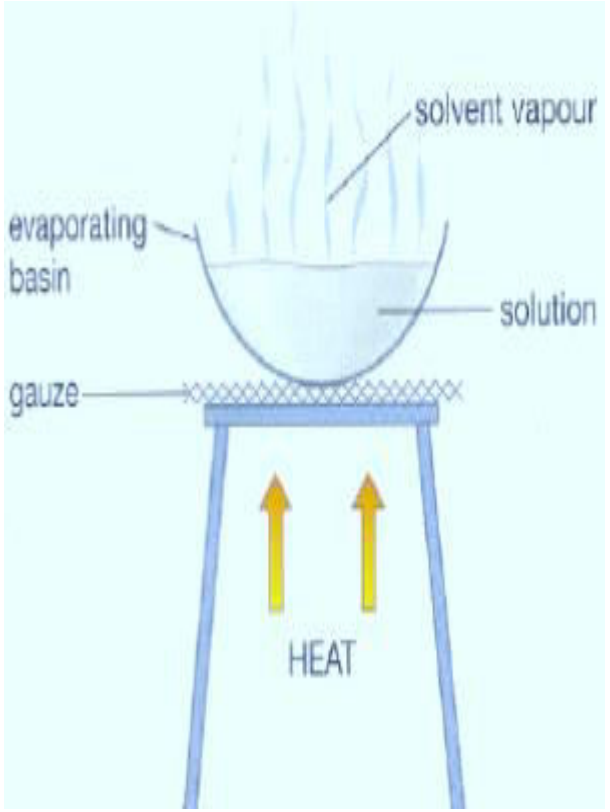
- Evaporation is type of vaporization that occurs on the surface of liquid as it changes into gas phase.
- This separation technique can be used to separate solutes that are dissolved in solvent, by boiling the solution. The solvent gets vaporized leaving back the solute.
- Factors affecting:

1. Concentration of the substance evaporating in the air.
2. Flow rate of air, pressure, temperature.
3. Surface area. Intermolecular forces.

- **Evaporation equipment:**
- Natural circulation evaporators: Evaporating still and pans. Evaporating short tubes. Forced circulation evaporators.
- Film evaporators: Climbing film evaporators. Horizontal film evaporators.

Applications/examples

- Recovering salts from solution.
- The use of evaporation to dry or concentrate samples is a common preparatory step for many laboratory analyses such as spectroscopy and chromatography.
- Demineralization of water
- In the case which we do not need to collect the solvent. The solvent is boiled off and escape into the air while the solute is left behind in the holding container. Note that this method is not suitable for use on solutes which can decomposed by heating (e.g. Copper II sulfate).



Distillation

- Distillation is an effective method to separate mixtures comprised of two or more pure liquids (called “components”). The separation of a mixture of liquids based on the physical property of boiling point. Distillation is a purification process where the components of a liquid mixture are vaporized and then condensed and isolated.
- The mixture is heated until one of the components boils (turns to a vapor). The vapor is then fed into a condenser, which cools the vapor and changes it back into a liquid that is called distillate. What remains in the original container is called the "residue".

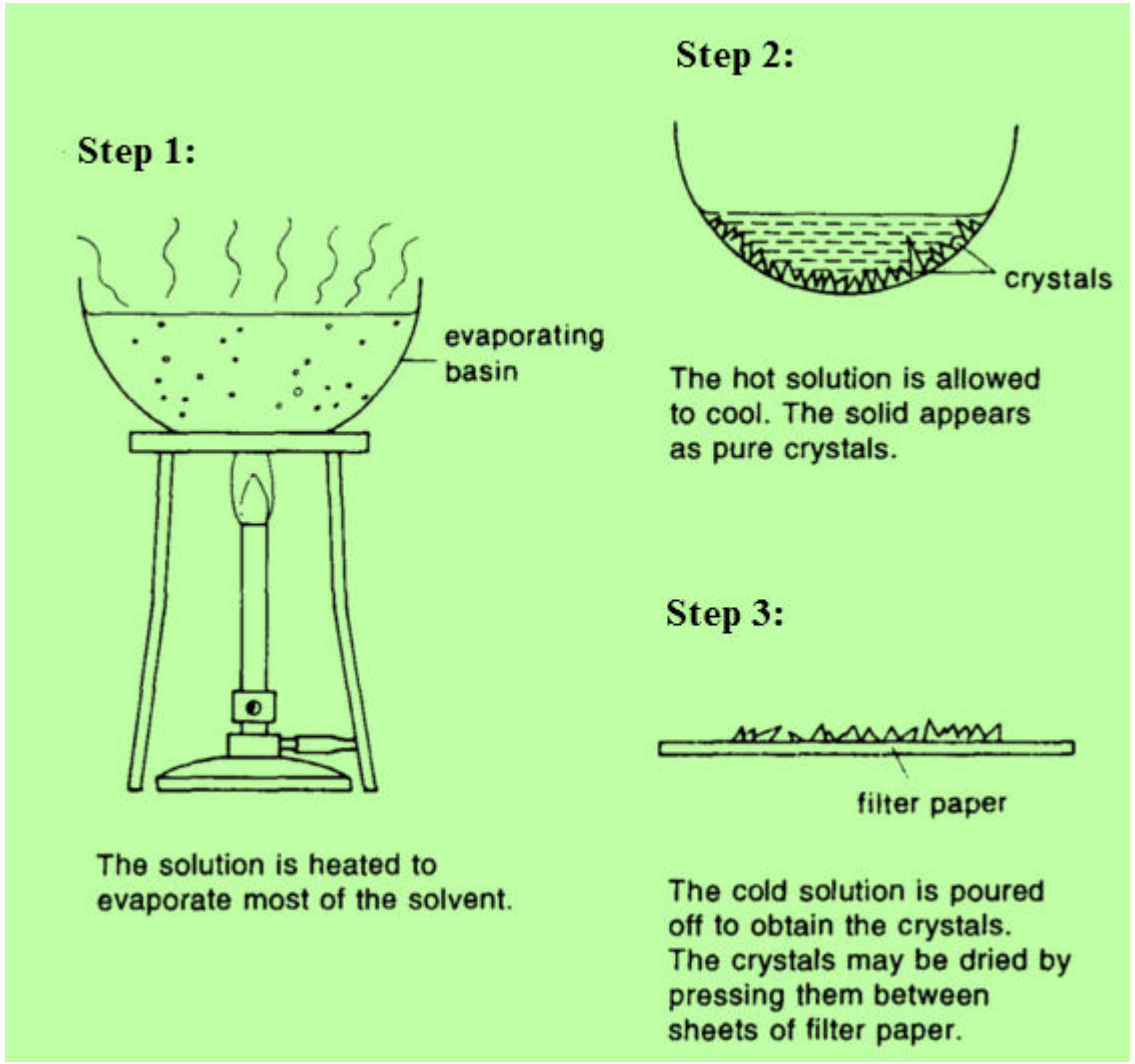
Applications

- Separation of different fractions from petroleum products.
- Separation of a mixture of methanol and ethanol.
- Separation of volatile oils.
- Separation of drugs obtained from plant or animal sources – vitamin A from fish liver oil.
- Purification of organic solvents.
- Manufacture of official preparations – spirit of nitrous ether, spirit of ammonia, distilled water, and water for injection.
- Refining petroleum products.

Crystallization

- Used to separate a dissolved heat-liable (will decompose upon heating and hence can sublime) solid (solute) from a solution.
- You will need a saturated solution to being with.
- A **saturated solution** is a solution that contains the maximum amount of solute dissolved in a given volume of solvent at a particular temperature.
- Do not mix this up with a **concentrated solution**, which is a solution that contains lots of solute dissolved in it.
- The amount of solute in a concentrated solution may/may not be the maximum amount which can be dissolved in the solution.

- First, you will need to heat to evaporate off most of the solvent from a solution to make a hot and nearly saturated solution.
- Else, if you already have a saturated solution, heat it up slightly such that the solution becomes hot.
- After which, allow the hot solution to cool naturally. The solubility of the solute decreases as the solution is cooled, and the excess solute which can no longer be dissolved in the saturated solution crystallizes out of the solution.
- The crystals which are formed can be separated from the remaining solution by filtration.

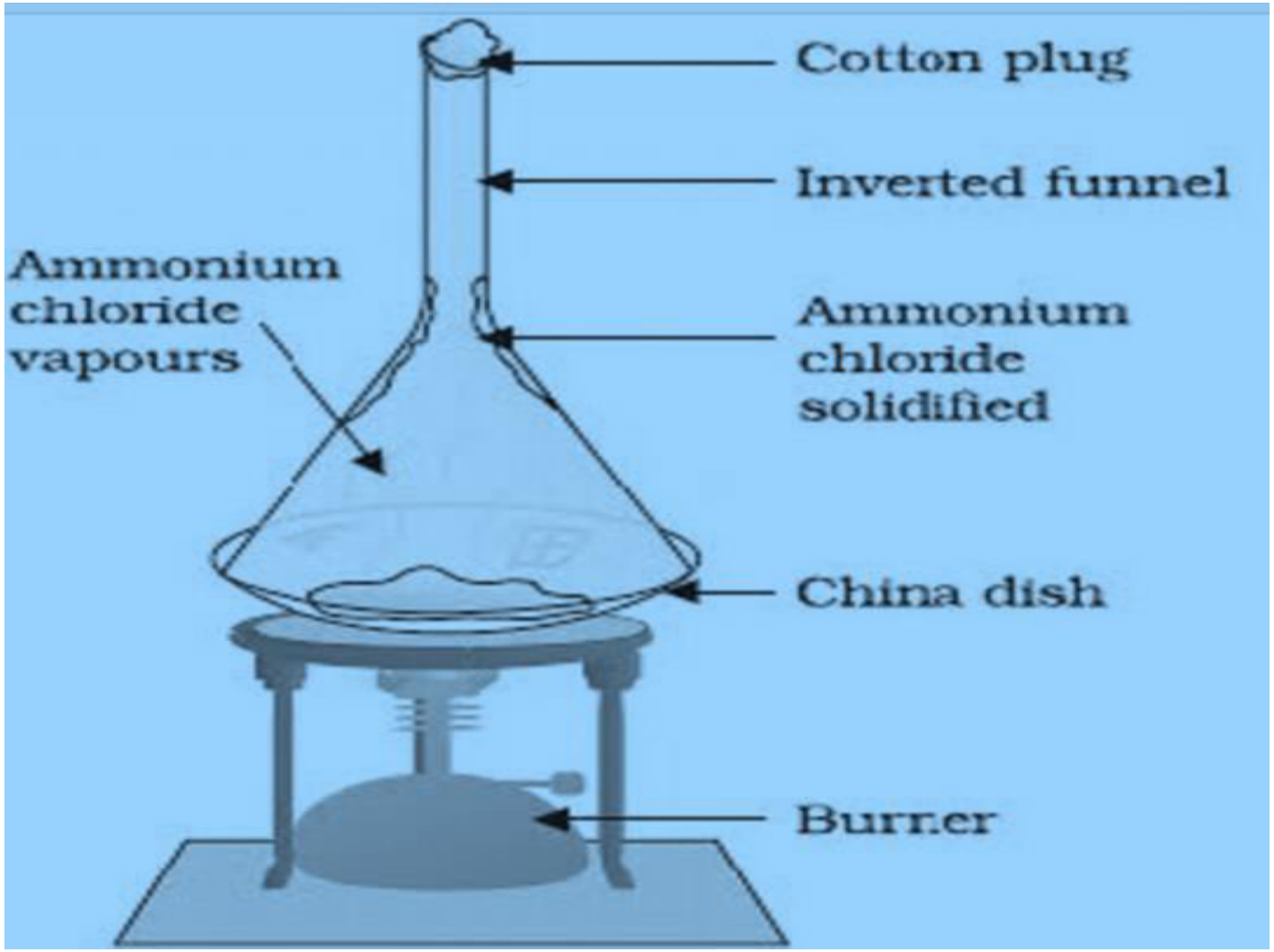


Sublimation Separation

- Sublimation is the transition of a substance directly from the solid to the gas phase, without passing through the intermediate liquid phase.
- This technique takes the advantage of substance's sublimely property.
- Separate a mixture of solids containing one which sublimes and one (or more than one) which does not, by heating the mixture.

Experiment showing the sublimation process

This experiment shows the separation of ammonium chloride from ammonium chloride salt.



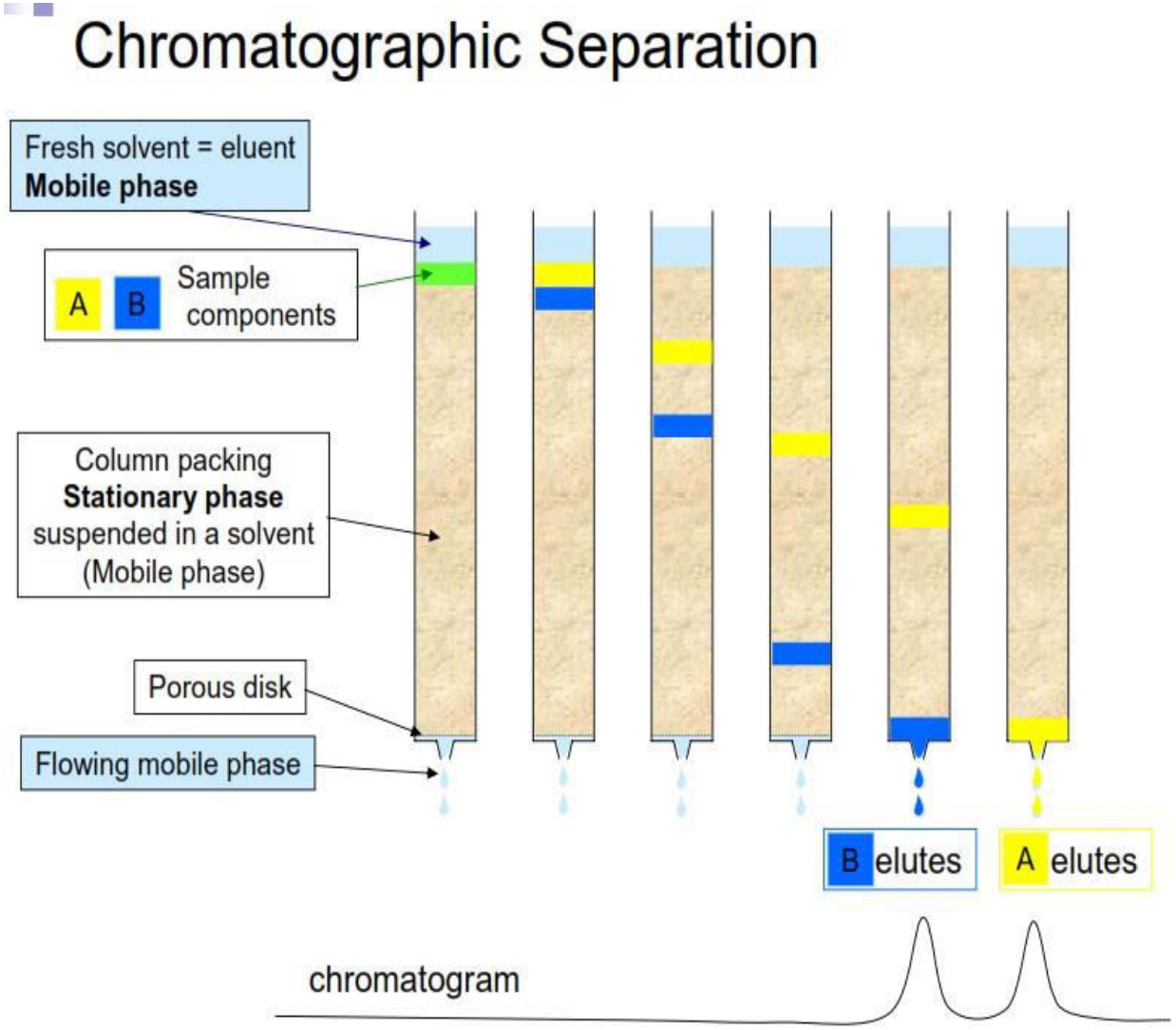
Applications

- Separate iodine from sand.
- Dye-sublimation printers help in rendering digital pictures in a detailed and realistic fashion which helps in the analysis of substances.
- Sublimation finds practical application in forensic sciences.
- Chemists usually prefer sublimation as a purification method to purify volatile compounds.

Chromatographic separation

- Chromatography is a separation technique used to separate the different components in a liquid mixture.
- The mixture is dissolved in a fluid called the mobile phase, which carries it through a structure holding another material called the stationary phase.
- Chromatography is vast separation technique which has many methods or has different principles of separation involved.

This picture shows the chromatographic separation of two samples showing yellow and blue color, stationary phase, mobile phase and porous disk allowing the elutes to move down for collection.



Applications

1. Food industry:

- Spoilage detection
- Additive detection
- Determining nutritional quality

2. Forensic

- Crime scenes testing
- Forensics pathology

3. Molecular biology studies

- Metabolomics and proteomics
- Nucleic acids research

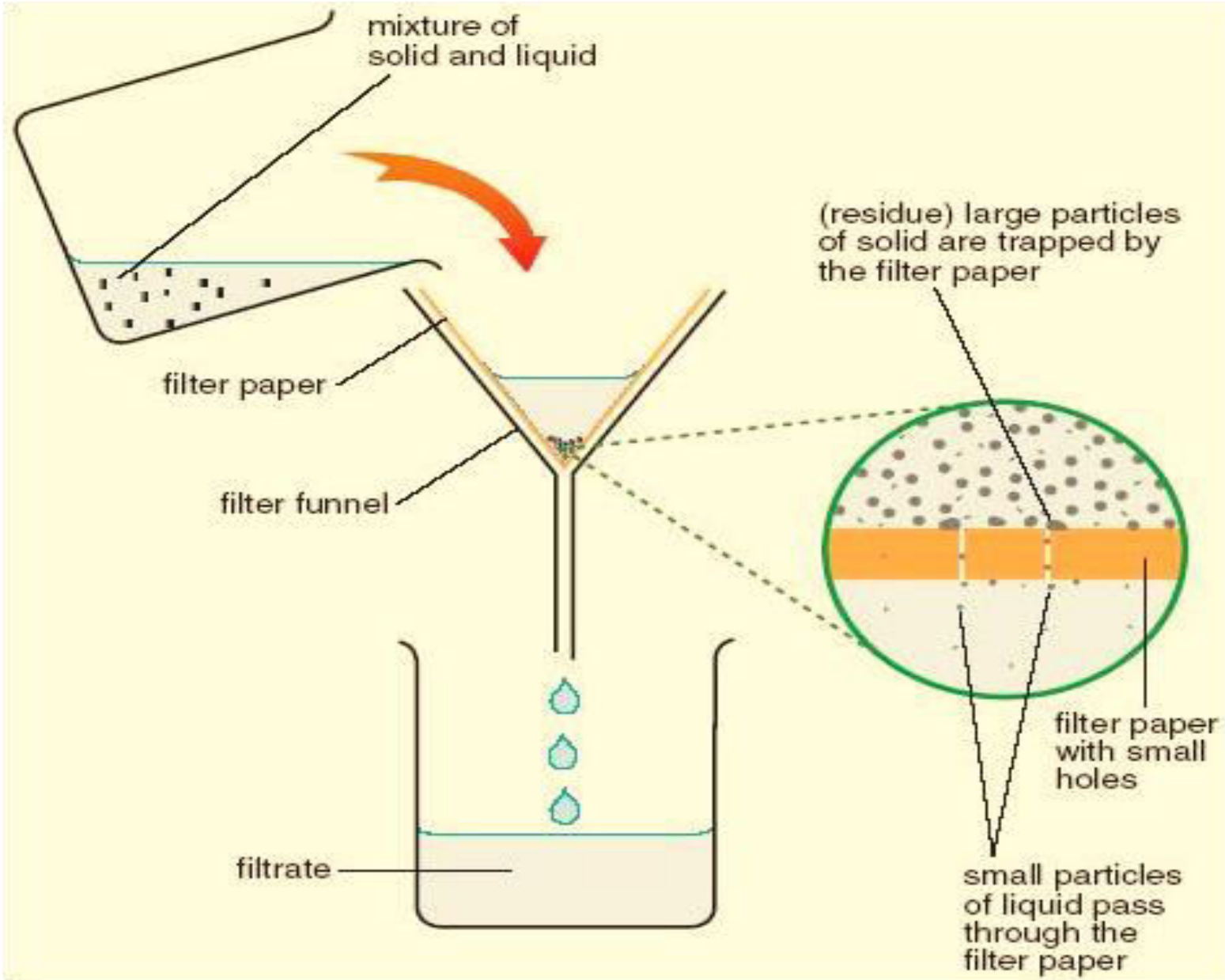
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- https://www.azolifesciences.com/article/Life-Science-Applications-of-Chromatography.aspx

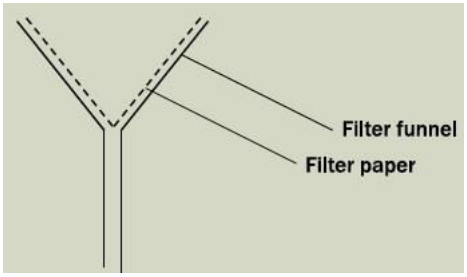
Lesson 43
FILTRATION

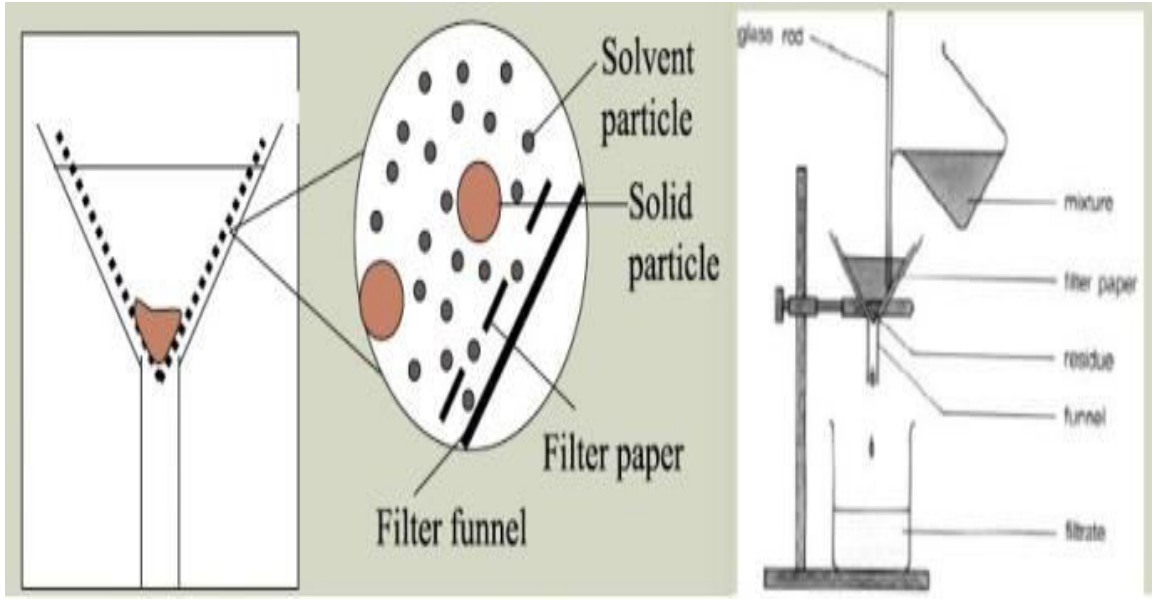
Filtration

- Filtration is the process of separating substances based on their different physical and chemical qualities. In this process, we refer to the collected solid material as the residue and the fluid material as the filtrate. Usually, a tool that contains some form of pores is used, which allows the fluid portion, but not the solid portion, to pass through. Different materials are used for the purpose of filtration, including paper, sand, and cloth. Filtration also occurs naturally in our bodies, for example in the kidneys, where the blood is filtered in a process called glomerular filtration.


mixture of
solid and liquid
filter paper
filter funnel
filtrate
(residue) large particles
of solid are trapped by
the filter paper
filter paper
with small
holes
small particles
of liquid pass
through the
filter paper

- The medium which we are using over here is the filter paper. The filter paper is folded and placed onto the filter funnel.
- The liquid-solid mixture is poured onto the filter paper. Using a filter paper with pores of a smaller size than the solid particles (and is larger than the size of the liquid molecules), the liquid (or solvent) should pass through the filter paper, and is collected by a collection container placed at the bottom of the filter funnel.
- The liquid that passes through the filter paper is called the filtrate while the solid left on the filter paper is called the residue

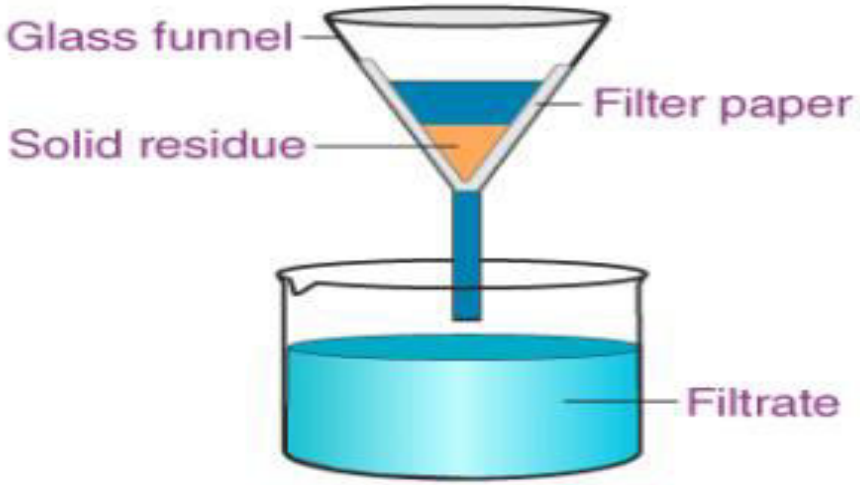




Filtration Examples

- The most common example is making tea. While preparing tea, a filter or a sieve is used to separate tea leaves from the water. Through the sieve pores, only water will pass. The liquid which has obtained after filtration is called the filtrate; in this case, water is the filtrate. The filter can be a paper, cloth, cotton-wool, asbestos, slag- or glass-wool, unglazed earthenware, sand, or any other porous material. Filtration is used in water treatment and sewage treatment.

Filtration Process



Filtration process

Filtration Process

- The mixtures are of two main types: homogeneous mixtures and heterogeneous mixtures. A homogeneous mixture is a mixture that is uniform throughout. A heterogeneous mixture is a mixture that is not uniform throughout, i.e., ingredients of the mixture are distributed unequally. Air is a homogeneous mixture of different gases, including oxygen, nitrogen, carbon dioxide, and water vapour.

Continue

- Homogeneous mixtures are sometimes also called solutions; especially when it is a mixture of a solid dissolved in a liquid. An example of a heterogeneous mixture is the mixture of sand in water. On shaking, sand will stay undissolved and are distributed unevenly. The sand particles floating around which will eventually settle to the bottom of the bottle makes it a heterogeneous mixture. Different types of filters are used to purify and for separation of mixtures from the contaminants. Based on the type of contaminant-large or small, filters of different pore sizes can be used, even at home.

REFERENCES

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LESSON 44

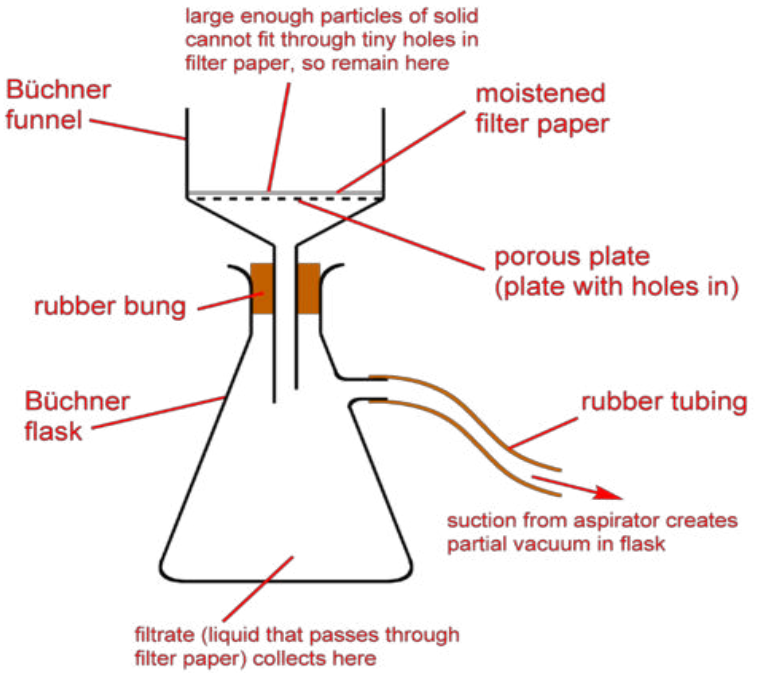
TYPES OF FILTERATION

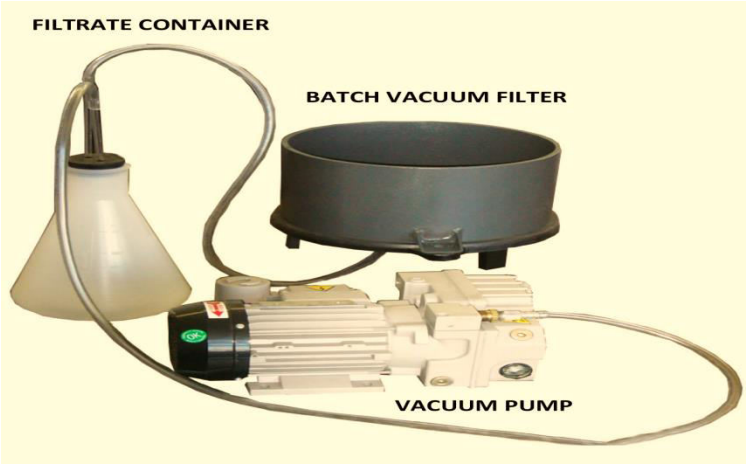
Types of filtration

- Vacuum Filtration
- Centrifugal filtration
- Gravity filtration
- Cold filtration
- Hot filtration
- Multilayer filtration

Vacuum Filtration

- In vacuum filtration, a vacuum pump is used to rapidly draw the fluid through a filter.
- Hirsch funnels and Buchner funnels, which are the same kind of funnel in two different sizes, are used along with filter paper.
- The funnels have a plate with holes in it, as we can see below, and they are usually used when the substance to be filtered is small in volume.



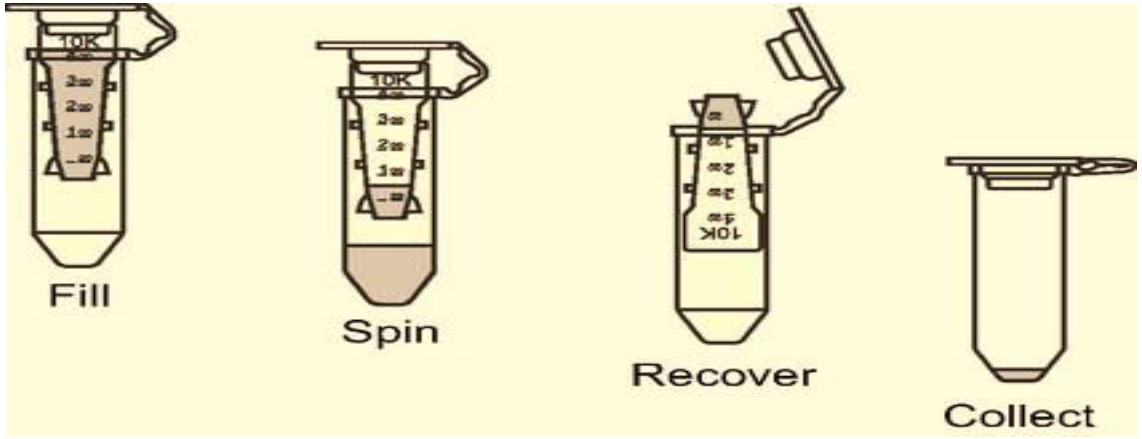


Multilayer filtration

- This can refer to multiple layers of different material, including sand, gravel, or charcoal, where the different layers contain different particle sizes of that material. In this type of filtration, a mixture of liquid and insoluble solid particles is poured over the layers, and the solid particles are caught throughout, resulting in a filtered liquid.

Centrifugal Filtration

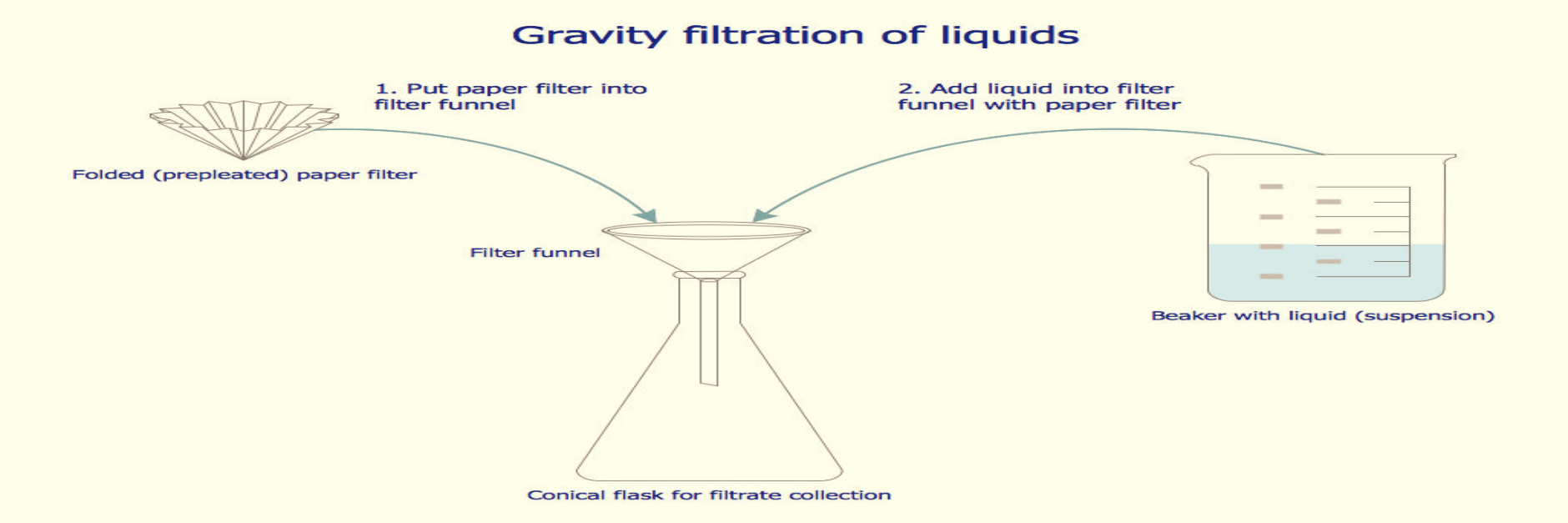
- This kind of filtration is done by rotating the substance to be filtered at very high speed.
- Due to the horizontal rotation, the more dense matter is separated from the less dense matter.





Gravity filtration

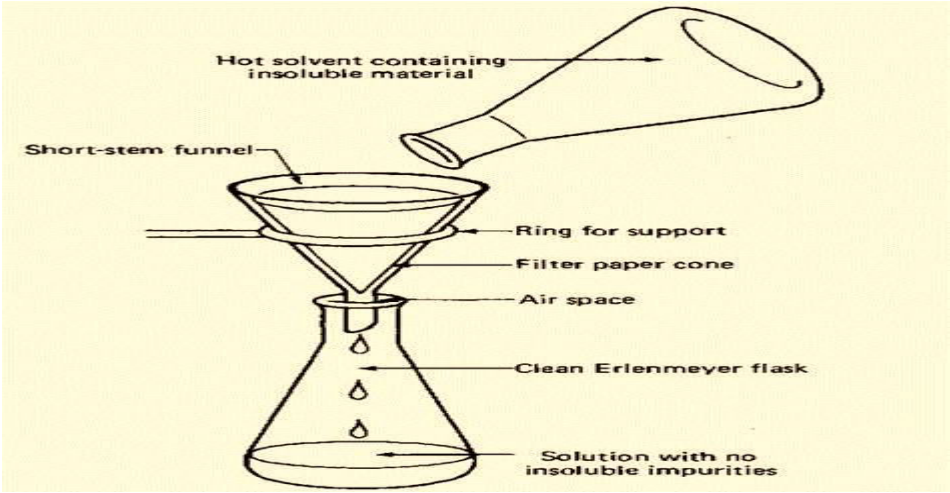
- This is where the mixture is poured from a higher point to a lower one. It is commonly done through simple filtration, using filter paper in a glass funnel, where the insoluble solid particles are captured by the filter paper and the liquid goes right through by gravity's pull. Depending on the volume of the substance at hand, filter cones, fluted filters, or filtering *pipets* can be used.



Cold filtration

- Cold filtration makes use of very low temperatures, often by using an ice bath. Some substances, such as fatty acid particles, become suspended in the mixture as they cool down, which then allows us to filter them out more easily.





Hot filtration

- This is often used for crystalline compounds that contain impurities. The way this filtration is done is by melting down the crystalline compound, removing the impurities as the substance is still in liquid form, and finally recrystallizing the now pure substance. Often, it is recommended that the apparatus used in this filtration be heated up so that the filtered substance doesn't crystallize in the funnel and block the flow.

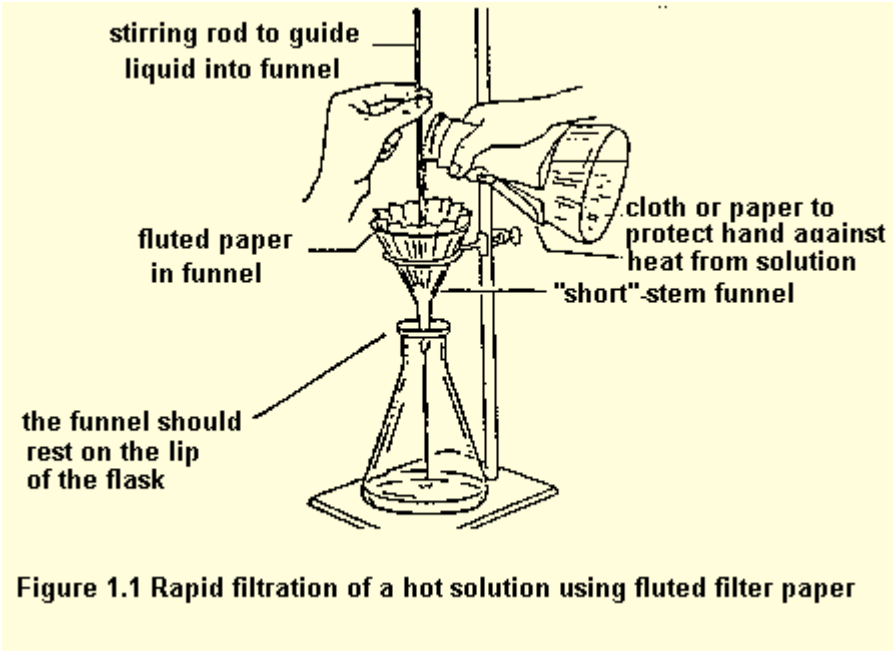


Figure 1.1 Rapid filtration of a hot solution using fluted filter paper

References

- https://biologydictionary.net/filtration/

LESSON# 45
Distillation

What is distillation?

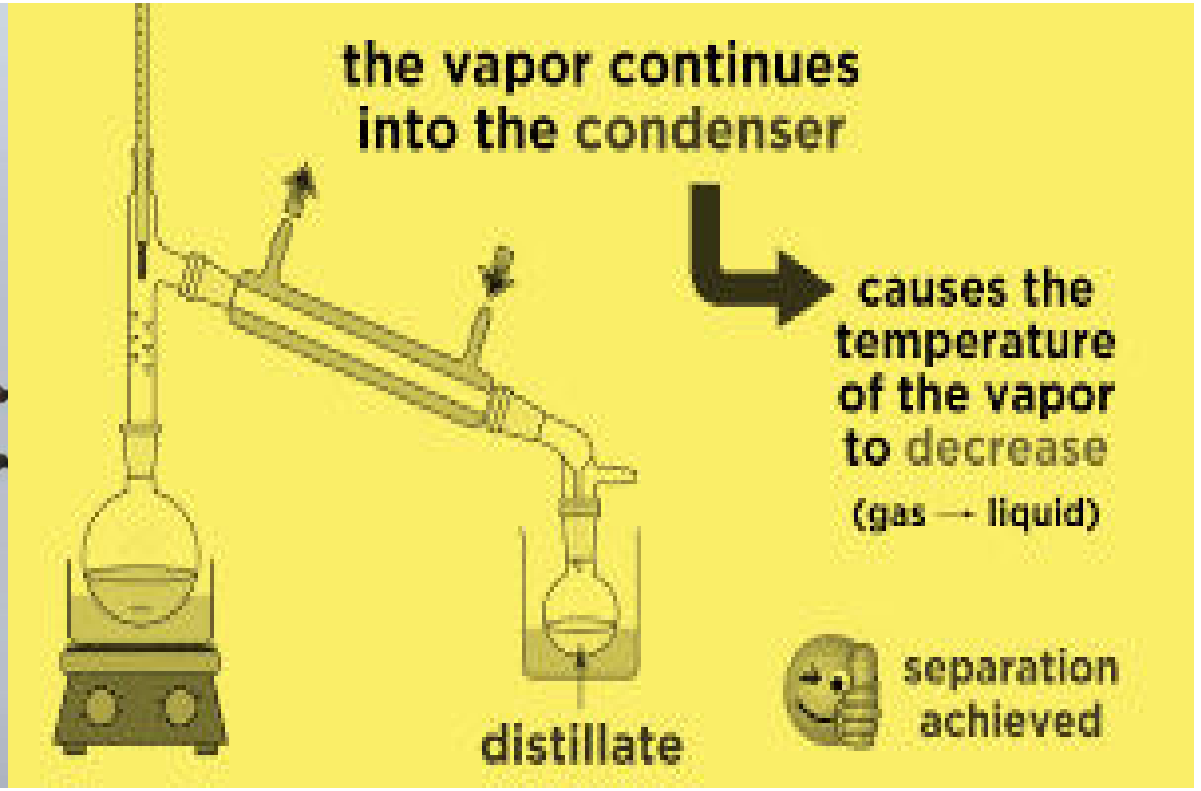
- Distillation is a widely used method for separating mixtures based on differences in the conditions required to change the phase of components of the mixture.
- To separate a mixture of liquids, the liquid can be heated to force components, which have different boiling points, into the gas phase. The gas is then condensed back into liquid form and collected. Repeating the process on the collected liquid to improve the purity of the product is called **double distillation**.

Although the term is most commonly applied to liquids, the reverse process can be used to separate gases by liquefying components using changes in temperature and/or pressure.

A plant that performs distillation is called a distillery. The apparatus used to perform distillation is called a still.

Plant for distillation





Applications of Distillation

Some important applications of distillation are listed below.

- Distillation plays an important role in many water purification techniques. Many desalination plants incorporate this method in order to obtain drinking water from seawater.
- Distilled water has numerous applications, such as in lead-acid batteries and low-volume humidifiers.
- Many fermented products such as alcoholic beverages are purified with the help of this method.
- Many perfumes and food flavorings are obtained from herbs and plants via distillation.

- Oil stabilization is an important type of distillation that reduces the vapor pressure of the crude oil, enabling safe storage and transportation.
- Air can be separated into nitrogen, oxygen, and argon by employing the process of cryogenic distillation.
- Distillation is also employed on an industrial scale to purify the liquid products obtained from chemical synthesis.

Processes involved in distillation

Distillation refining consists of two processes of distilling and condensation reflux. Distillation is usually carried out in distillation column, the gas-liquid two-phase flow through the countercurrent contact, the phase heat and mass transfer.

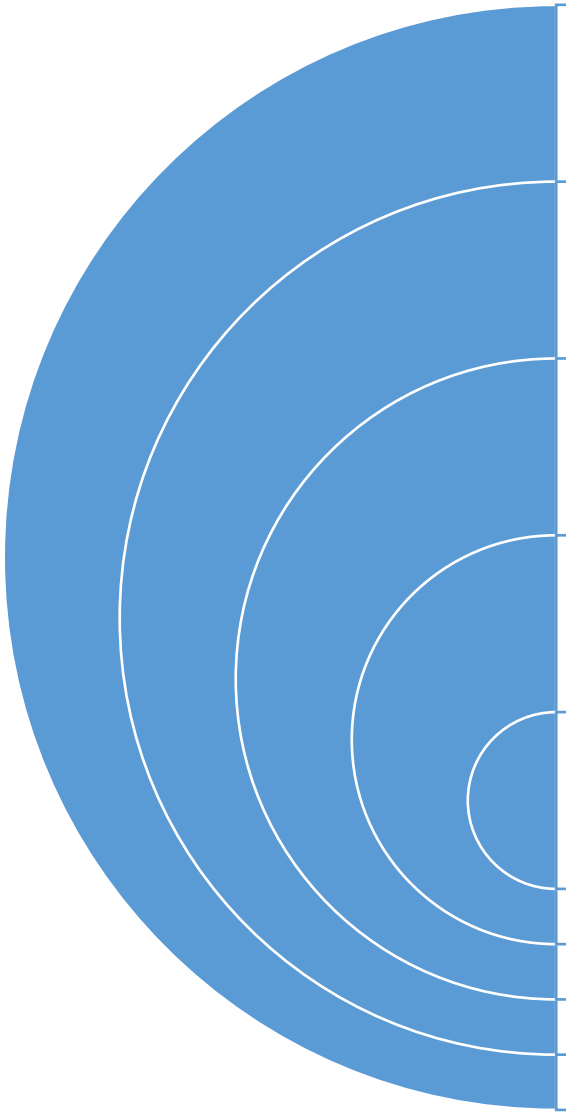
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LESSON 47

Chromatography, Types and their Principle

What is chromatography?



Chromatography is an important biophysical technique that enables the separation, identification, and purification of the components of a mixture for qualitative and quantitative analysis.

The Russian botanist Mikhail Tswett coined the term chromatography in 1906.

The first analytical use of chromatography was described by James and Martin in 1952, for the use of gas chromatography for the analysis of fatty acid mixtures.

A wide range of chromatographic procedures makes use of differences in size, binding affinities, charge, and other properties to separate materials.

It is a powerful separation tool that is used in all branches of science and is often the only means of separating components from complex mixtures

History

Chromatography gets its name from a technique first used in the late 19th century to separate pigments in a complex mixture.

If a sheet of paper or cloth contacts a container filled with water or alcohol in which a complex pigment is dissolved, capillary action will carry the mixture up the paper or cloth, but the components of the pigment will not all travel at the same rate.

The largest molecules of the mixture will travel more slowly while the smallest ones race ahead, causing the stationary phase to develop discrete bands of color corresponding to each component of the mixture. This gives the technique the name "chromatography" or "writing color.

Principle of Chromatography (how does chromatography work)

•Chromatography is based on the principle where molecules in mixture applied onto the surface or into the solid, and fluid stationary phase (stable phase) is separating from each other while moving with the aid of a mobile phase.

•The factors effective on this separation process include molecular characteristics related to adsorption (liquid-solid), partition (liquid-solid), and affinity or differences among their molecular weights.

•Because of these differences, some components of the mixture stay longer in the stationary phase, and they move slowly in the chromatography system, while others pass rapidly into the mobile phase, and leave the system faster.

Three components thus form the basis of the chromatography technique.

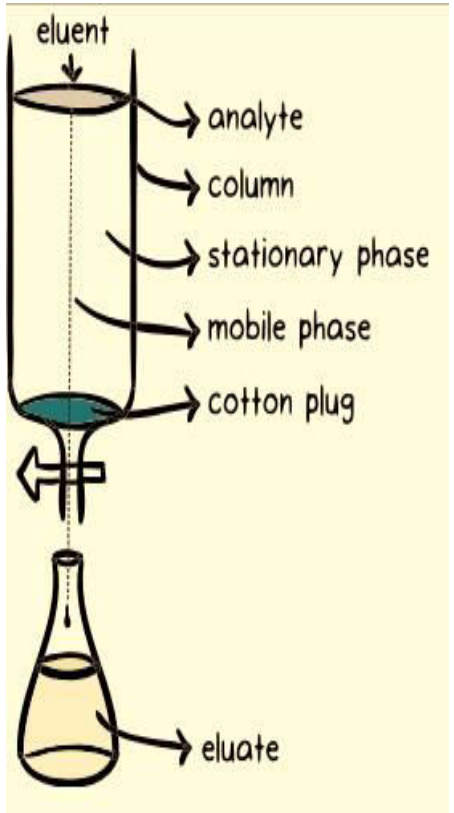
1.Stationary phase: This phase is always composed of a “solid” phase or “a layer of a liquid adsorbed on the surface solid support”.

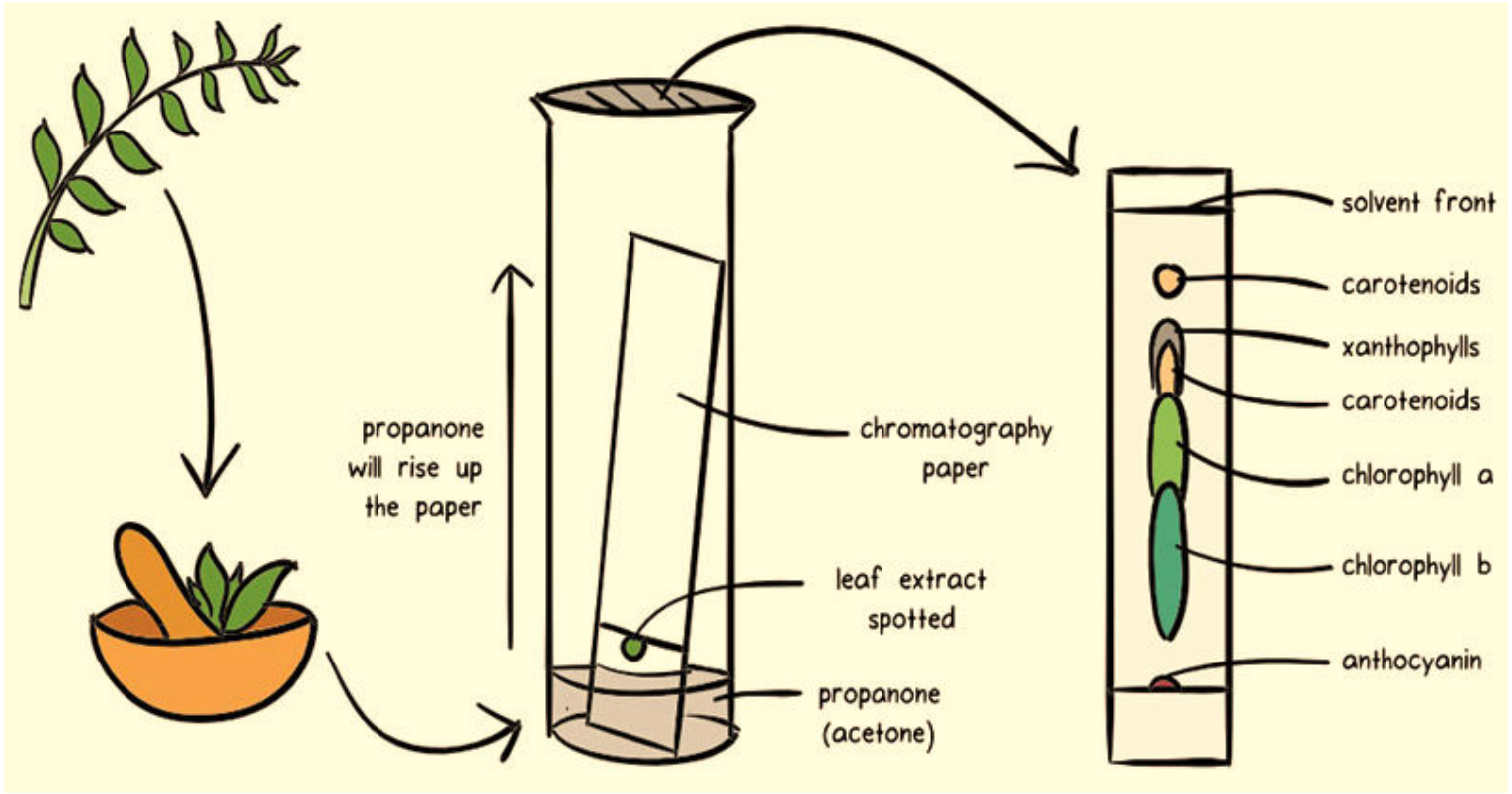
2.Mobile phase: This phase is always composed of “liquid” or a “gaseous component.”

3.Separated molecules

The type of interaction between the stationary phase, mobile phase, and substances contained in the mixture is the basic component effective on the separation of molecules from each other.

Experimental elaboration of chromatography



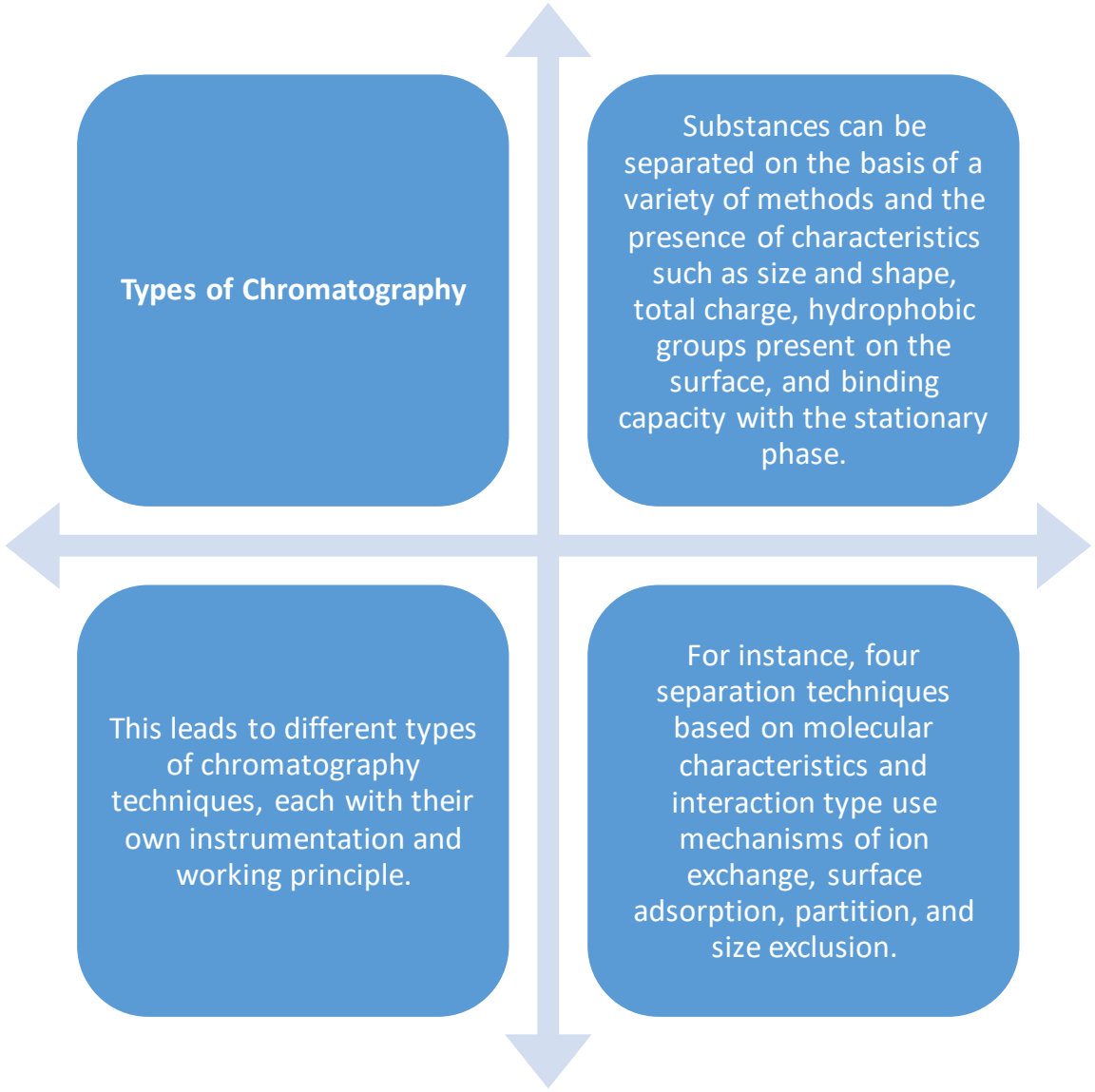


Components of Chromatography

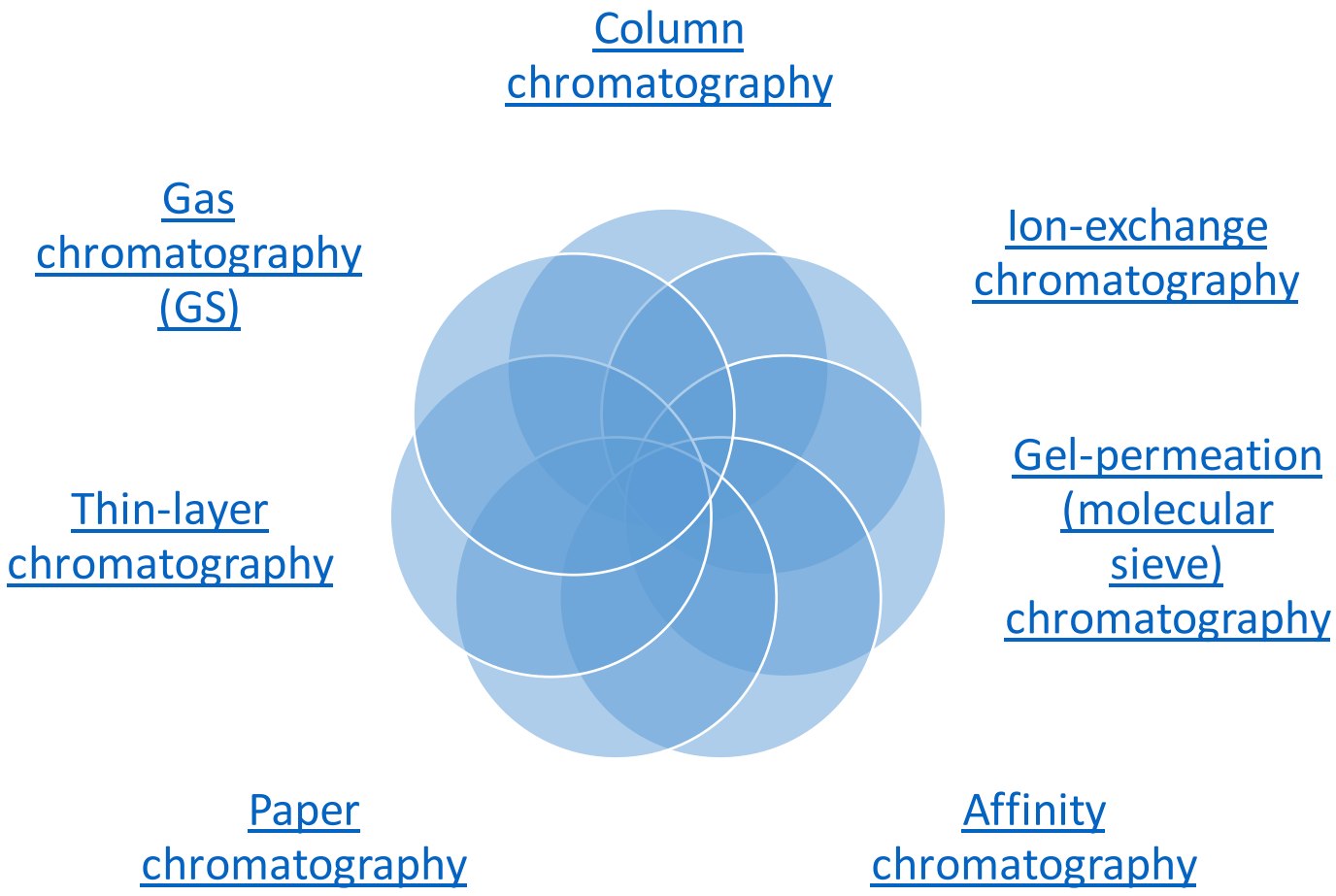
- Three components thus form the basis of the chromatography technique.
- **Stationary phase:** This phase is always composed of a “solid” phase or “a layer of a liquid adsorbed on the surface solid support”.
- **Mobile phase:** This phase is always composed of “liquid” or a “gaseous component.”
- **Separated molecules**
- The type of interaction between the stationary phase, mobile phase, and substances contained in the mixture is the basic component effective on the separation of molecules from each other

Types of chromatography

- Gas chromatography
- Liquid chromatography
- Thin layer chromatography
- Ion-exchange chromatography


Types of Chromatography
Substances can be separated on the basis of a variety of methods and the presence of characteristics such as size and shape, total charge, hydrophobic groups present on the surface, and binding capacity with the stationary phase.
This leads to different types of chromatography techniques, each with their own instrumentation and working principle.
For instance, four separation techniques based on molecular characteristics and interaction type use mechanisms of ion exchange, surface adsorption, partition, and size exclusion.

Commonly employed chromatography techniques include:

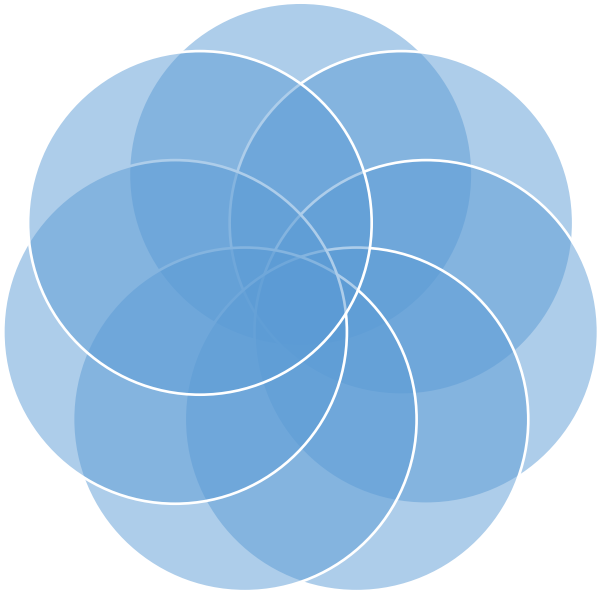


Applications of Chromatography

Pharmaceutical sector

To identify and analyze samples for the presence of trace elements or chemicals.

In testing water samples and also checks air quality.



Separation of compounds based on their molecular weight and element composition.

Chemical industry

In drug development.

Detects the unknown compounds and purity of mixture.

Forensic Science

•In forensic pathology and crime scene testing like analyzing blood and hair samples of crime place.

Molecular Biology Studies

•Various hyphenated techniques in chromatography such as EC-LC-MS are applied in the study of metabolomics and proteomics along with nucleic acid research.

•HPLC is used in Protein Separation like Insulin Purification, Plasma Fractionation, and Enzyme Purification and also in various departments like Fuel Industry, biotechnology, and biochemical processes.

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LESSON 48
PAPER CHROMATOGRAPHY

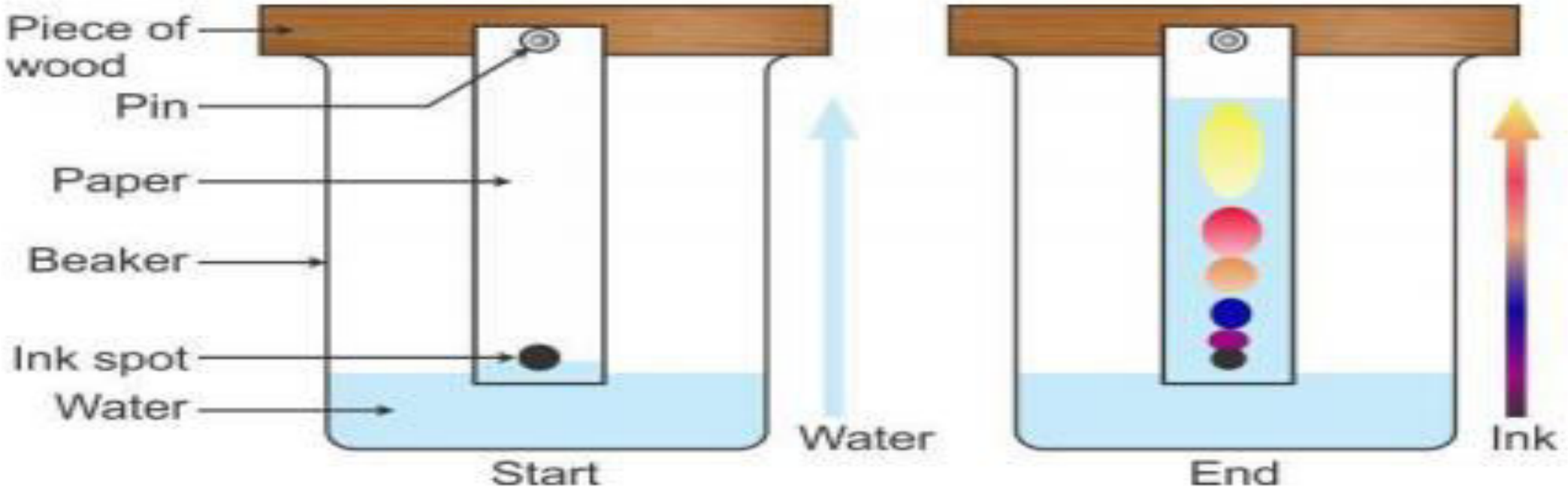
PAPER CHROMATOGRAPHY

- Chromatography technique that uses paper sheets or strips as the adsorbent being the stationary phase through which a solution is made to pass is called paper chromatography. It is an inexpensive method of separating dissolved chemical substances by their different migration rates across the sheets of paper. It is a powerful analytical tool that uses very small quantities of material. Paper chromatography was discovered by Synge and Martin in the year 1943.

Paper Chromatography Principle

- The principle involved can be partition chromatography or adsorption chromatography. Partition chromatography because the substances are partitioned or distributed between liquid phases. The two phases are water held in pores of the filter paper and the other phase is a mobile phase which passes through the paper. When the mobile phase moves, the separation of the mixture takes place. The compounds in the mixture separate themselves based on the differences in their affinity towards stationary and mobile phase solvents under the capillary action of pores in the paper. Adsorption chromatography between solid and liquid phases, wherein the solid surface of the paper is the stationary phase and the liquid phase is the mobile phase.

Paper Chromatography Diagram



Paper Chromatography

Types of paper chromatography

- Ascending Paper Chromatography – The techniques goes with its name as the solvent moves in an upward direction.
- Descending Paper Chromatography – The movement of the flow of solvent due to gravitational pull and capillary action is downwards hence the name descending paper chromatography.
- Ascending – Descending Paper Chromatography – In this version of paper chromatography movement of solvent occurs in two directions after a particular point. Initially, the solvent travels upwards on the paper which is folded over a rod and after crossing the rod it continues with its travel in the downward direction.

Types

Radial or Circular Paper Chromatography – The sample is deposited at the centre of the circular filter paper. Once the spot is dried, the filter paper is tied horizontally on a Petri dish which contains the solvent.

Two Dimensional Paper Chromatography – Substances which have the same rf values can be resolved with the help of two-dimensional paper chromatography.

Applications of paper chromatography

- There are various applications of paper chromatography. Some of the uses of Paper Chromatography in different fields are discussed below:
- To study the process of fermentation and ripening.
- To check the purity of pharmaceuticals.
- To inspect cosmetics.
- To detect the adulterants.
- To detect the contaminants in drinks and foods.
- To examine the reaction mixtures in biochemical laboratories.
- To determine dopes and drugs in humans and animals

LESSON 49
Thin-layer chromatography

Thin-layer chromatography

Thin-layer chromatography is a “solid-liquid adsorption” chromatography. In this physical method of separation, the components to be separated are distributed between two phases, one of which is stationary (stationary phase) while the other (the mobile phase) moves in a definite direction. Depending upon the stationary phase and mobile phase chosen, they can be of different types.

Thin Layer Chromatography can be defined as a method of separation or identification of a mixture of components into individual components by using finely divided adsorbent solid / (liquid) spread over a plate and liquid as a mobile phase.

Principle of Thin Layer Chromatography (TLC)

- Thin-layer chromatography is performed on a sheet of glass, plastic, or aluminium foil, which is coated with a thin layer of adsorbent material, usually silica gel, aluminium oxide (alumina), or cellulose. This layer of adsorbent is known as the stationary phase.
- After the sample has been applied on the plate, a solvent or solvent mixture (known as the mobile phase) is drawn up the plate via capillary action. Because different analytes ascend the TLC plate at different rates, separation is achieved.
- It is thus based on the principle of adsorption chromatography or partition chromatography or combination of both, depending on adsorbent, its treatment and nature of solvents employed. The components with more affinity towards stationary phase travels slower. Components with less affinity towards stationary phase travels faster.
- Once separation occurs, the individual components are visualized as spots at a respective level of travel on the plate. Their nature or character is identified by means of suitable detection techniques.

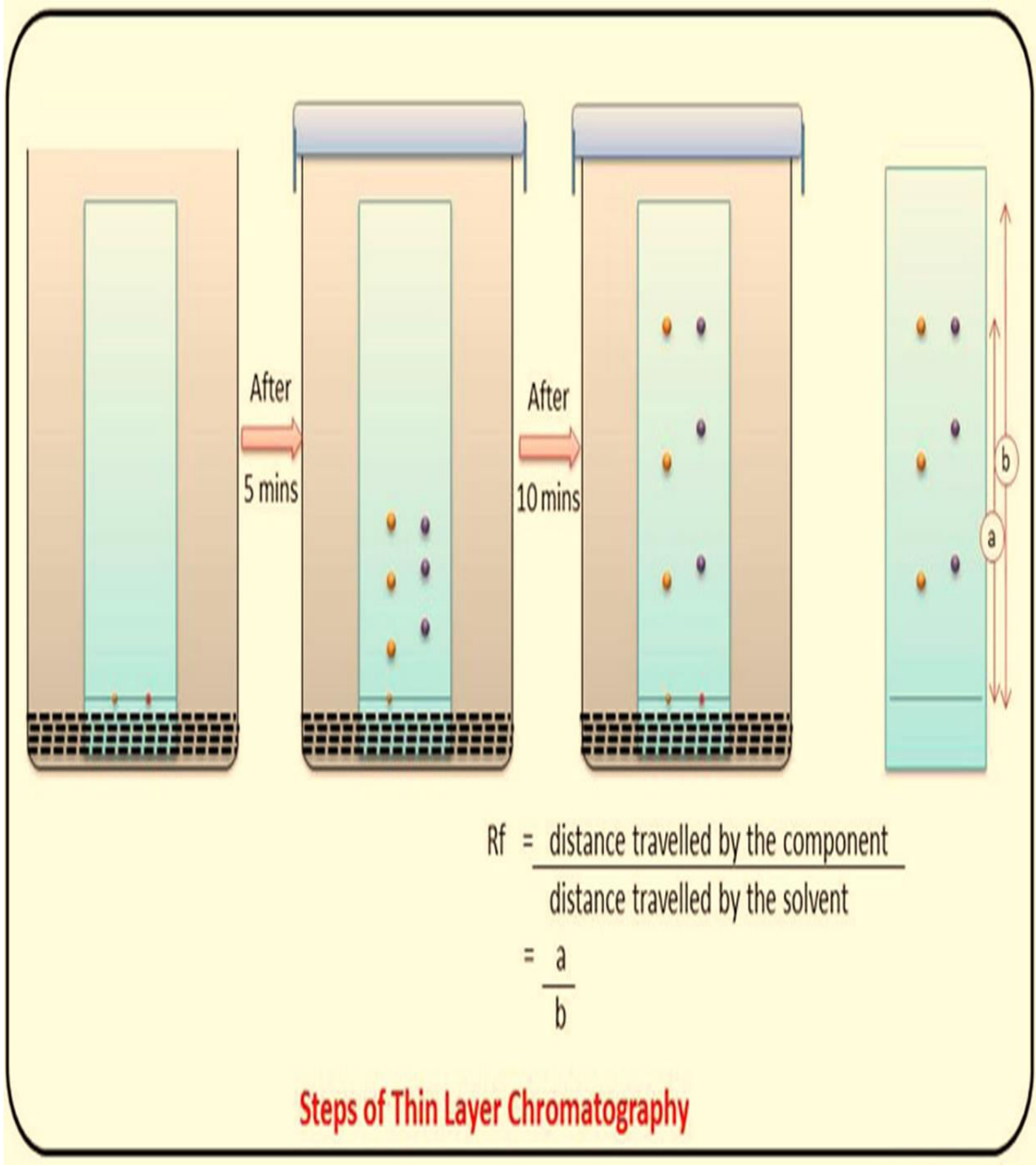
TLC components

- TLC system components consists of:
- **TLC plates,** preferably ready made with a stationary phase: These are stable and chemically inert plates, where a thin layer of stationary phase is applied on its whole surface layer. The stationary phase on the plates is of uniform thickness and is in a fine particle size.
- **TLC chamber-** This is used for the development of TLC plate. The chamber maintains a uniform environment inside for proper development of spots. It also prevents the evaporation of solvents, and keeps the process dust free.
- **Mobile phase-** This comprises of a solvent or solvent mixture The mobile phase used should be particulate-free and of the highest purity for proper development of TLC spots. The solvents recommended are chemically inert with the sample, a stationary phase.
- **A filter paper-** This is moistened in the mobile phase, to be placed inside the chamber. This helps develop a uniform rise in a mobile phase over the length of the stationary phase.

Procedure of Thin Layer Chromatography (TLC)

- The stationary phase is applied onto the plate uniformly and then allowed to dry and stabilize. These days, however, ready-made plates are more commonly used.
- With a pencil, a thin mark is made at the bottom of the plate to apply the sample spots.
- Then, samples solutions are applied on the spots marked on the line in equal distances.
- The mobile phase is poured into the TLC chamber to a leveled few centimeters above the chamber bottom.
- A moistened filter paper in mobile phase is placed on the inner wall of the chamber to maintain equal humidity (and also thereby avoids edge effect).
- Now, the plate prepared with sample spotting is placed in TLC chamber so that the side of the plate with the sample line is facing the mobile phase. Then the chamber is closed with a lid.

- The plate is then immersed, such that the sample spots are well above the level of mobile phase (but not immersed in the solvent) for development.
- Sufficient time is given for the development of spots.
- The plates are then removed and allowed to dry.
- The sample spots are then seen in a suitable UV light chamber, or any other methods as recommended for the given sample.



Some common techniques for visualizing the results of a TLC plate include

UV light
Iodine Staining: is very useful in detecting carbohydrates since it turns black on contact with Iodine
KMnO4 stain (organic molecules)
Ninhydrin Reagent: often used to detect amino acids and proteins

Retention Factor (R_f) Value

The behaviour of a compound on a TLC is usually described in terms of its relative mobility or Rf value.
Rf or Retention factor is a unique value for each compound under the same conditions.
The Rf for a compound is a constant from one experiment to the next only if the chromatography conditions below are also constant:
solvent system
adsorbent
thickness of the adsorbent
amount of material spotted
temperature
Since these factors are difficult to keep constant from experiment to experiment, relative Rf values are generally considered.
Relative Rf" means that the values are reported relative to a standard.

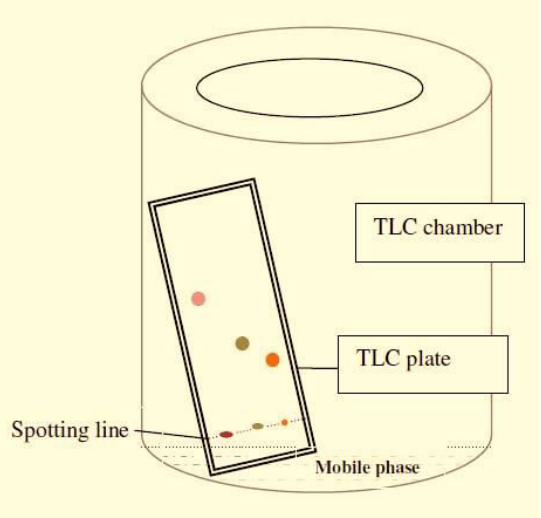
Applications of Thin Layer Chromatography (TLC)

In monitoring the progress of reactions
Identify compounds present in a given mixture
Determine the purity of a substance.
Analyzing ceramides and fatty acids
Detection of pesticides or insecticides in food and water
Analyzing the dye composition of fibers in forensics
Assaying the radiochemical purity of radiopharmaceuticals
Identification of medicinal plants and their constituents

In cases where molecules of the sample are colorless, florescence, radioactivity or a specific chemical substance can be used to produce a visible colored reactive product so as to identify their positions on the chromatogram.

Formation of a visible color can be observed under room light or UV light. The position of each molecule in the mixture can be measured by calculating the ratio between the distances travelled by the molecule and the solvent.

This measurement value is called relative mobility, and expressed with a symbol R_f. R_f. value is used for qualitative description of the molecules

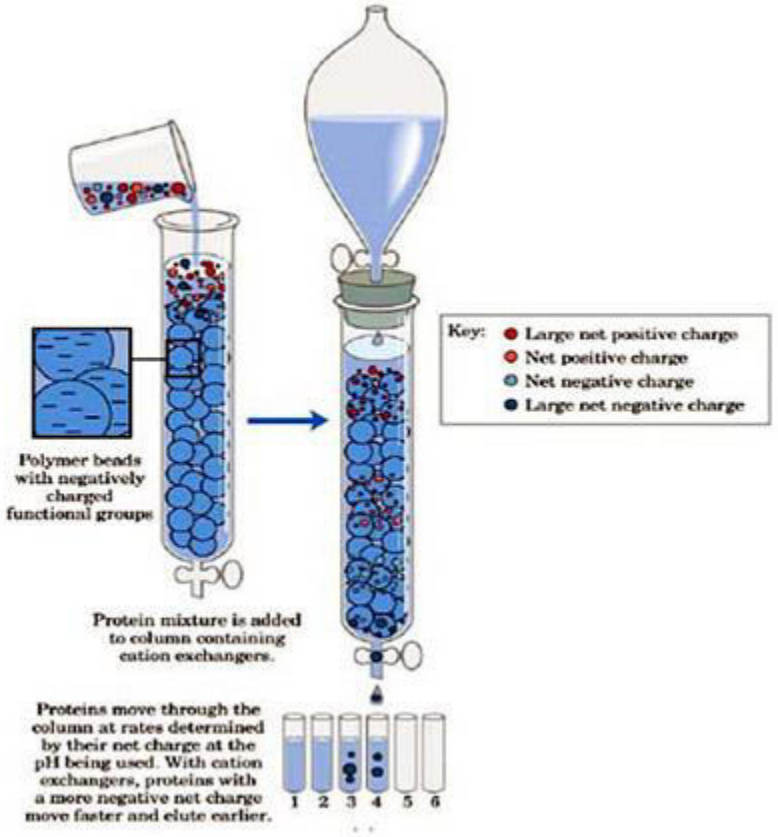


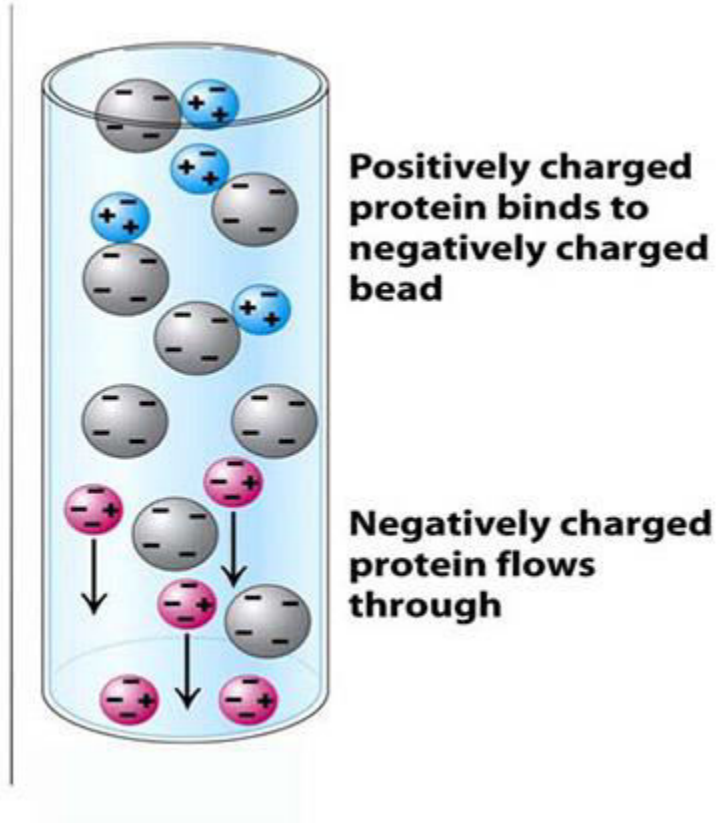
LESSON 50

Ion exchange chromatography

Ion exchange chromatography (or ion chromatography) is a process that allows the separation of ions and polar molecules based on their affinity to ion exchangers. The principle of separation is thus by reversible exchange of ions between the target ions present in the sample solution to the ions present on ion exchangers. In this process two types of exchangers i.e., cationic and anionic exchangers can be used.

Ion exchange chromatography

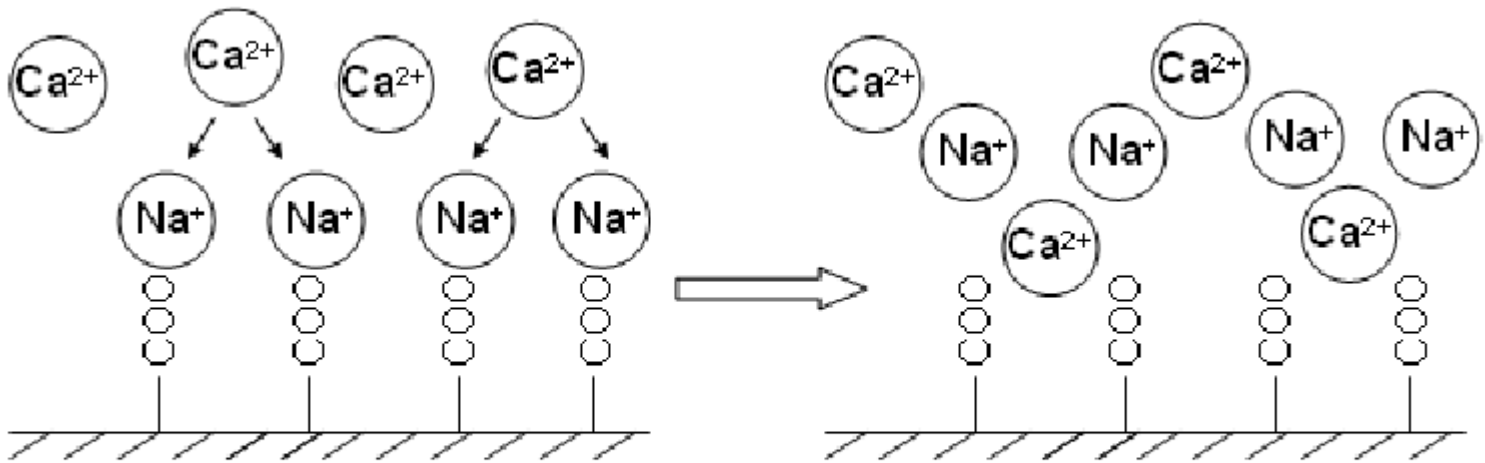




Cationic exchangers

- Cationic exchangers possess negatively charged group, and these will attract positively charged cations. These exchangers are also called “Acidic ion exchange” materials, because their negative charges result from the ionization of acidic group.

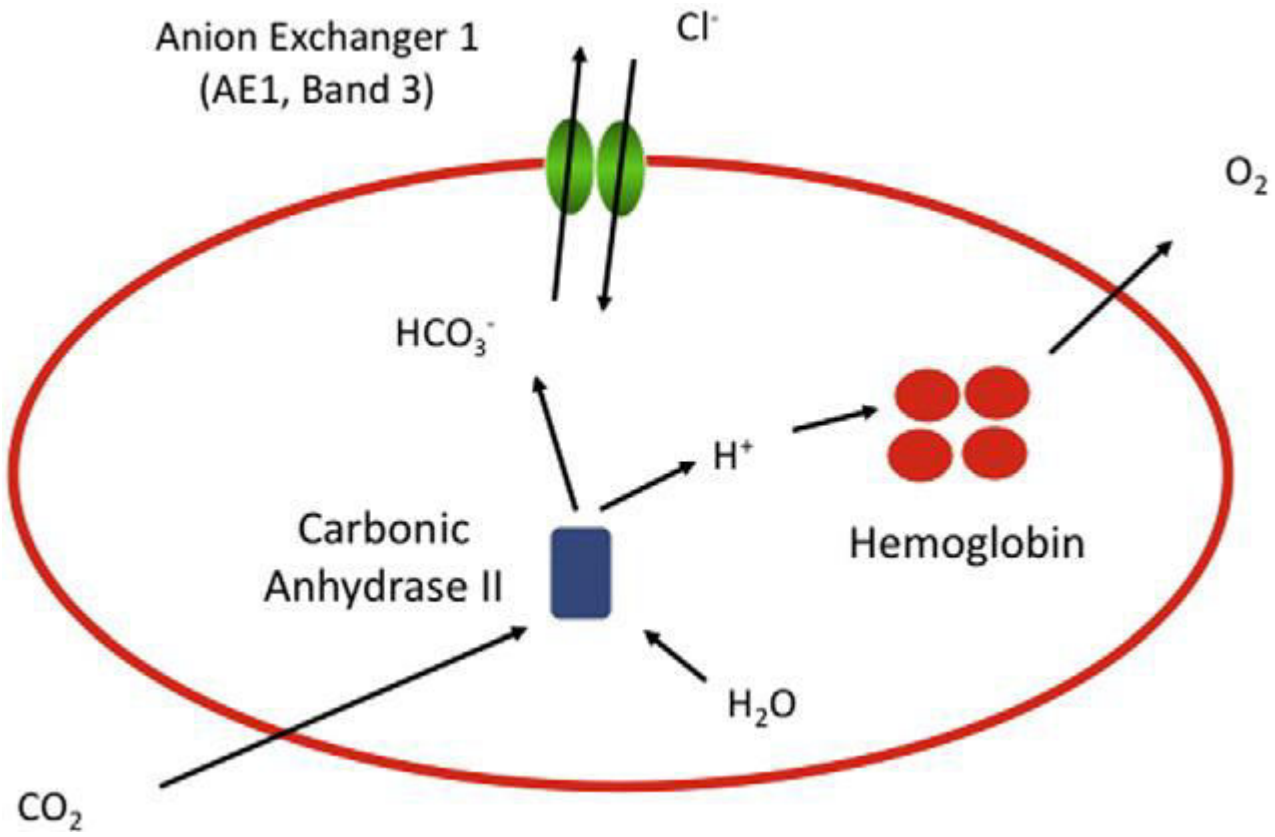
Mechanism of cation exchange



Anionic exchangers

Anionic exchangers have positively charged groups that will attract negatively charged anions. These are also called “Basic ion exchange” materials. Ion exchange chromatography is most often performed in the form of column chromatography. However, there are also thinlayer chromatographic methods that work basically based on the principle of ion exchange.

Mechanism of cation exchange



Working Principle of ion exchange chromatography

This form of chromatography relies on the attraction between oppositely charged stationary phase, known as an ion exchanger, and analyte. The ion exchangers basically contain charged groups covalently linked to the surface of an insoluble matrix.

The charged groups of the matrix can be positively or negatively charged.

When suspended in an aqueous solution, the charged groups of the matrix will be surrounded by ions of the opposite charge.

In this “ion cloud”, ions can be reversibly exchanged without changing the nature and the properties of the matrix

Applications of ion exchange chromatography

- An important use of ion-exchange chromatography is in the routine analysis of amino acid mixtures.
- The 20 principal amino acids from blood serum or from the hydrolysis of proteins are separated and used in clinical diagnosis.
- This is most effective method for water purification. Complete deionization of water (or) a non-electrolyte solution is performed by exchanging solute cations for hydrogen ions and solute anions for hydroxyl ions. This is usually achieved by method is used for softening of drinking water.

Applications

- In the analysis of products of hydrolysis of nucleic acids. In this way, information is gained about the structure of these molecules and how it relates to their biological function as carriers of hereditary information.
- Chelating resins are used to collect trace metals from seawater.
- To analyze lunar rocks and rare trace elements on Earth.

Limitations of ion exchange chromatography

- Only charged molecules can be separated
- Buffer Requirement

Lesson 51
High performance liquid chromatography (HPLC)

High performance liquid chromatography

- HPLC stands for “High performance liquid chromatography”(sometimes referred to as High pressure chromatography).

- High performance liquid chromatography is a powerful tool in analysis, it yields high performance and high speed compared to traditional columns chromatography because of the forcibly pumped mobile phase.

- HPLC is a chromatographic technique that can separate a mixture of compounds.
- It is used in biochemistry and analytical chemistry to identify, quantify and purify the individual.

HPLC

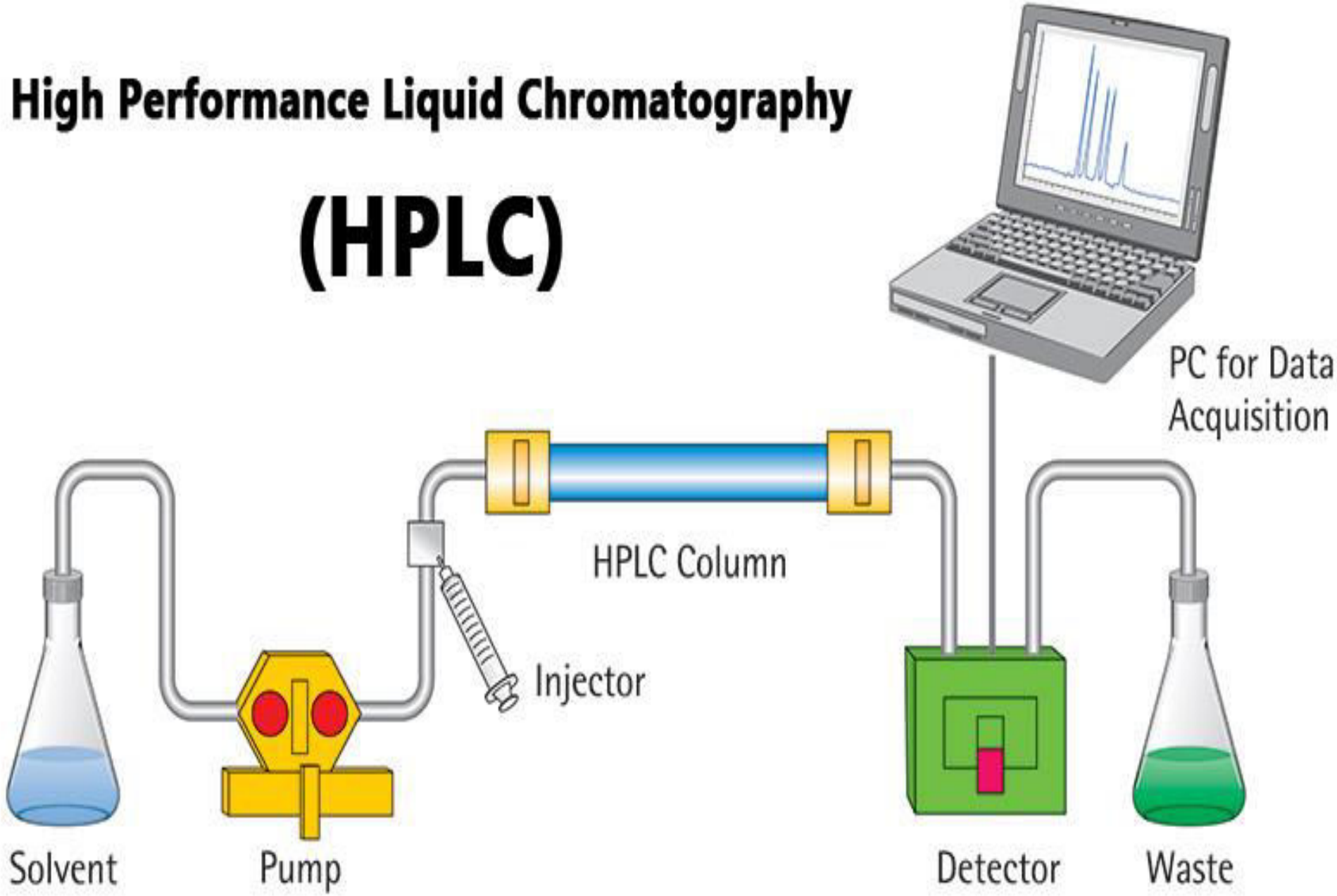
- **Chromatography :** physical method in which separation of components take place between two phases –a stationary phase and a mobile phase.

- **Stationary phase:-** The substance on which phase adsorption of analyte(the substance to be separated during chromatography) take place.IT can be a solid ,a gel, or a solid liquid combination

- **Mobile phase:-** Solvent which carries the analyte (a liquid or a gas)

High Performance Liquid Chromatography

(HPLC)



Principle of High-Performance Liquid Chromatography (HPLC)

- The purification takes place in a separation column between a stationary and a mobile phase.
- The stationary phase is a granular material with very small porous particles in a separation column.
- The mobile phase, on the other hand, is a solvent or solvent mixture which is forced at high pressure through the separation column.

Principle

❖Via a valve with a connected sample loop, i.e. a small tube or a capillary made of stainless steel, the sample is injected into the mobile phase flow from the pump to the separation column using a syringe.

❖Subsequently, the individual components of the sample migrate through the column at different rates because they are retained to a varying degree by interactions with the stationary phase.

❖After leaving the column, the individual substances are detected by a suitable detector and passed on as a signal to the HPLC software on the computer.

❖At the end of this operation/run, a chromatogram in the HPLC software on the computer is obtained.

Types of High-Performance Liquid Chromatography (HPLC)

- **Normal phase :** Column packing is polar (e.g. silica) and the mobile phase is non-polar. It is used for water-sensitive compounds, geometric isomers, cistrans isomers, and chiral compounds.
- **Reverse phase :**The column packing is non-polar (e.g. C18), the mobile phase is water+ miscible solvent (e.g. methanol). It can be used for polar, non-polar, ionizable and ionic samples.

Types

❖**Ion exchange:** Column packing contains ionic groups and the mobile phase is buffer. It is used to separate anions and cations.

❖**Size exclusion :** Molecules diffuse into pores of a porous medium and are separated according to their relative size to the pore size. Large molecules elute first and smaller molecules elute later.

Applications of High-Performance Liquid Chromatography (HPLC)

- The HPLC has developed into a universally applicable method so that it finds its use in almost all areas of chemistry, biochemistry, and pharmacy.
- Analysis of drugs
- Analysis of synthetic polymers
- Analysis of pollutants in environmental analytics
- Determination of drugs in biological matrices
- Isolation of valuable products
- Product purity and quality control of industrial products and fine chemicals

Applications of High-Performance Liquid Chromatography (HPLC)

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Applications

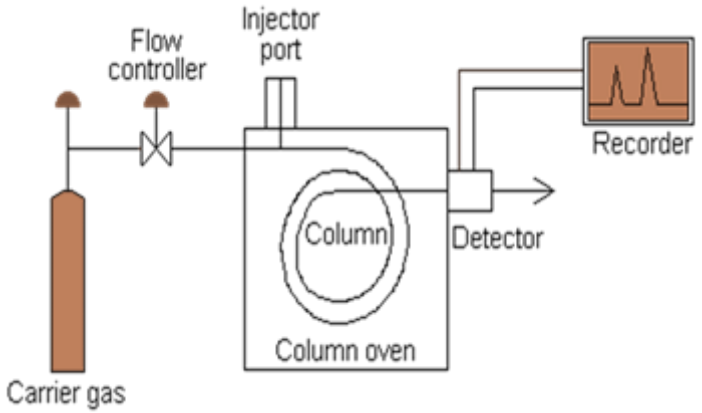
- Separation and purification of biopolymers such as enzymes or nucleic acids
- Water purification
- Pre-concentration of trace components
- Ligand-exchange chromatography
- Ion-exchange chromatography of proteins
- High-pH anion-exchange chromatography of carbohydrates and oligosaccharides

LESSON 52

Gas Chromatography

Gas Chromatography

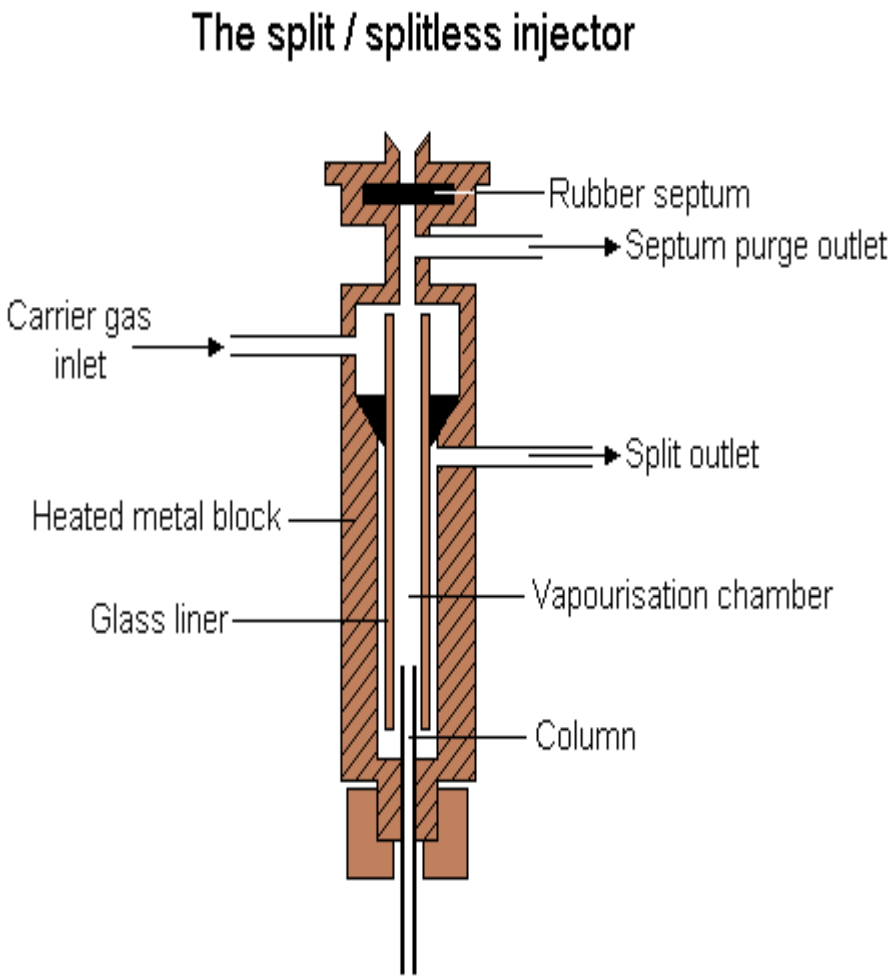
- In this method stationary phase is a column which is placed in the device, and contains a liquid stationary phase which is adsorbed onto the surface of an inert solid. Gas chromatography is a "gas-liquid" chromatography.
- Its carrier phase consists of gases as He or N_2. Mobile phase which is an inert gas is passed through a column under high pressure. The sample to be analyzed is vaporized, and enters into a gaseous mobile phase .
- The components contained in the sample are dispersed between mobile phase, and stationary phase on the solid support.
- Gas chromatography is a simple, multifaceted, highly sensitive, and rapidly applied technique for the extremely excellent separation of very minute molecules.



Have a look at this schematic diagram of a gas chromatograph

- **Instrumental components**
- **Carrier gas:** The carrier gas must be chemically inert. Commonly used gases include nitrogen, helium, argon, and carbon dioxide. The choice of carrier gas is often dependent upon the type of detector which is used. The carrier gas system also contains a molecular sieve to remove water and other impurities.
- **Sample injection port:** For optimum column efficiency, the sample should not be too large, and should be introduced onto the column as a "plug" of vapor - slow injection of large samples causes band broadening and loss of resolution.
- The most common injection method is where a micro syringe is used to inject sample through a rubber septum into a flash vaporizer port at the head of the column. The temperature of the sample port is usually about 50°C higher than the boiling point of the least volatile component of the sample. For packed columns, sample size ranges from tenths of a microliter up to 20 microliters.

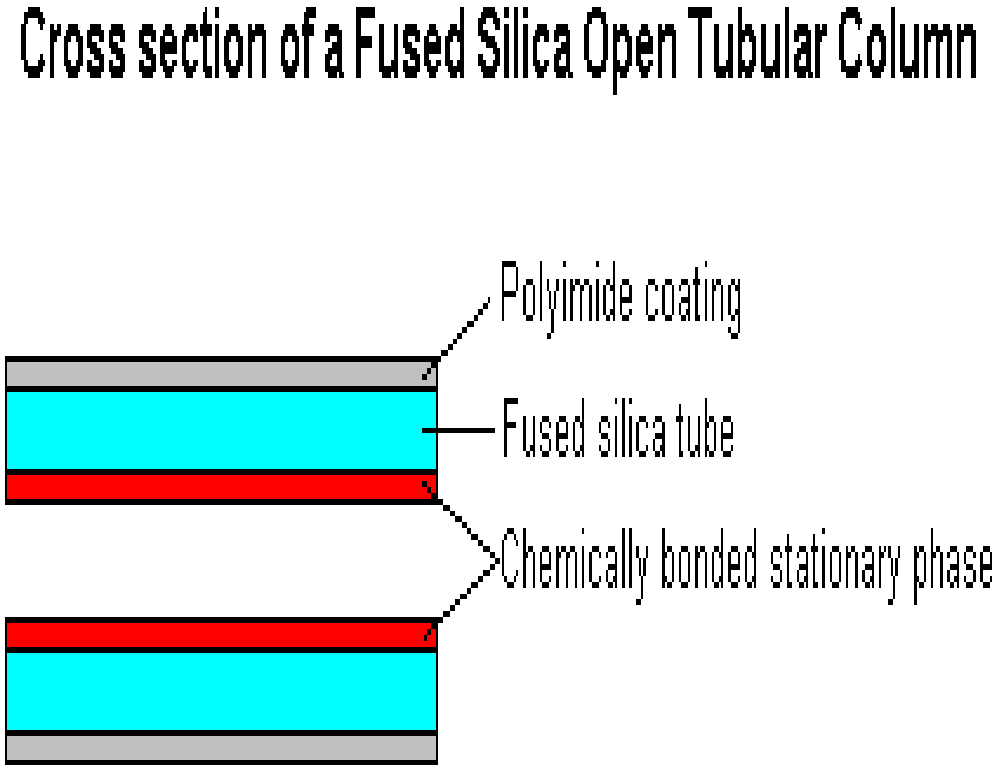
- Capillary columns, on the other hand, need much less sample, typically around 10^{-3} ml. For capillary GC, split/split less injection is used. Have a look at this diagram of a split/split less injector.
- The injector can be used in one of two modes; split or split less. The injector contains a heated chamber containing a glass liner into which the sample is injected through the septum.
- The carrier gas enters the chamber and can leave by three routes (when the injector is in split mode). The sample vaporizes to form a mixture of carrier gas, vaporized solvent and vaporized solutes.
- A proportion of this mixture passes onto the column, but most exits through the split outlet. The septum purge outlet prevents septum bleed components from entering the column.



Columns

- There are two general types of column, packed and capillary (also known as open tubular). Packed columns contain a finely divided, inert, solid support material (commonly based on diatomaceous earth) coated with liquid stationary phase.
- Most packed columns are 1.5 - 10m in length and have an internal diameter of 2 - 4mm.
- Capillary columns have an internal diameter of a few tenths of a millimeter. They can be one of two types; wall-coated open tubular (WCOT) or support-coated open tubular (SCOT).
- Wall-coated columns consist of a capillary tube whose walls are coated with liquid stationary phase.
- In support-coated columns, the inner wall of the capillary is lined with a thin layer of support material such as diatomaceous earth, onto which the stationary phase has been adsorbed.
- SCOT columns are generally less efficient than WCOT columns. Both types of capillary column are more efficient than packed columns.

- These have much thinner walls than the glass capillary columns, and are given strength by the polyimide coating. These columns are flexible and can be wound into coils. They have the advantages of physical strength, flexibility and low reactivity.



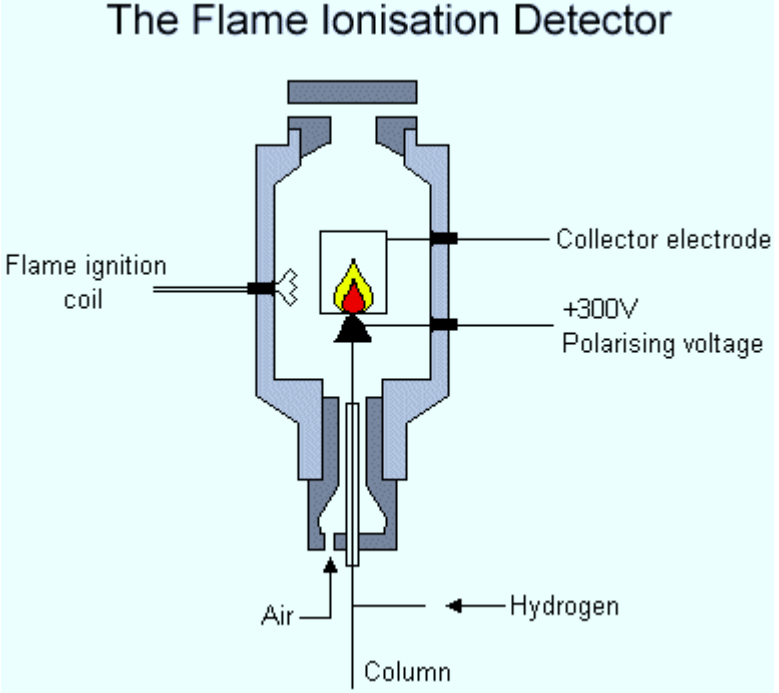
Column temperature

- For precise work, column temperature must be controlled to within tenths of a degree.
- The optimum column temperature is dependant upon the boiling point of the sample. As a rule of thumb, a temperature slightly above the average boiling point of the sample results in an elution time of 2 - 30 minutes.
- Minimal temperatures give good resolution, but increase elution times. If a sample has a wide boiling range, then temperature programming can be useful.
- The column temperature is increased (either continuously or in steps) as separation proceeds.

Detectors

- There are many detectors which can be used in gas chromatography. Different detectors will give different types of selectivity. A non-selective detector responds to all compounds except the carrier gas, a selective detector responds to a range of compounds with a common physical or chemical property and a specific detector responds to a single chemical compound. Detectors can also be grouped into concentration dependant detectors and mass flow dependant detectors.

- The signal from a concentration dependent detector is related to the concentration of solute in the detector, and does not usually destroy the sample Dilution of with make-up gas will lower the detectors response.
- Mass flow dependent detectors usually destroy the sample, and the signal is related to the rate at which solute molecules enter the detector. The response of a mass flow dependent detector is unaffected by make-up gas.



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LESSON 53
Use of Soxhalet and Rotary evaporator for extraction

Definition of Soxhlet extractor

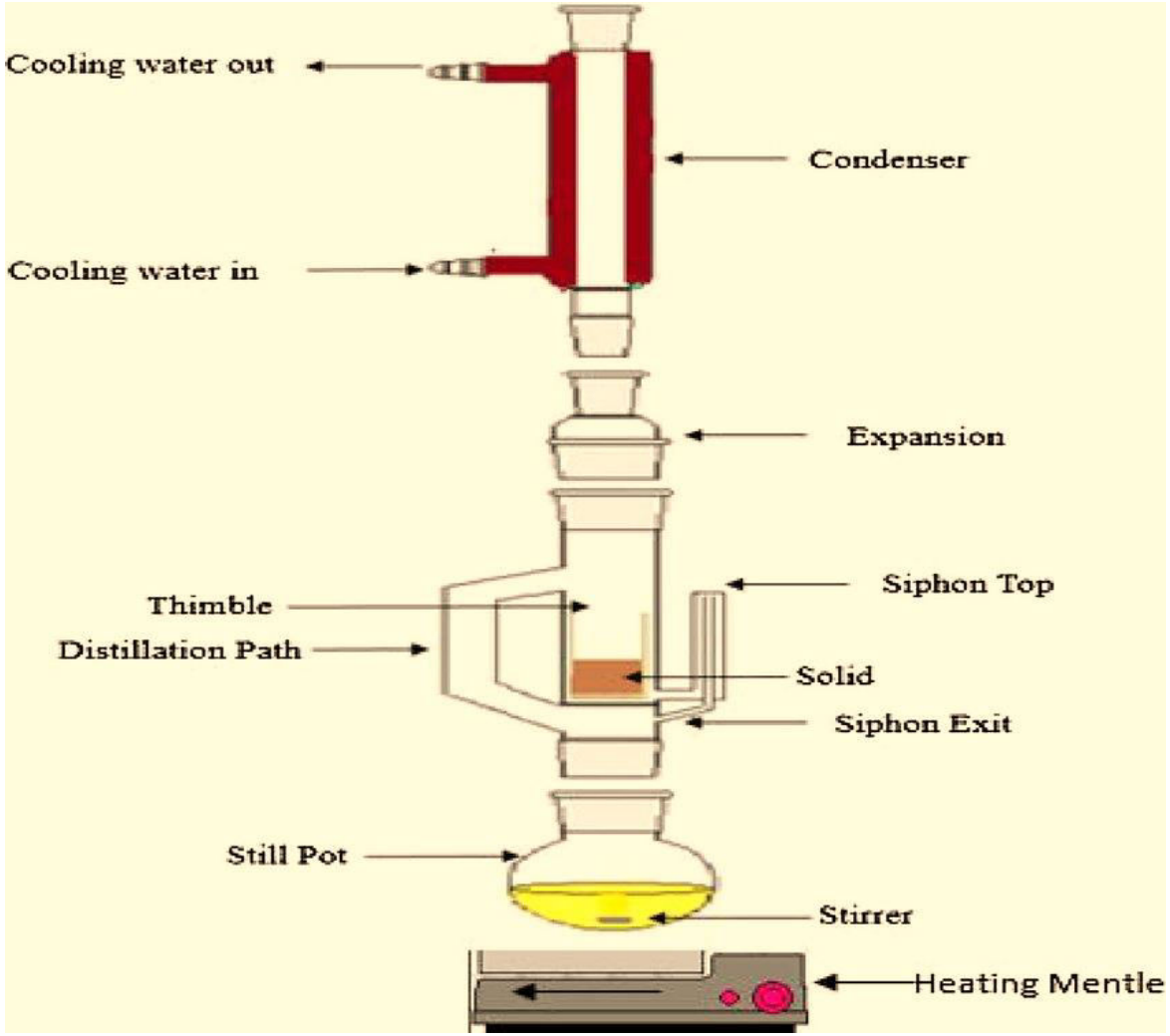
- A **Soxhlet extractor** is a piece of laboratory apparatus invented in 1879 by Franz von Soxhlet.
- It was originally designed for the extraction of a lipid from a solid material. Typically, Soxhlet extraction is used when the desired compound has a *limited* solubility in a solvent, and the impurity is insoluble in that solvent.
- It allows for unmonitored and unmanaged operation while efficiently recycling a small amount of solvent to dissolve a larger amount of material.

Description

A Soxhlet extractor has three main sections:

1. a percolator (boiler and reflux) which circulates the solvent,
2. a thimble (usually made of thick filter paper) which retains the solid to be extracted,
3. a siphon mechanism, which periodically empties the thimble.

- The source material containing the compound to be extracted is placed inside the thimble.
- The thimble is loaded into the main chamber of the Soxhlet extractor.
- The extraction solvent to be used is placed in a distillation flask. The flask is placed on the heating element.
- The Soxhlet extractor is placed atop the flask.
- A reflux condenser is placed atop the extractor.



Working Principle

- The solvent vapors travels up a distillation arm and floods into the chamber housing the thimble of solid.
- The condenser ensures that any solvent vapors cool, and drips back down into the chamber housing the solid material. The chamber containing the solid material slowly fills with warm solvent.
- Some of the desired compound will then dissolve in the warm solvent.
- When the Soxhlet chamber is almost full, the chamber is automatically emptied by a siphon side arm, with the solvent running back down to the distillation flask.
- The thimble ensures that the rapid motion of the solvent does not transport any solid material to the still pot. This cycle may be allowed to repeat many times, over hours or days.

- During each cycle, a portion of the non-volatile compound dissolves in the solvent. After many cycles the desired compound is concentrated in the distillation flask.
- The advantage of this system is that instead of many portions of warm solvent being passed through the sample, just one batch of solvent is recycled.
- After extraction the solvent is removed, typically by means of a rotary evaporator, yielding the extracted compound.
- The non-soluble portion of the extracted solid remains in the thimble, and is usually discarded.

Effort By Amaan Khan

LESSON 54
Use of Rotary evaporator for extraction

Rotary evaporator for extraction

- Rotary evaporator (or rotavap/rotovap) is a device used in chemical laboratories for the efficient and gentle removal of solvents from samples by evaporation.
- When referenced in the chemistry research literature, description of the use of this technique and equipment may include the phrase "rotary evaporator", though use is often rather signaled by other language (e.g., "the sample was evaporated under reduced pressure").
- Rotary evaporators are also used in molecular cooking for the preparation of distillates and extracts.
- A simple rotary evaporator system was invented by Lyman C. Craig. It was first commercialized by the Swiss company Büchi in 1957.In research the most common form is the 1L bench-top unit, whereas large scale (e.g., 20L-50L) versions are used in pilot plants in commercial chemical operations.



Design

The main components of a rotary evaporator are:

- A motor unit that rotates the evaporation flask or vial containing the user's sample.
- A vapor duct that is the axis for sample rotation, and is a vacuum-tight conduit for the vapor being drawn off the sample.
- A vacuum system, to substantially reduce the pressure within the evaporator system.
- A heated fluid bath (generally water) to heat the sample.
- A condenser with either a coil passing coolant, or a "cold finger" into which coolant mixtures such as dry ice and acetone are placed.
- A condensate-collecting flask at the bottom of the condenser, to catch the distilling solvent after it re-condenses.
- A mechanical or motorized mechanism to quickly lift the evaporation flask from the heating bath.



- The vacuum system used with rotary evaporators can be as simple as a water aspirator with a trap immersed in a cold bath (for non-toxic solvents), or as complex as a regulated mechanical vacuum pump with refrigerated trap.
- Glassware used in the vapor stream and condenser can be simple or complex, depending upon the goals of the evaporation, and any propensities the dissolved compounds might give to the mixture (e.g., to foam or "bump").
- Commercial instruments are available that include the basic features, and various traps are manufactured to insert between the evaporation flask and the vapor duct.
- Modern equipment often adds features such as digital control of vacuum, digital display of temperature and rotational speed, and vapor temperature sensing

Theory

- Vacuum evaporators as a class function because lowering the pressure above a bulk liquid lowers the boiling points of the component liquids in it.
- Generally, the component liquids of interest in applications of rotary evaporation are research solvents that one desires to remove from a sample after an extraction, such as following a natural product isolation or a step in an organic synthesis.
- Liquid solvents can be removed without excessive heating of what are often complex and sensitive solvent-solute combinations.
- Rotary evaporation is most often and conveniently applied to separate "low boiling" solvents such a n-hexane or ethyl acetate from compounds which are solid at room temperature and pressure.
- However, careful application also allows removal of a solvent from a sample containing a liquid compound if there is minimal co-evaporation (azeotropic behavior), and a sufficient difference in boiling points at the chosen temperature and reduced pressure.

- Solvents with higher boiling points such as water (100 °C at standard atmospheric pressure, 760 torr or 1 bar), dimethylformamide (DMF, 153 °C at the same), or dimethyl sulfoxide (DMSO, 189 °C at the same), can also be evaporated if the unit's vacuum system is capable of sufficiently low pressure. (For instance, both DMF and DMSO will boil below 50 °C if the vacuum is reduced from 760 torr to 5 torr [from 1 bar to 6.6 mbar]) However, more recent developments are often applied in these cases (e.g., evaporation while centrifuging or vortexing at high speeds).
- Rotary evaporation for high boiling hydrogen bond-forming solvents such as water is often a last recourse, as other evaporation methods or freeze-drying (lyophilization) are available. This is partly due to the fact that in such solvents, the tendency to "bump" is accentuated. The modern centrifugal evaporation technologies are particularly useful when one has many samples to do in parallel, as in medium- to high-throughput synthesis now expanding in industry and academia.

The key advantages in use of a rotary evaporator are

- The centrifugal force and the frictional force between the wall of the rotating flask and the liquid sample result in the formation of a thin film of warm solvent being spread over a large surface.
- The forces created by the rotation suppress bumping. The combination of these characteristics and the conveniences built into modern rotary evaporators allow for quick, gentle evaporation of solvents from most samples, even in the hands of relatively inexperienced users.
- Solvent remaining after rotary evaporation can be removed by exposing the sample to even deeper vacuum, on a more tightly sealed vacuum system, at ambient or higher temperature (e.g., on a Schlenk line or in a vacuum oven).

Disadvantages

- A key disadvantage in rotary evaporations, besides its single sample nature, is the potential of some sample types to bump, e.g. ethanol and water, which can result in loss of a portion of the material intended to be retained.
- Even professionals experience periodic mishaps during evaporation, especially bumping, though experienced users become aware of the propensity of some mixtures to bump or foam, and apply precautions that help to avoid most such events.
- In particular, bumping can often be prevented by taking homogeneous phases into the evaporation, by carefully regulating the strength of the vacuum (or the bath temperature) to provide for an even rate of evaporation, or, in rare cases, through use of added agents such as boiling chips (to make the nucleation step of evaporation more uniform).
- Rotary evaporators can also be equipped with further special traps and condenser arrays that are best suited to particular difficult sample types, including those with the tendency to foam or bump.

Lesson 55

Centrifugation

Centrifugation

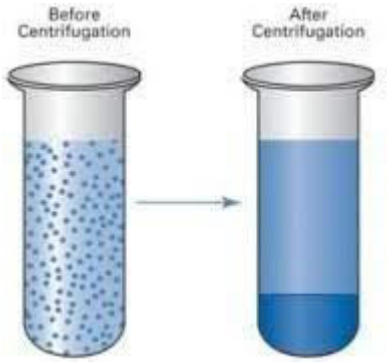


Centrifuge

Centrifuge is a device for separating particles from a solution according to their size, shape, density, medium viscosity of the medium.

Centrifugation

Centrifugation is a mechanical process which involves the use of the centrifugal force for the sedimentation of heterogeneous mixtures with a centrifuge.
This process is used to separate two immiscible liquids.



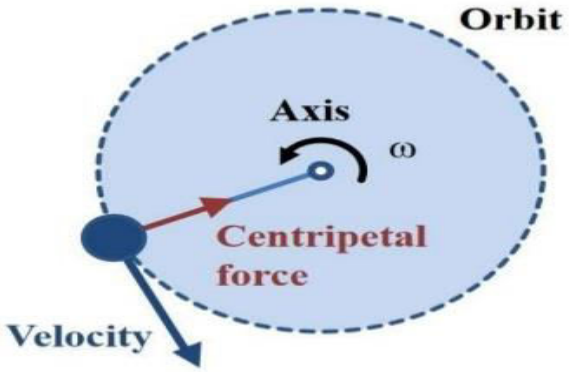
History

The first analytical ultracentrifuge was developed by Svedberg in 1920



Principle of Centrifugation

- Centrifugation is a technique of separating substances which involves the application of centrifugal force. In a solution, particles whose density is higher than that of the solvent sink (sediment), and particles that are lighter than it floats to the top.
- The greater the difference in density, the faster they move. If there is no difference in density ,the particles stay steady.
- To take advantage of even tiny differences in density to separate various particles in a solution, gravity can be replaced with the much more powerful “centrifugal force” provided by a centrifuge.



Components of Centrifuge

- Motor- which give rotator motion.
- Drive shaft- attached to motor.
- Rotor- placed on drive shaft having space to place tubes.
- Centrifuge rotor: a rotor is the rotating unit of the centrifuge, which has fixed holes drilled at an angle. Test tubes are placed inside these holes and the rotor spins to aid in the separation of the materials.

Swing Bucket Rotor



Fixed Angle Rotor



Vertical Rotor



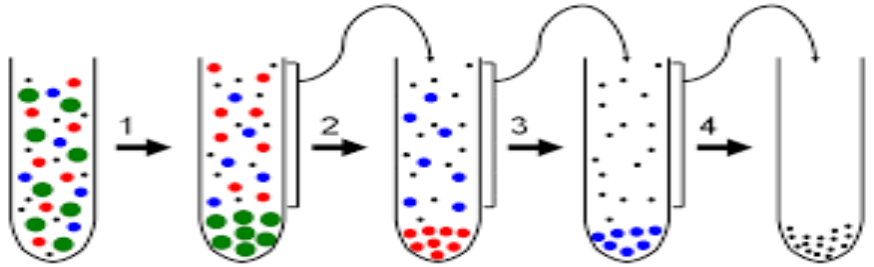
Lesson 56
Types Of Centrifugation

Types of Centrifugation

- There are two main types of Centrifugation.

1. Differential Pelleting (differential centrifugation)

- It is the most common type of centrifugation employed.
- Tissue such as the liver is homogenized at 32 degrees in a sucrose solution that contains buffer.
- The homogenate is then placed in a centrifuge and spun at constant centrifugal force at a constant temperature.
- After some time a sediment forms at the bottom of a centrifuge called pellet and an overlying solution called supernatant.
- The overlying solution is then placed in another centrifuge tube which is then rotated at higher speeds in progressing steps

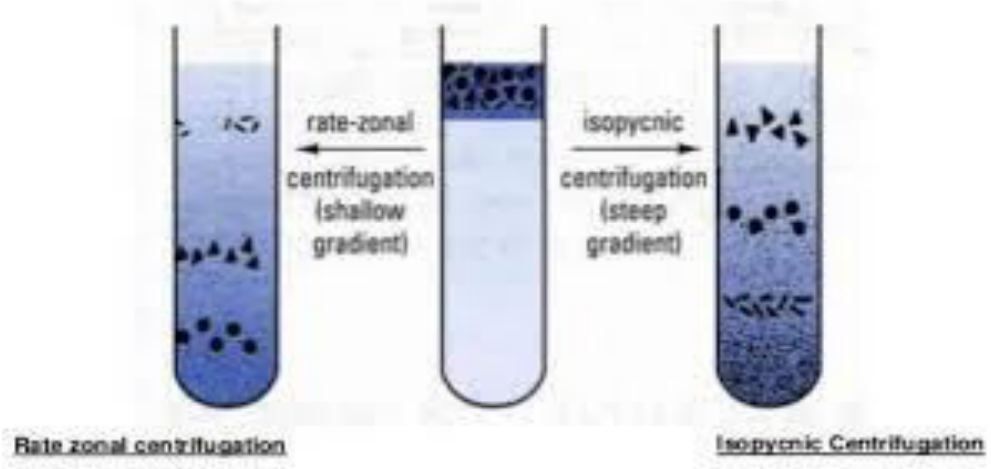


2. Density Gradient Centrifugation

- This type of centrifugation is mainly used to purify viruses, ribosomes, membranes, etc.
- A sucrose density gradient is created by gently overlaying lower concentrations of sucrose on higher concentrations in centrifuge tubes
- The particles of interest are placed on top of the gradient and centrifuge in ultracentrifuges.
- The particles travel through the gradient until they reach a point at which their density matches the density of surrounding sucrose.
- The fraction is removed and analyzed.

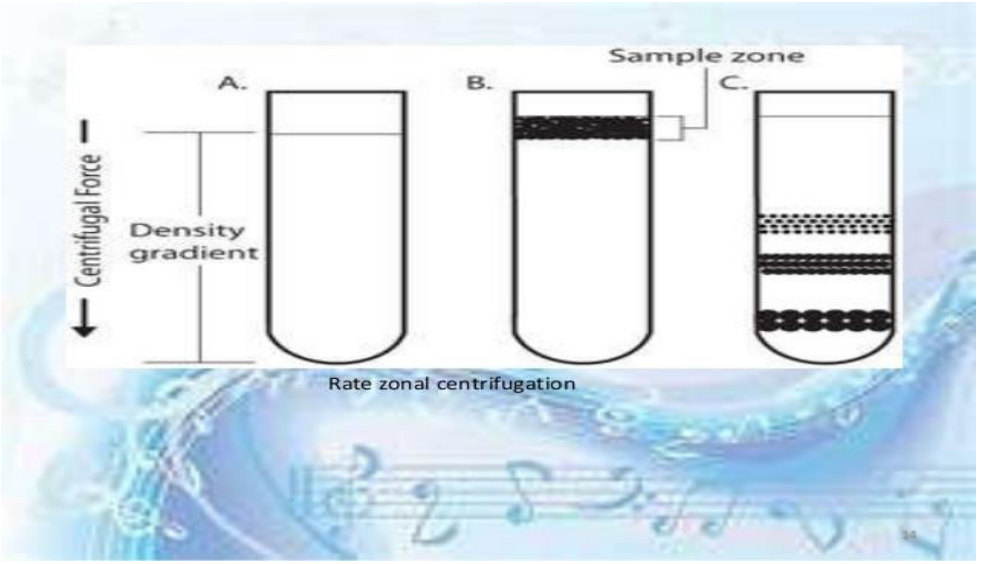
Two types of this:

(i) Rate-Zonal Density-Gradient Centrifugation
(ii) Isopynic Centrifugation



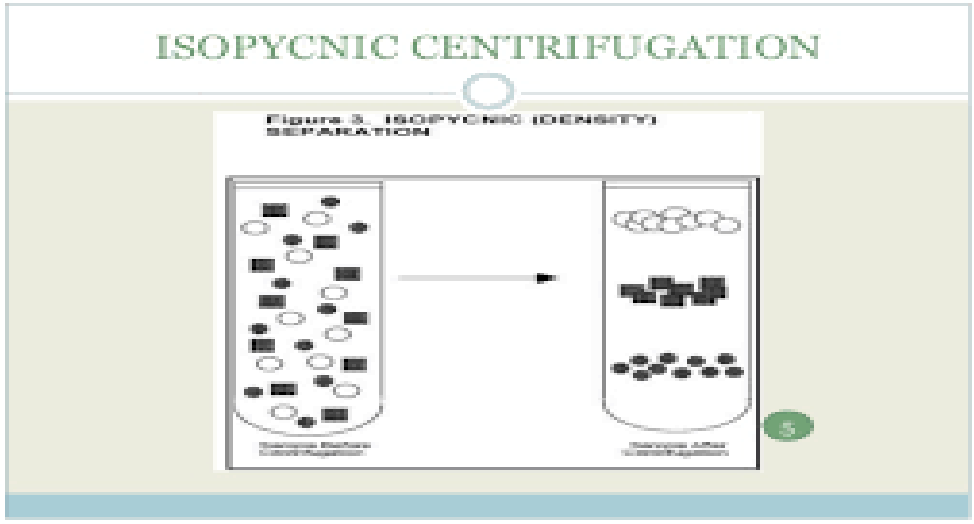
(i) Rate-Zonal Density-Gradient Centrifugation

- Zonal centrifugation is also known as band or gradient centrifugation
- It relies on the concept of sedimentation coefficient (i.e. movement of sediment through the liquid medium)
- In this technique, a density gradient is created in a test tube with sucrose and high density at the bottom.
- The sample of protein is placed on the top of the gradient and then centrifuged.
- With centrifugation, faster-sedimenting particles in sample move ahead of slower ones i.e. sample separated as zones in the gradient.
- The protein sediment according to their sedimentation coefficient and the fractions are collected by creating a hole at the bottom of the tube.



(ii) Isopynic Centrifugation

- The sample is loaded into the tube with the gradient-forming solution (on top of or below pre-formed gradient, or mixed in with self-forming gradient)
- The solution of the biological sample and cesium salt is uniformly distributed in a centrifuge tube and rotated in an ultracentrifuge.
- Under the influence of centrifugal force, the cesium salts redistribute to form a density gradient from top to bottom.
- Particles move to point where their buoyant density equals that part of gradient and form bands. This is to say the sample molecules move to the region where their density equals the density of gradient.
- It is a "true" equilibrium procedure since depends on bouyant densities, not velocities.
- Eg: CsCl, NaI gradients for macromolecules and nucleotides – "self-forming" gradients under centrifugal force.



Lesson 57

Ultracentrifugation

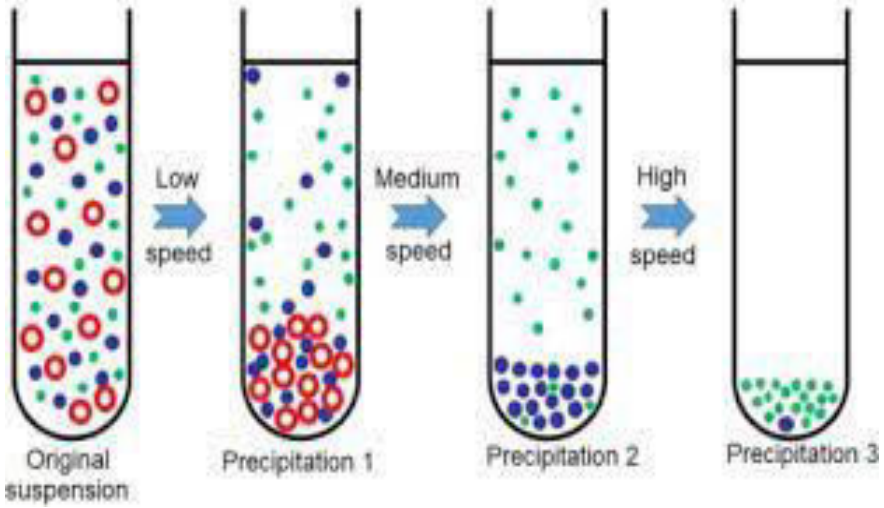
Ultracentrifugation

A high-speed centrifuge able to separate out colloidal and other small particles and used especially in determining the sizes of such particles or the molecular weights of large molecules.

Example;

Red cells separated from plasma of blood, nuclei from mitochondria in cell homogenates, one protein from another in complex mixtures. And also isolation of macromolecules such as DNA, RNA, Lipids etc.

- Its rotational speed up to 150,000 rpm.
- It is creating a centrifugal force unto 900,000 x g.



The Physics of Ultracentrifugation

1. Centrifugal force:-

- The outward force experienced by a particle in circular motion
- The tube containing the suspension of particles is rotated at a high speed, which exerts a centrifugal force directed from the centre of the rotor towards the bottom of the tube.

Centrifugal force:

$$F=M\,\omega^2 r$$

M: mass of particle r: radius of rotation (cm) (i.e. distance of particle from axis of rotation) ω :Average angular velocity (radians/sec)

Centrifugal field :-

• field where centrifugal force is experienced.

• Depends on the radical distance of the particle from the rotation axis and the square of the angular velocity.

$$G=r\omega^2 \longrightarrow G= 4\pi^2 (rev\ min^{-1})^2\ r/3600$$

Angular Velocity:-

• Rate of rotation around an axis

• Detect to revolution per minute (r.p.m)

$$\omega = 2\pi\ rev\ min^{-1}\ /60$$

2. Sedimentation rate:-

This force acts on the suspended particles pushing them towards the bottom of the tube at a rate determined by the velocity of the spinning rotor.

Rate of Sedimentation: $dr/dt=M(1- \nu p)/N_A f)\omega^2 r$

Where,
r = radius at which the organelle is located
t = time
M = molecular weight
ν = partial specific volume of the molecule; inverse of the density
ρ = density of the solvent
f = translational frictional coefficient
ω = angular velocity
N = Avagadro's number

3. Sedimentation coefficient:-

Centrifugation separates particles in a suspension based on differences in size, shape and density that together define their sedimentation coefficient.

Sedimentation Coefficient: $S=dr/dt\ (1/\omega^2 r)$

This is known as the Svedberg equation and is usually expressed in Svedberg units,

S (= 10-13 second).

This equation indicates that 'S' is dependent upon the molecular weight, the density and the frictional coefficient.

Types of Ultracentrifugation

1. **Analytical ultracentrifugation:-**
   The aim of Analytical ultracentrifugation is use to study molecular interactions between macromolecules or to analyse the properties of sedimenting particles such as their apparent molecular weight.

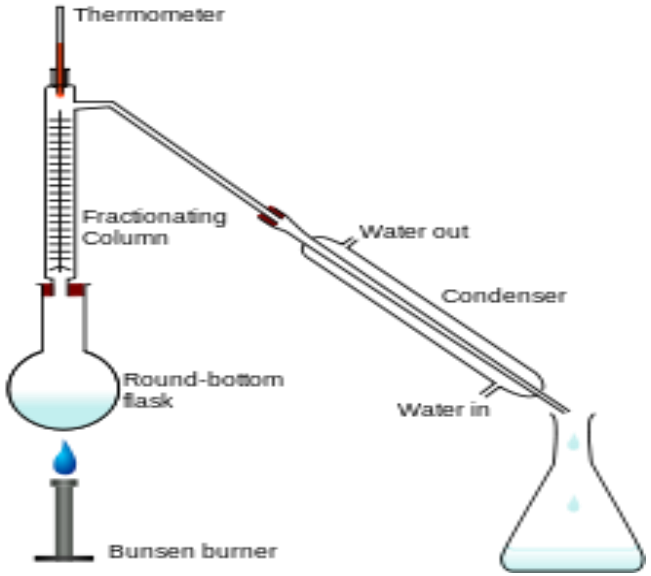
2. **Preparative ultracentrifugation:-**
   The aim of Preparative ultracentrifugation to isolate and purify specific particles such as subcellular organelles.

Lesson 58

Fractionation

Fractionation

- Fractionation is a separation process in which a certain quantity of a mixture (gas, solid, liquid, enzymes, suspension, or isotope) is divided during a phase transition, into a number of smaller quantities (fractions) in which the composition varies according to a gradient.
- Fractions are collected based on differences in a specific property of the individual components.



Fractional distillation apparatus using a Liebig condenser. A conical flask is used as a receiving flask. Here the distillation head and fractionating column are combined in one piece.

- A common trait in fractionations is the need to find an optimum between the amount of fractions collected and the desired purity in each fraction.
- Fractionation makes it possible to isolate more than two components in a mixture in a single run. This property sets it apart from other separation techniques.
- Fractionation is widely employed in many branches of science and technology. Mixtures of liquids and gases are separated by fractional distillation by difference in boiling point.
- Fractionation of components also takes place in column chromatography by a difference in affinity between stationary phase and the mobile phase.
- In fractional crystallization and fractional freezing, chemical substances are fractionated based on difference in solubility at a given temperature.
- In cell fractionation, cell components are separated by difference in mass.

1. Fractionation of natural samples:

(i) Bioassay-guided fractionation

A typical protocol to isolate a pure chemical agent from natural origin is, step-by-step separation of extracted components based on differences in their bioassay-guided fractionation physicochemical properties, and assessing the biological activity, followed by next round of separation and assaying. Typically, such work is initiated after a given crude extract is deemed "active" in a particular in vitro assay.

(ii) Blood fractionation

The process of blood fractionation involves separation of blood into its main components. Blood fractionation refers generally to the process of separation using a centrifuge (centrifugation), after which three major blood components can be visualized: plasma, buffy coat and erythrocytes(blood cells). These separated components can be analyzed and often further separated.

(iii) Of food

Fractionation is also used for culinary purposes, as coconut oil, palm oil, and palm kernel oil are fractionated to produce oils of different viscosities, that may be used for different purposes. These oils typically use fractional crystallization (separation by solubility at temperatures) for the separation process instead of distillation. Mango oil is an oil fraction obtained during the processing of mango butter.

Milk can also be fractionated to recover the milk protein concentrate or the milk basic proteins fraction.

2. Isotope fractionation

Isotope fractionation describes fractionation processes that affect the relative abundance of isotopes, phenomena which are taken advantage of in isotope geochemistry and other fields. Normally, the focus is on stable isotopes of the same element. Isotopic fractionation can be measured by isotope analysis, using isotope-ratio mass spectrometry or cavity ring-down spectroscopy to measure ratios of isotopes, an important tool to understand geochemical and biological systems. For example, biochemical processes cause changes in ratios of stable carbon isotopes incorporated into biomass.

Types

There are four types of isotope fractionation:

Equilibrium fractionation

Kinetic fractionation

Mass-independent fractionation (or Non-mass-dependent fractionation)

Transient kinetic isotope fractionation

Example:

Isotope fractionation occurs during a phase transition, when the ratio of light to heavy isotopes in the involved molecules changes. When water vapor condenses (an equilibrium fractionation), the heavier water isotopes (^{18}O and ^{2}H) become enriched in the liquid phase while the lighter isotopes (^{16}O and ^{1}H) tend toward the vapor phase.

Lesson 59

Spectrophotometery

Spectrophotometery

Spectrophotometry is a method to measure how much a chemical substance absorbs light, by measuring the intensity of light, as a beam of light passes through sample solution.

Spectrophotometry uses photometers that can measure a light beam's intensity as a function of its color (wavelength) known as spectrophotometers.

A spectrophotometer measures either the amount of light reflected from a sample object or the amount of light that is absorbed by the sample object.

Spectrophotometer

A spectrophotometer is employed to measure the amount of light that a sample absorbs. The instrument operates by passing a beam of light through a sample and measuring the intensity of light reaching a detector.



History of Spectrophotometer

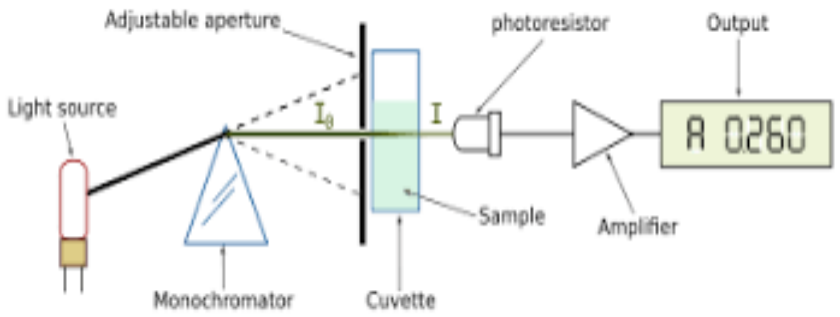
In the 1930s, vitamin research indicated that several vitamins, particularly vitamin A, absorb ultraviolet (UV) light. Spurred by the American government's interest in measuring vitamin content in soldiers' rations using ultraviolet and visible (UV-Vis) light, this research culminated in the commercial launch of UV-Vis spectrophotometers in the early 1940s. The spectrophotometer was invented in 1940, by Arnold J. Beckman and his colleagues at National Technologies Laboratories, the company Beckman had started in 1935. They were led by project leader Howard H. Cary. The spectrophotometer was the company's greatest discovery.

Spectrophotometery

Spectrophotometry is a scientific method based on the absorption of light by a substance, and takes advantage of two laws of light absorption.

Beer-Lambert Law (also known as Beer's Law) states that there is a linear relationship between the absorbance and the concentration of a sample.

Beer–Lambert law can be applied to the analysis of a mixture by Spectrophotometry, without the need for extensive pre-processing of the sample.



Lambert's Law

The proportion of light absorbed by a medium is independent of the intensity of the incident light. A sample which absorbs 75% (25% Transmittance) of the light will always absorb 75% of the light, no matter the strength of the light source.

Lambert's law is expressed as $I/I_o = T$

Where,

I = Intensity of transmitted light

Io = Intensity of Incident light

T = Transmittance

This allows different spectrophotometers with different light sources to produce comparable absorption readings independent of the power of the light source.

Beer's Law

The absorbance of light is directly proportional to both the concentration of the absorbing medium and the thickness of the medium. In Spectrophotometry the thickness of the medium is called the path length.

The path length is measured in centimeters, because a standard spectrophotometer uses a cuvette that is 1 cm in width, or is always assumed to equal 1 cm. Beer's law allows us to measure samples of differing path length, and compare the results directly with each other.

Beer Lambert Law

Beer-Lambert Law (also known as Beer's Law) states that there is a linear relationship between the absorbance and the concentration of a sample.

$$A = \log 10\ I/Io = \epsilon\ lc$$

where A is the measure of absorbance (amount of light absorbed by a sample.)

ϵ is the molar extinction coefficient or molar absorptivity or absorption coefficient (It is defined as a measure of a chemical's ability to absorb light at a specified wavelength.)

l is the path length, and
c is the concentration.

The absorbance is directly proportional to the concentration (c) of the solution of the sample used in the experiment.

$A \propto c$

The absorbance is directly proportional to the length of the light path (l), which is equal to the width of the cuvette.

$A \propto l$

Combining the two relationships,

$A \propto c\, l$

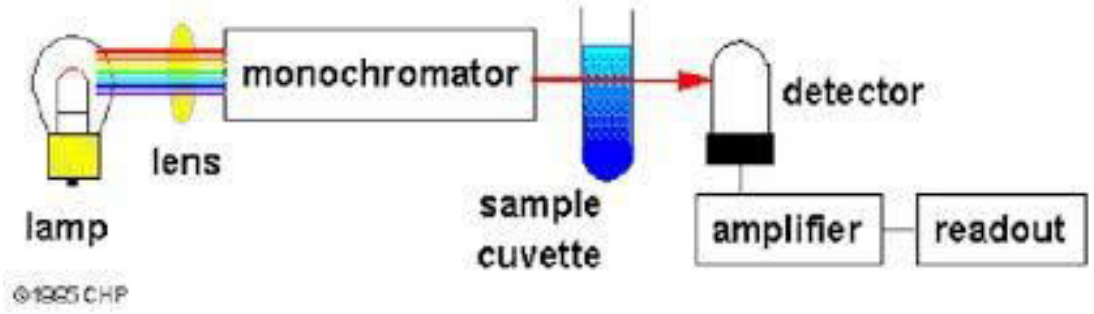
This proportionality can be converted into an equation by including a constant.

$A = \epsilon\, l\, c$

Types of Spectrophotometery

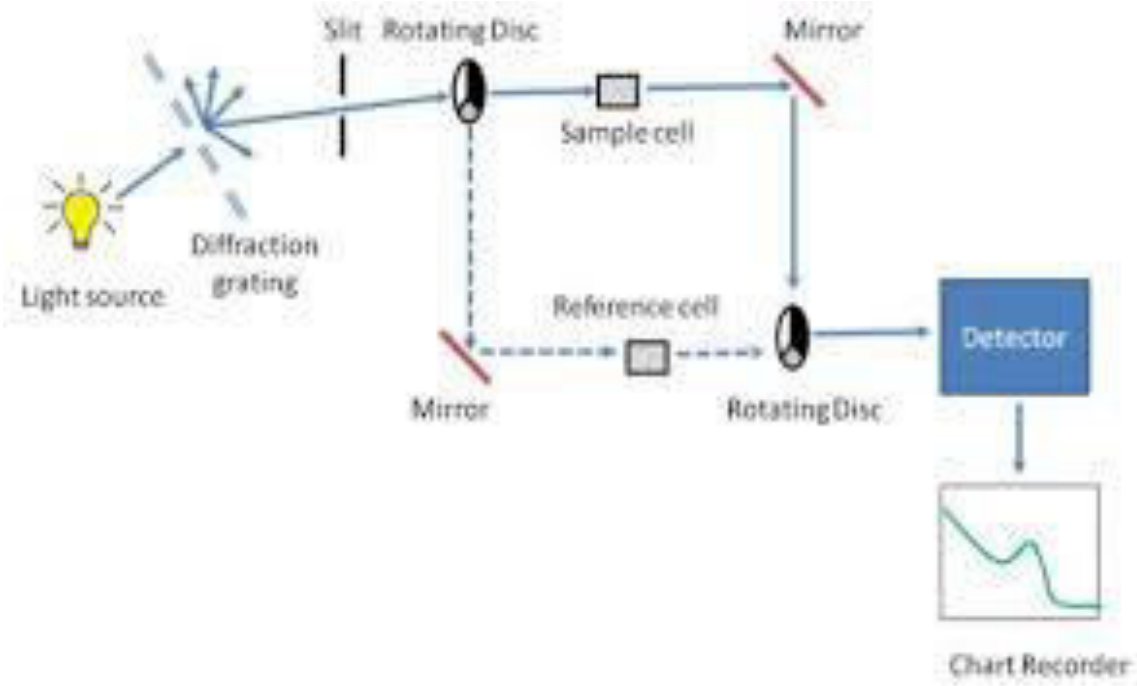
1. Single Beam Spectrophotometer

This is an instrument composed of a Light source, a Wavelength selector (monochromator), a Sample compartment, and a Detector. The type of light source will depend on the desired wavelength. The wavelength selector splits the light spectrum into its component colors and selects a narrow band of this spectrum. Either a prism or a grating may be used as a dispersing element to split the light. The wavelength of the selected light may be adjusted by changing the angle of the dispersing element, so that the desired wavelength passes through the slit. The selected light passes into the sample compartment, through the sample and to the detector (photomultiplier). By measuring transmittance with (I) and without (Io) the sample present, absorbance can be determined. This is how a single beam spectrophotometer operates.



2. Double Beam Spectrophotometer

This instrument has a chopper motor which alternately deflects the light beam through a reference cell and the sample cell. This is done many times per second, and the average ratio between the two readings gives the transmittance.



Lesson 60

UV-Visible Spectrophotometery

UV-Visible Spectrophotometery

Ultraviolet and visible (UV-Vis) absorption spectroscopy is the measurement of the attenuation of a beam of light after it passes through a sample or after reflection from a sample surface.

Absorption measurements can be at a single wavelength or over an extended spectral range.

Principle of UV-Visible Spectrophotometery

Ultraviolet absorption spectra arise from transition of electron with in a molecule from a lower level to a higher level.

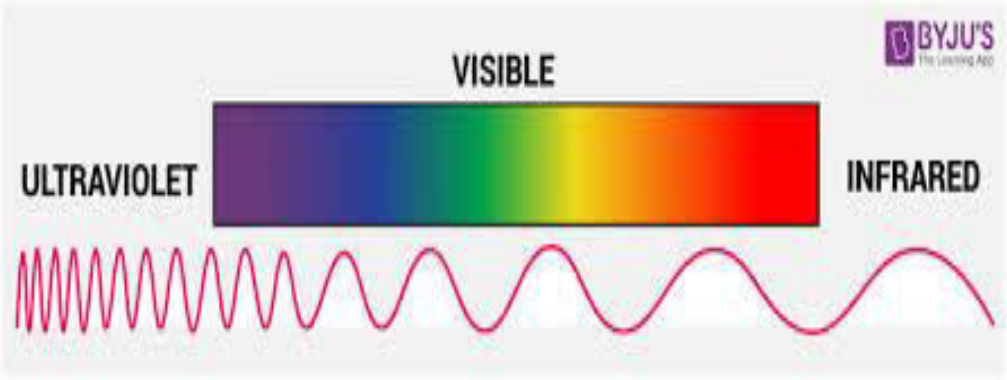
A molecule absorb ultraviolet radiation of frequency (ν), the electron in that molecule undergo transition from lower to higher energy level.

The energy can be calculated by the equation, $E=h\nu$

$E1-Eo=h\nu$

UV Radiation

The region beyond red is called infra-red while that beyond violet is called as ultra –violet. The wavelength range of uv radiation starts at blue end of visible light(4000A°) & ends at 2000A°.



The Absorption Spectrum

When a sample is exposed to light energy that matches the energy difference between a possible electronic transition within the molecule, a fraction of the light energy would be absorbed by the molecule and the electrons would be promoted to the higher energy state orbital. A spectrometer records the degree of absorption by a sample at different wavelengths and the resulting plot of absorbance (A) versus wavelength (λ) is known as a spectrum.

The significant features:

λmax (wavelength at which there is a maximum absorption)

ϵmax (The intensity of maximum absorption)

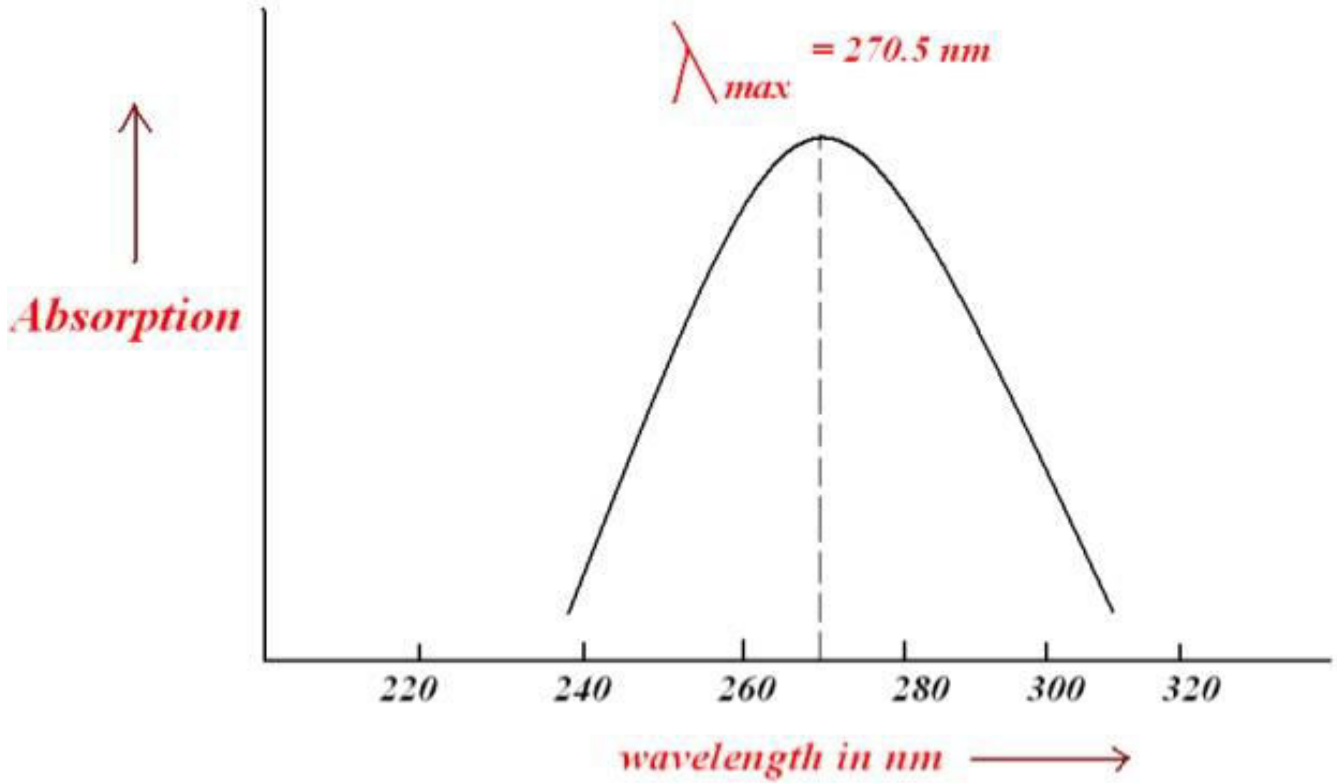


fig:- UV spectrum of acetone

Types of Transitions

In U.V spectroscopy molecule undergo electronic transition involving σ, π and n electrons.

Four types of electronic transition are possible.

i. σ→ σ* transition
ii. n → σ* transition
iii. n → π* transition
iv. π→π* transition

i. σ →σ* transition

An electron in a bonding σ orbital of a molecule is excited to the corresponding anti-bonding orbital by the absorption of radiation.

To induce a σ→ σ* transition it required large energy.

Ex: Methane

Methane contain only single C-H bonds it undergo only σ → σ* transition only, it gives absorption maximum at 125nm.

ii. n →σ* transition

In this type saturated compounds containing atoms with unshared electron pairs are undergo n →σ* transition. It require less energy than the σ → σ* type. Most of the absorption peaks appearing below 200nm. In the presence of polar solvents the absorption maximum tend to shift shorter wavelength Ex: Water , ethanol.

In this the peaks in U.V region relatively small. Ex: Methyl chloride , Oxygen, Nitrogen.

iii. n → π* and π → π* transition

Most organic compounds are undergo transitions for n → π* and π → π* transition. Because energies required for processes bring the absorption peaks into spectral region.

Both transition require the presence of an unsaturated functional group to the ′∏′ orbitals.

Ex: For π → π* → Alkenes, compounds, alkynes carbonyl

For n → π* → carbonyl compounds.

LESSON 61

APPLICATIONS OF SPECTROPHOTOMETERY

Applications of Spectrophotometery

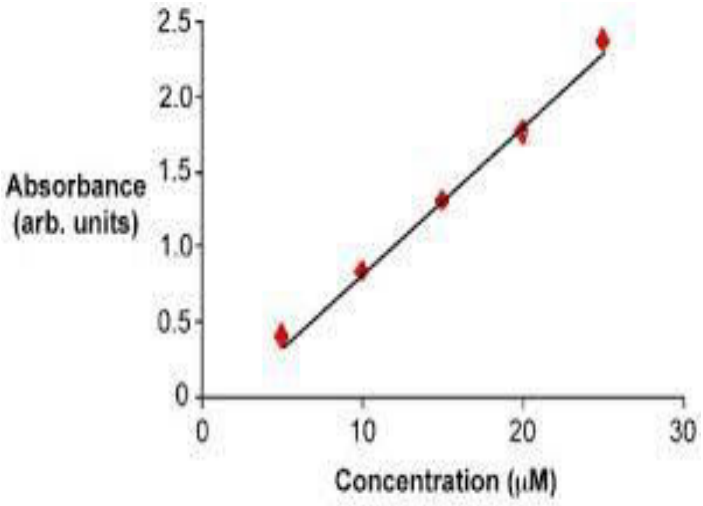
1. Concentration measurement

-Prepare samples

-Make series of standard solutions of known concentrations

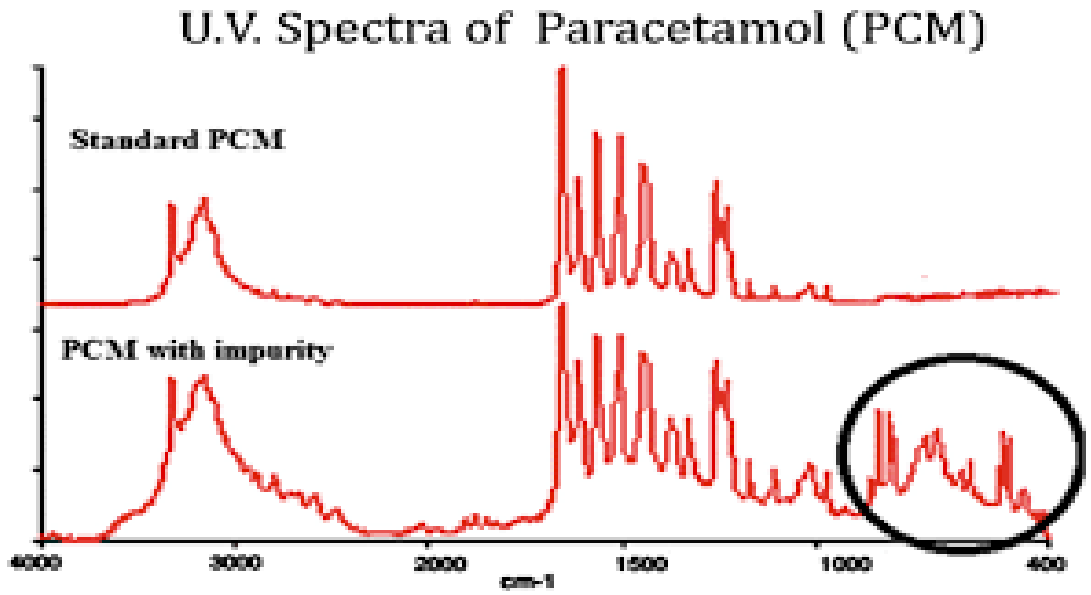
– Set spectrophotometer to the λ of maximum light absorption

– Measure the absorption of the unknown, and from the standard plot, read the related concentration



2. Detection of Impurities

- UV absorption spectroscopy is one of the best methods for determination of impurities inorganic molecules.
- Additional peaks can be observed due to impurities in the sample and it can be compared with that of standard raw material.



3. Structure elucidation of organic compounds

From the location of peaks and combination of peaks UV spectroscopy elucidate structure of organic molecules: the presence or absence of unsaturation, the presence of hetero atoms.

4. Chemical kinetics

Kinetics of reaction can also be studied using UV spectroscopy. The UV radiation is passed through the reaction cell and the absorbance changes can be observed.

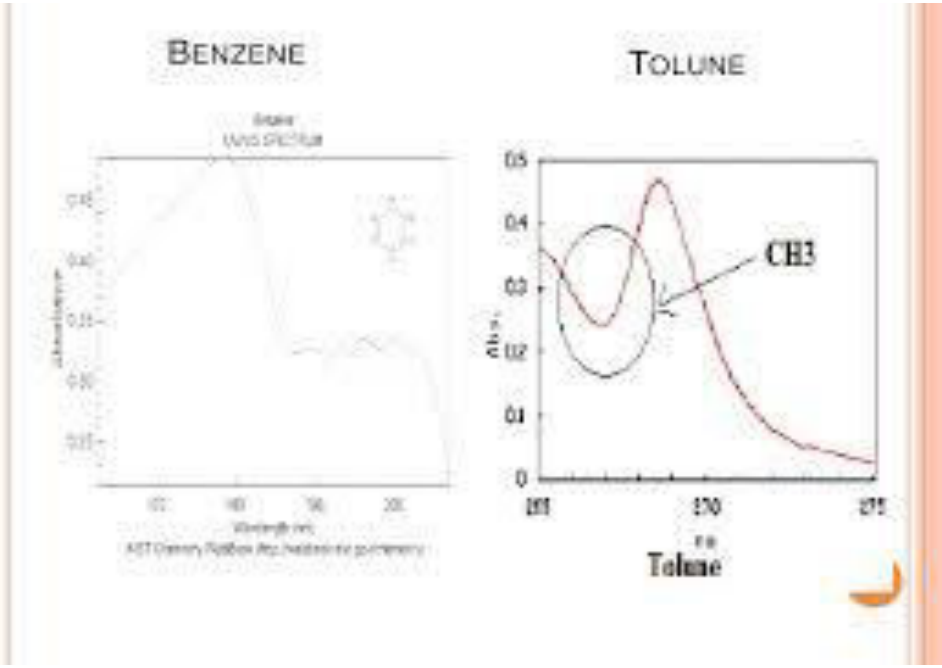
5. Molecular weight determination

Molecular weights of compounds can be measured spectrophotometrically by preparing the suitable derivatives of these compounds.

For example, if we want to determine the molecular weight of amine then it is converted in to amine picrate

6. Detection of Functional Groups

Absence of a band at particular wavelength regarded as an evidence for absence of particular group.



Lesson 62
STANDARD SYSTEM FOR

LENGTH
VOLUME
WEIGHT

CONTENTS :

01. Metric system
02. Prefixes
03. Length
04. Volume
05. Weight
06. Conversion

Metric system:

- A widely used system of measurement that is based on the decimal system and multiples of 10.
- The metric system uses units such as meter, liter, gram and newton to measure length, liquid volume, mass and weight.
- The metric system is based on joining one of a series of prefixes, including kilo-, hecto-, deka-, deci-, centi-, and milli-, with a base unit of measurement, such as meter, liter, or gram.

Prefixes in metric system:

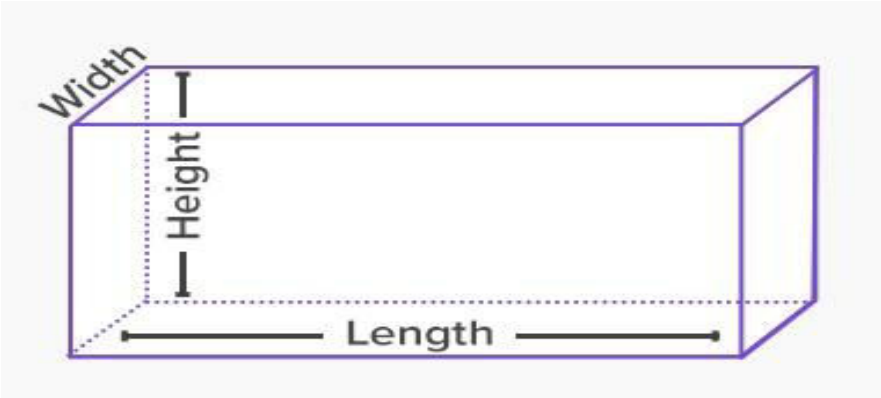
- The names of metric units are formed by adding a prefix to the basic unit of measurement.
- The prefixes have the same meanings whether they are attached to the units of length (meter), mass (gram), or volume (liter).

Prefixes in the Metric System						
kilo-	*hecto-*	*deka-*	meter gram liter	*deci-*	*centi-*	*milli-*
1,000 times **larger** than base unit	100 times **larger** than base unit	10 times **larger** than base unit	base units	10 times **smaller** than base unit	100 times **smaller** than base unit	1,000 times **smaller** than base unit

- Using this table as a reference, you can see the following:
- · A kilogram is 1,000 times larger than one gram **(so 1 kilogram = 1,000 grams)**.
- · A centimeter is 100 times smaller than one meter **(so 1 meter = 100 centimeters).**
- · A dekaliter is 10 times larger than one liter **(so 1 dekaliter = 10 liters)**.

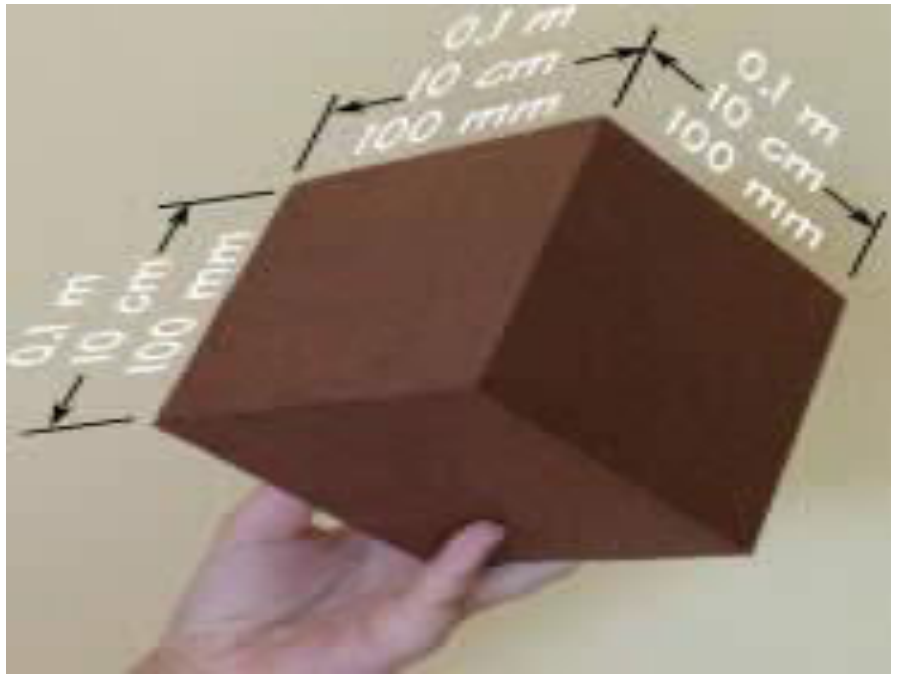
Length:

- Length is a measure of how long an object is or the distance between two points.
- The standard unit of length based on the metric system is a meter (m)
- According to the length that needs to be measured, we can convert a meter into various units like millimeters (mm), centimeter (cm), and kilometer (km).



Volume:

- Volume is the amount of space occupied by a sample of matter. The SI unit of volume is the cubic meter (m 3), which is the volume occupied by a cube that measures 1 m on each side.
- According to metric system there are two basic units used for volume measurements : LITRE(=dm^3) and MILLILITERS.(=cm^3)



Weight:

- In metric system unit **newton** is used to measure weight and **gram / kilogram** is used as standard for mass.
- **Weight is a measure of the pull of gravity on an object.**
- **Mass measures the amount of substance in an object.**
- However, in laboratory measurements , we normally use gram /kilogram to represent as the standard unit of mass and weight



Conversion of units:

- Units in the metric system are all related by a power of 10, which means that each successive unit is 10 times larger than the previous one. This makes converting one metric measurement to another a straightforward process, and is often as simple as moving a decimal point. It is always important, though, to consider the direction of the conversion.
- If you are converting a smaller unit to a larger unit, then the decimal point has to move to the left (making your number smaller); if you are converting a larger unit to a smaller unit, then the decimal point has to move to the right (making your number larger).

Example

Problem **Convert 1 centimeter to kilometers.**

km	hm	dam	m	dm	cm	mm
^					^	

Identify locations of kilometers and centimeters.

Kilometers (km) are larger than centimeters (cm), so you expect there to be less than one km in a cm.

÷10 ÷10 ÷10 ÷10 ÷10

km	hm	dam	m	dm	cm	mm
^					^	

← ← ← ← ←

Cm is 10 times smaller than a dm; a dm is 10 times smaller than a m, etc.

Since you are going from a smaller unit to a larger unit, divide.

1 cm ÷ 10 ÷ 10 ÷ 10 ÷ 10 ÷ 10 = 0.00001 km

Divide: 1 ÷ 10 ÷ 10 ÷ 10 ÷ 10 ÷ 10, to find the number of kilometers in one centimeter.

Answer 1 centimeter (cm) = 0.00001 kilometers (km).

Volume

10 **milliliters** (ml) =	1 **centiliter** (cl)
10 **centiliters** =	1 **deciliter** (dl) = 100 milliliters
1 **liter** =	1000 **milliliters**
1 **milliliter** =	1 **cubic centimeter**
1 **liter** =	1000 **cubic centimeters**
100 **centiliter** =	1 **liter** (L) = 1,000 milliliters
1000 **liters** =	1 **kiloliter** (km)

If you have this	Do this	To get this
milliliters (ml) or CCs	Divide by 10 (ml/10)	centiliter (cl)
centiliters (cl)	Multiply by 10 (cl * 10)	Milliliters (ml)
Liters (L)	Multiply by 1000 (L * 100)	Milliliters (mL) or CCs
Liters (L)	Multiply by 100 (L * 100)	Centiliters (cl)
Centiliters (cl)	Divide by 100 (cl/100)	Liters (L)
Milliliters (ml) or CCs	Divide by 1000 (ml/1000)	Liters (L)

Weight:

1 milligram = 0.001 gram
1 centigram = 0.01 gram
1 decigram = 0.1 gram
1 kilogram = 1000 grams
1 gram = 1000 milligrams

Lesson 63
Weighing Balance

WEIGHING BALANCE

- A weighing scale is a device for measuring weight
- Balances measure the mass of an object and are used in science
- In many industrial and commercial applications, scales and balances to determine the weight and/or mass of things ranging from feathers to loaded tractor-trailers.

WEIGHING BALANCE



WEIGHING PRINCIPLE

- The basis of the rapid and exact working method of our Weigh Cells is the Principle of **Electro Magnetic Force Restoration (EMFR).**
- The basic principle is comparable to a simple beam balance. The **weight** is laid on one side of the beam (coil arm). The result is that the coil attached to the other side of the **beam tries to move** out of the magnetic field of the magnet.

WEIGHING PRINCIPLE

- A zero indicator (**photoelectric beam**) recognizes any minimal deviation, and immediately **so much current is sent through the coil** via an electrical regulator circuit that the balance beam hardly moves and **remains in its neutral position**.
- The deviation occurring is merely a matter of a **few nanometres.**

DIFFERENT TYPES OF BALANCES

- **Analytical balance**
- Micro balance
- Gold and carat balance
- Spring balance

Analytical balances

- Very accurate balances, called analytical balances, are used in scientific fields such as chemistry.
- An analytical balance is a class of balance designed to measure small mass in the sub-milligram range.





Analytical balances/lab balance

- Analytical balances are found throughout most laboratories.
- They are mostly used to weigh substances and **samples between 0.01 to 500 milligrams.**
- These units' measuring pans are usually encased in a **glass box** so **as to prevent any dust particles** settling in the pan.
- An analytical balance measures masses to within **0.0001 g.**
- Use these balances when you need **this high degree of precision.**



Mechanical analytical balance



Automated analytical balance



Micro balance

- Microbalances are generally designed to measure, weigh and provide data on the **tiniest of samples**.
- Most models can effectively provide data for samples weighing between **6 and 0.0001** milligrams.
- These types of balances are generally used to **weigh highly valuable substances in minute quantities.**
- These units typically come standard with **draft shields** so that dust and other foreign particles do not make their way into the dish and corrupt data and materials being worked on.

Gold and Carat balance

- Gold and carat balances, as the name implies, are designed to **weigh gold** as well as give carat values.
- These are more commonly found in jewellery design workshops and retail jeweller outlets, however, they are used in some laboratory exercises too.
- These units are not limited to weighing gold; they are also commonly used to measure and provide carat values for both **precious and semiprecious stones and metals.**

Spring Balance

- Spring balances or spring scales measure **weight (force) by balancing the force due to gravity** against the force on a spring,
- whereas a balance or **pair of scales** using a balance beam compares masses by **balancing the weight to the unknown mass of an object against the known weight mass** or masses.

Reference :

https://www.slideshare.net/mobile/AdhiyamaanRaj/weighing-balance

Lesson 64
Size Measurement

- To measure the size of the specimen under the microscope, the ocular micrometer is used.
- Stage micrometers do not measure specimens directly but measure the objective lens's magnification error to enable more accurate measurement by ocular micrometers.
- The magnified image of the specimen is formed on the ocular micrometer and that the micrometer scale and the sample image can be viewed simultaneously through eyepiece lenses. At this time, the specimen size can be measured by determining the single pitch width on a micrometer scale. The single pitch width on the micrometer scale depends on the objective lens's magnification, and it can be calculated as follows;
- **Actual size = Measured size / Magnification**

How to Use an Ocular & Stage Micrometer for Calibration?

- Objective lenses are allowed to have a slight magnification error.
  The purpose of the stage micrometer is to measure this magnification error in advance for more precise measurements.
- **Example of Magnification Error Measurement for Objective Lenses :**
- Place the micrometers in a way that the ocular and the stage micrometer scale can be seen parallel to each other when looking through the eyepiece.
  You can then calculate the exact magnification from the error.
- The figure below shows an example of an stage micrometer **NOB1** (1mm/100 div/pitch=10μm) and an ocular micrometer **S11** (10mm /100 div / pitch = 100μm) used under an objective lens at **20x**.

- If the magnification is correct, ten pitches on the stage micrometer should be equivalent to 20 pitches on the ocular micrometer.
  (2000 μm = 10 μm <one pitch of stage micrometer> x 10 pitch x 20x objective)
  However, in this case, the pitch reads 21, so this objective's magnification is 21x.
- In the actual measurement, the ocular micrometer's value multiplied by the
  error rate calculated 0.95 (≒20÷21) would be the more accurate measurement value.

Scale bar

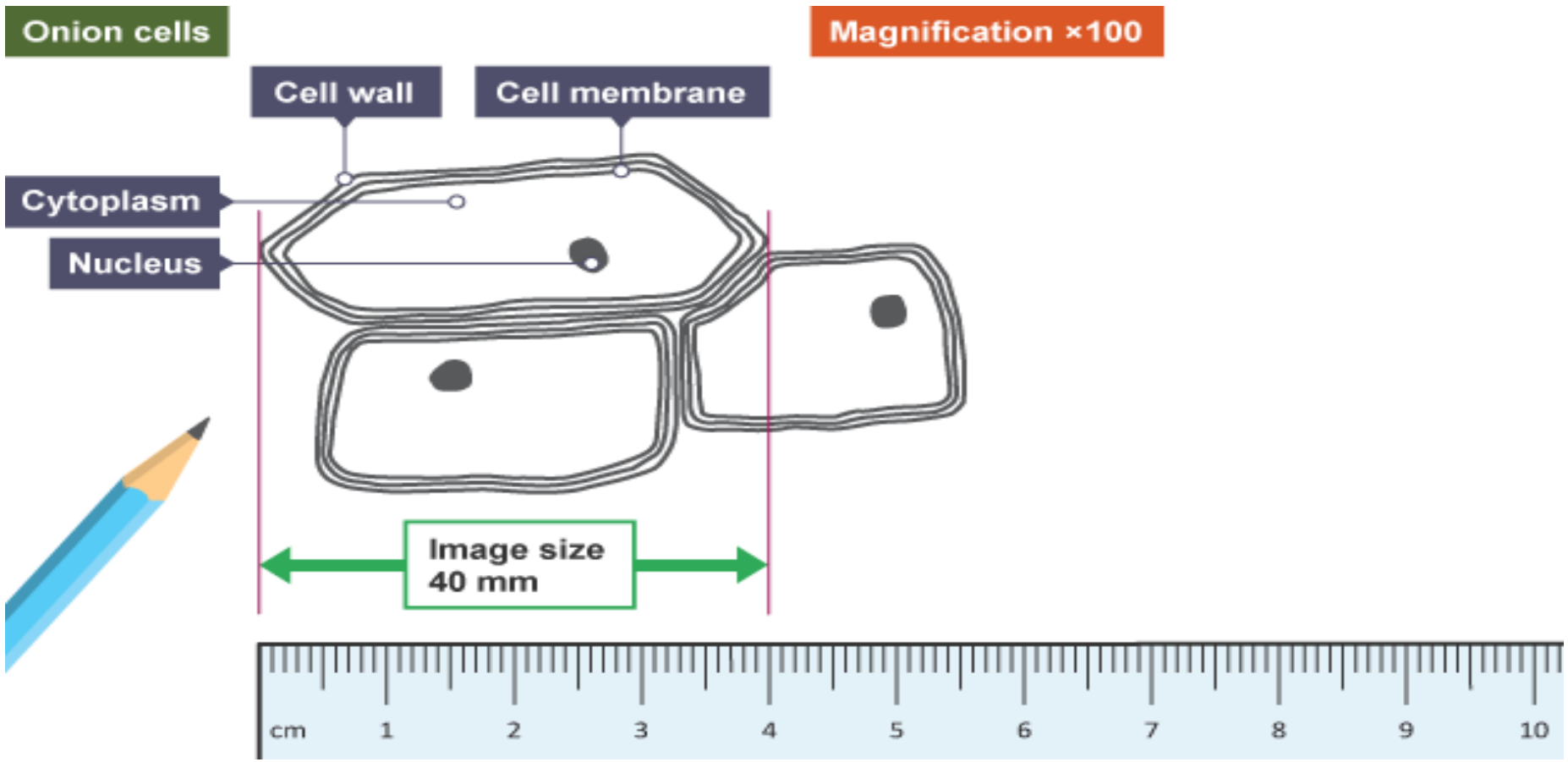
Magnification can be calculated using a scale bar. This is a line drawn near the photograph or drawing which has a label showing the actual length of the bar before being magnified.

Working out magnification:

Measure the scale bar image (beside drawing) in mm.

Convert to μm (multiply by 1000).

Magnification = scale bar image divided by actual scale bar length (written on the scale bar).


Onion cells
Magnification ×100
Cell wall
Cell membrane
Cytoplasm
Nucleus
Image size
40 mm
cm
1
2
3
4
5
6
7
8
9
10

Diameter measurement

- Step 1: Use a clear ruler with a cm/mm scale to measure the diameter of your viewing field at scanning (40x). On our scopes, we estimated the viewing field to be about 4 mm across.
- Step 2: Repeat the process on low power (100x). We estimate our low power viewing field to be about 2 mm across.
- Convert mm to microns for both the scanning and low power. There are 1000 microns in a millimeter. So… Scanning = 4000 microns; Low Power = 1000 microns.
- You can at this point use that measurement to measure anything in your viewing field that you can see with low or scanning power. The measurement is an estimation though, and probably not very accurate.
- In reality, you cannot place your slide over the ruler, so you have to make a guess based on how large your microscope's viewing field is.

Calculating High Power Field of View

- Measurements on High Power can be a little more complicated. If you try to use the clear ruler technique, you'll find that you cannot see the individual ruler marks. This is where math comes in, the values you estimated above can be used to solve a ratio problem and determine the size of your viewing field on high power.

$$\frac{\text{High Power Field of View}}{\text{Low Power Field of View}} = \frac{\text{Low Power magnification}}{\text{High Power Magnification}}$$

- Solving for "High Power Field of View" ….
- X / 1000 = 100/400
- X (high power field of view) = 250 μ
- Now that you have 250 microns as an estimate of your viewing field, any object you are viewing under high power can be estimated based on that.

Using a Stage (Slide) Micrometer

- These are slides you buy that contain a tiny ruler on them. The ruler is marked as .01 mm, which means each little etch mark on the ruler is .01. The total length of the scale on most stage micrometers is 1 mm, which then means that there are 1000 microns measured by that scale.

LESSON 65

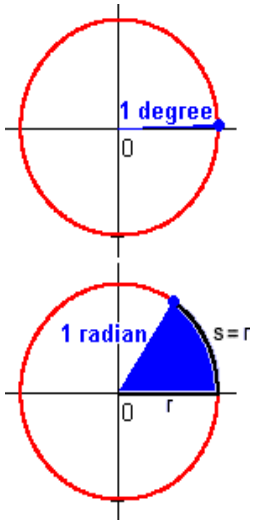
- **CIRCULAR OR ANGULAR MEASURE**
- **CONCENTRATION**
- **PERCENT – VOLUME**
- **PPM**
- **PPT**

Circular measure

- A circular measure was used in comparing circular cross-sections, e.g., of wires, etc. A circular unit of the Ares is the area of the circle whose diameter is one linear unit.
- For example, 1 circular mil is equivalent to 0.7854 square mil in area, 1 circular millimeter = 1550 circular mils = 0.7854 square millimeter. Here {\displaystyle 0.7854=\pi /4.}

Angular Measure

- Just as we have various **unit systems** for measuring lengths (inches, feet, meters, light years, etc.), we have various unit systems for measuring angles. There are two systems in common use.
- The oldest system (dating back to the Babylonians) is the **degree system**. In this system, a full circle is divided into 360 equal degrees. One degree, therefore, is only a very small part of a circle.



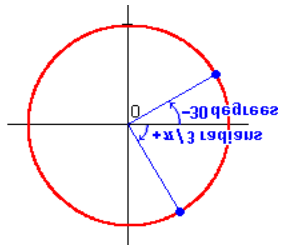
- A more natural system of angle measurement – one that is based on the geometry of the circle itself – is the **radian system**. One radian is the angle formed when one radius is layed, like a piece of wet spaghetti, along the arc s of a circle's perimeter. Since the entire perimeter of any circle is only 2p (a bit more than six) times the radius, there are far fewer radians than degrees in a full circle.
- To convert an angular measure from one system to the other, we need only remember that when a degree measure is some fraction of 360, the corresponding radian measure is the same fraction of 2p :

$$\frac{\text{degrees}}{360} = \frac{\text{radians}}{2\pi}$$

- We can plug in for either "degrees" or "radians" and solve for the other. In this way we find, for example, that 1 radian is approximately 57.3 degrees.
- The convenience of the radian system comes from the simple relationship between angle measure and **arc length** on the circle. By definition, one radian cuts off an arc length of one radius, two radians cuts off an arc length of two radii, and so forth. The two measures are proportional, with the radius as the constant of proportionality.
- Review of **proportionality**:
- Thus, if r is the circle's radius, q is the angle in radians, and s is the arc length cut off by the angle, then s= r q

.

- When the angle is measured in degrees, the relationship between s and q is still a proportional one, but the constant of proportionality is not quite so simple as r . (What would it be?)
- It is often convenient to give angles signs, to indicate direction. The convention is to measure **positive angles** in a counter-clockwise direction from some reference line (usually the x-axis), and **negative angles** in a clockwise direction.



Concentration

- In chemistry, concentration is the abundance of a constituent divided by the total volume of a mixture. Several types of mathematical description can be distinguished: mass concentration, molar concentration, number concentration, and volume concentration. A concentration can be any kind of chemical mixture, but most frequently solutes and solvents in solutions. The molar (amount) concentration has variants such as normal concentration and osmotic concentration.

- There are four quantities that describe concentration:

- **Mass concentration**

- The mass concentration **pi** is defined as the mass of a constituent of **mi** divided by the volume of the mixture V.
- **pi= mi /V**
- The SI unit is kg/m^3 (equal to g/L).

- **Molar concentration**
- The molar concentration **cì** is defined as the amount of a constituent **ni** (in moles) divided by the volume of the mixture V.
- **ci= ni /V**
- The **SI unit** is mol/m^3. However, more commonly the unit mol/L (= mol/dm^3) is used

- **Number concentration**
- The number concentration **Ci** is defined as the number of entities of a constituent **Ni** in a mixture divided by the volume of the mixture **V.**
- **Ci= Ni / V** The SI unit is $1/m^3$.
- **Volume concentration** **:** The **volume concentration**(σi (not to be confused with volume fraction is defined as the volume of a constituent **Vi** divided by the volume of the mixture V.
- $\sigma i = Vi / V$
- Being dimensionless, it is expressed as a number, e.g., 0.18 or 18%; its unit is 1.

Volume percent

- **Volume Percent**: The volume percent is used to express the concentration of a solution when the volume of a solute and the volume of a solution is given
- Volume percent = [(volume of solute)/(volume of solution)] x 100%
- **Example 1**
- Determine the volume/volume percent solution made by combining 25 mL of ethanol with enough water to produce 200 mL of the solution.

Cont..

Solution :

- Given parameters are
- Volume of solute is 25 mL
- Volume of the solution is 200 mL
- Substitute the values in the given formula,
- Volume percent = volume of solute /volume of solution x 100%
- = {25 mL / 200 mL }x 100%
- Volume percent = 12.5 %

Ppm

Parts per Million:

- A concentration of a solution that contained 1 g solute and 1000000 mL solution (same as 1 mg solute and 1 L solution) would create a very small percentage concentration. Because a solution like this would be so dilute, the density of the solution is well approximated by the density of the solvent; for water that is 1 g/mL (but would be different for different solvents). So, after doing the math and converting the milliliters of solution into grams of solution
- assuming water is the solvent):1 g solute/1000000 mL solution × 1 mL/1 g. = 1 g solute1000000 g solution

- We get (1 g solute)/(1000000 g solution). Because both the solute and the solution are both now expressed in terms of grams, it could now be said that the solute concentration is 1 part per million (ppm).1 ppm=1 mg Solute / 1 L Solution
- The ppm unit can also be used in terms of volume/volume (v/v)

Ppt

- **Parts per Trillion**:
- Just like ppb, the idea behind parts per trillion (ppt) is similar to that of ppm. However, 1 ppt is 1000-fold more dilute than 1 ppb and 1000000-fold more dilute than 1 ppm.
- 1 ppt=1 ng Solute / 1 L Solution

Reference

- https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Physical_Properties_of_Matter/Solutions_and_Mixtures/Solution_Basics/Units_Of_Concentration

Lesson 66

Preparation of stock solution

SOLUTION PREPARATION

A solution is a homogeneous mixture created by dissolving one or more solutes in a solvent. The chemical present in a smaller amount, the solute, is soluble in the solvent (the chemical present in a larger amount). Solutions with accurately known concentrations can be referred to as standard (stock) solutions. These solutions are bought directly from the manufacturer or formed by dissolving the desired amount of solute into a volumetric flask of a specific volume. Stock solutions are frequently diluted to solutions of lesser concentration for experimental use in the laboratory.

Preparing a Standard Solution from a Solid

A solution of known concentration can be prepared from solids by two similar methods. Although inherent errors exist with each of the methods, with careful technique either will suffice for making solutions in General Chemistry Laboratory.

In the first method, the solid solute is weighed out on weighing paper or in a small container and then transferred directly to a volumetric flask (commonly called a "vol flask"). A funnel might be helpful when transferring the solid into the slim neck of the vol flask. A small quantity of solvent is then added to the vol flask and the contents are swirled gently until the substance is completely dissolved. More solvent is added until the meniscus of the liquid reaches the calibration mark on the neck of the vol flask (a process called “diluting to volume”). The vol flask is then capped and inverted several times until the contents are mixed and completely dissolved. The disadvantage of this method is that some of the weighed solid may adhere to the original container, weighing paper, or funnel. Also, solid may be spilled when it is transferred into the slim neck of the vol flask.

- In the second method the solid is weighed out first in a small beaker. A small amount of solvent is added to the beaker and the solution is stirred until the solid is dissolved. The solution is then transferred to the vol flask. Again, a funnel may need to be inserted into the slim neck of the vol flask. Before adding additional solvent to the flask, the beaker, stirring rod, and funnel must be rinsed carefully and the washings added to the vol flask making sure all remaining traces of the solution have been transferred. Finally, the vol flask is diluted to volume (additional solvent is added to the flask until the liquid level reaches the calibration mark). The flask is capped and inverted as before until the contents are thoroughly mixed. The disadvantage to this method is that some of the solution may adhere to the beaker, stirring rod, or funnel if not washed thoroughly. Also, a possibility of contamination exists from the beaker, rod, or funnel if they have not been washed carefully.
-

- In general chemistry **molarity** is the most commonly used concentration unit:
-
- (1) Molarity = $\frac{\text{moles of solute}}{\text{liters of solution}}$ = $\frac{\text{grams of solute}}{\text{molar mass solute x liters of solution}}$
-
- *Example:* A student weighs 0.563 g of $FeCl_3$ and dissolves it in enough deionized (DI) water to make 100.0 mL of solution. ($FeCl_3$ is the solute and water is the solvent; the mixture of $FeCl_3$ and water is called the solution.) The molarity of the $FeCl_3$ (aq) solution is:
-
- $$\frac{0.563 \text{ g } FeCl_3}{162.2 \text{ g/mol } FeCl_3 \text{ x } 0.1000 \text{ L}} = 3.47 \text{ x } 10^{-2} \text{ M}$$
-

- ***Diluting a Solution of Known Concentration***
- Dilution is the addition of more solvent to produce a solution of reduced concentration. Most often a diluted solution is created from a small volume of a more concentrated stock solution. To make such a solution, a volumetric pipet is used to deliver an exact amount of the stock solution into a clean vol flask, which is then diluted to volume.To prevent extra dilution or contamination, prerinse the vol pipet with the stock solution to remove any water droplets or impurities.

Example:

- A student pipets exactly 5.00-mL of 3.47×10^{-2} M $FeCl_3$ solution into a vol flask and adds enough water to make 250.-mL of solution. What is the concentration of the diluted solution? Answer: Let M_2 be the concentration of the new solution. By using equation (4) and substituting known values for M_1, V_1 and V_2, solve for M_2:
-
- $$M_2 = \frac{M_1 V_1}{V_2} = \frac{(3.47 \times 10^{-2}\ M)(5.00\ mL)}{(250\ mL)} = 6.94 \times 10^{-4}\ M$$

- *Caution:* This procedure is reversed if the addition of the concentrated solution to solvent causes heating (an exothermic reaction). A notable example is the dilution of a concentrated acid. NEVER add water to concentrated acid. The reaction is very exothermic, heating the solution and potentially causing splattering. Always add the concentrated acid to water slowly with stirring. Place the beaker or flask in an ice bath to help cool the resulting solution and prevent spattering.
-
- The diluted solution's molarity is less than the stock solution it was created from. The moles present in the volume of stock solution delivered by the volumetric pipet is equal to the moles present in the diluted solution created:

- Moles of solute)before dilution = (Moles of solute) after dilution
-
- The moles of solute is also equal to the molarity (M) of the solution times the volume (V) of the solution (note that the volume units cancel):
-
- Moles of solute = M x V = mol/liter x liter
-

So equation (2) can be rewritten:

$M_1V_1 = M_2V_2$ (where 1 = "before dilution" and 2 = "after dilution")

Reference:

http://faculty.sites.uci.edu/chem1l/files/2013/11/RDGsolnprep.pdf

Lesson 67
Chemical Molarity, Normality

Chemical Molarity

- Molarity is defined as the number of moles of solute (the material dissolved) per liter of solution.
- Number of moles of sloute contained in one liter of solution.

Example

Adding salt(Nacl) in some volume of water.

Calculation

Two information are needed:

The moles of solute present in the solution.

The volume of solution(in liters) containing the solute.

Formula

- Molarity = number of moles of solute/number of liters of solution
- M = mol solute/ L solution

Example Problem

- What is the molarity of a 0.40 moles of Nacl dissolved in 0.250 L of water ?
- Solution:
- M = mole solute/ L solution
- M = 0.40mol/0.250L
- M = 1.6 moles/ L

Formula for moles

- Moles = mass in grams/molar mass
- Molarity = moles/volume

Normality

The strength of solution measured in terms of gram equivalent per litre is called normality.

It is denoted by N.

A solution having 1 gram equivalent of dissolved solute in 1 litre of its solution is called normal solution.

Normality Formula

- Suppose N1 and V1 are normality and volume of one solution and N2 and V2 are the normality and volume of another solution , according to the law of equivalent,
- $N_1V_1 = N_2V_2$
- This is called normality formula.

Lesson 68

Temperature- Celsius, centigrade, Fahrenheit

Scales of Temperature

- There are three different temperature scales.
- Celsius scale of Temperature
- Fahrenheit Scale of Temperature
- Kelvin Scale of Temperature

Celsius scale of Temperature

- This scale was introduced by a Celsius (Swedish Astronomer) and is known after them.
- It is also called centigrade scale.
- According to this scale, the freezing point of water is marked as 0°C and the boiling point of water is 100 °C.
- The interval between these points is divided into 100 equal intervals.
- Each division on scale is called one degree centigrade or Celsius and is denoted by °C.

Fahrenheit Scale of Temperature

- This scale was introduced by Fahrenheit and is known after him.
- In this scale the ice point is marked as 32°F and the steam point as 212°F.
- The interval between these points is divide3d into 180 parts.
- Each part is called a Fahrenheit degree and is denoted by °F.

Kelvin Scale of Temperature

- According to this scale, the ice point is assigned a numerical value of 273 K and boiling point of water as 373 K.
- The length between these two points is divided into 100 equal parts.
- The zero of this Kelvin scale is called absolute zero of temperature.
- At absolute zero temperature all atomic or molecular motion ceases.
- On Kelvin's Scale there does not exist any temperature below the absolute zero temperature i.e., there is no negative temperature on the Kelvin scale.
- It has been adopted as a standard in international system(SI).

Relationship between different scales of Temperature

- °F =9/5°C + 32
- °C = 5/9(°F – 32)
- K = °C + 273

Numerical

- Convert 373 K to Celsius and Fahrenheit scales.
- Solution:
- As we know that;
- $K = C + 273$
- $C = K - 273$
- $373 - 273 = 100K = 100°C$
- Using relation $F = 9/5(K - 273) + 32$
- $F = 9/5(373 - 273) + 32$
- $F = 212$

LESSON 69
Polymerase chain reaction

Polymerase chain reaction (PCR)

Polymerase chain reaction (PCR) is a method widely used to rapidly make millions to billions of copies of a specific DNA sample, allowing scientists to take a very small sample of DNA and amplify it to a large enough amount to study in detail. PCR was invented in 1983 by the American biochemist Kary Mullis at Cetus Corporation. It is fundamental to much of genetic testing including analysis of ancient samples of DNA and identification of infectious agents. Using PCR, copies of very small amounts of DNA sequences are exponentially amplified in a series of cycles of temperature changes. PCR is now a common and often indispensable technique used in medical laboratory research for a broad variety of applications including biomedical research and criminal forensics.

The majority of PCR methods rely on thermal cycling. Thermal cycling exposes reactants to repeated cycles of heating and cooling to permit different temperature-dependent reactions – specifically, DNA melting and enzyme-driven DNA replication. PCR employs two main reagents – primers (which are short single strand DNA fragments known as oligonucleotides that are a complementary sequence to the target DNA region) and a DNA polymerase. In the first step of PCR, the two strands of the DNA double helix are physically separated at a high temperature in a process called nucleic acid denaturation

In the second step, the temperature is lowered and the primers bind to the complementary sequences of DNA. The two DNA strands then become templates for DNA polymerase to enzymatically assemble a new DNA strand from free nucleotides, the building blocks of DNA. As PCR progresses, the DNA generated is itself used as a template for replication, setting in motion a chain reaction in which the original DNA template is exponentially amplified.

Principles

PCR amplifies a specific region of a DNA strand (the DNA target). Most PCR methods amplify DNA fragments of between 0.1 and 10 kilo base pairs (kbp) in length, although some techniques allow for amplification of fragments up to 40 kbp. The amount of amplified product is determined by the available substrates in the reaction, which becomes limiting as the reaction progresses.

A basic PCR set-up requires several components and reagents, including:

- a *DNA template* that contains the DNA target region to amplify
- a *DNA polymerase*; an enzyme that polymerizes new DNA strands; heat-resistant *Taq* polymerase is especially common, as it is more likely to remain intact during the high-temperature DNA denaturation process
- two DNA *primers* that are complementary to the 3′ (three prime) ends of each of the sense and anti-sense strands of the DNA target (DNA polymerase can only bind to and elongate from a double-stranded region of DNA; without primers, there is no double-stranded initiation site at which the polymerase can bind); specific primers that are complementary to the DNA target region are selected beforehand, and are often custom-made in a laboratory or purchased from commercial biochemical suppliers

- *deoxynucleoside triphosphates*, or dNTPs (sometimes called "deoxynucleotide triphosphates"; nucleotides containing triphosphate groups), the building blocks from which the DNA polymerase synthesizes a new DNA strand
- a *buffer solution* providing a suitable chemical environment for optimum activity and stability of the DNA polymerase
- *bivalent cations*, typically magnesium (Mg) or manganese (Mn) ions; Mg^{2+} is the most common, but Mn^{2+} can be used for PCR-mediated DNA mutagenesis, as a higher Mn^{2+} concentration increases the error rate during DNA synthesis; and *monovalent cations*, typically potassium (K) ions

The reaction is commonly carried out in a volume of 10–200 µL in small reaction tubes (0.2–0.5 mL volumes) in a thermal cycler. The thermal cycler heats and cools the reaction tubes to achieve the temperatures required at each step of the reaction (see below). Many modern thermal cyclers make use of the Peltier effect, which permits both heating and cooling of the block holding the PCR tubes simply by reversing the electric current. Thin-walled reaction tubes permit favorable thermal conductivity to allow for rapid thermal equilibrium. Most thermal cyclers have heated lids to prevent condensation at the top of the reaction tube. Older thermal cyclers lacking a heated lid require a layer of oil on top of the reaction mixture or a ball of wax inside the tube

Lesson 70

PCR Procedure

PCR Procedure

Typically, PCR consists of a series of 20–40 repeated temperature changes, called thermal cycles, with each cycle commonly consisting of two or three discrete temperature steps (see figure below). The cycling is often preceded by a single temperature step at a very high temperature (>90 °C (194 °F)), and followed by one hold at the end for final product extension or brief storage. The temperatures used and the length of time they are applied in each cycle depend on a variety of parameters, including the enzyme used for DNA synthesis, the concentration of bivalent ions and dNTPs in the reaction, and the melting temperature (T_m) of the primers. The individual steps common to most PCR methods are as follows:

- *Initialization*: This step is only required for DNA polymerases that require heat activation by hot-start PCR. It consists of heating the reaction chamber to a temperature of 94–96 °C (201–205 °F), or 98 °C (208 °F) if extremely thermostable polymerases are used, which is then held for 1–10 minutes.
- *Denaturation*: This step is the first regular cycling event and consists of heating the reaction chamber to 94–98 °C (201–208 °F) for 20–30 seconds. This causes DNA melting, or denaturation, of the double-stranded DNA template by breaking the hydrogen bonds between complementary bases, yielding two single-stranded DNA molecules.

- *Annealing*: In the next step, the reaction temperature is lowered to 50–65 °C (122–149 °F) for 20–40 seconds, allowing annealing of the primers to each of the single-stranded DNA templates. Two different primers are typically included in the reaction mixture: one for each of the two single-stranded complements containing the target region. The primers are single-stranded sequences themselves, but are much shorter than the length of the target region, complementing only very short sequences at the 3′ end of each strand.
- It is critical to determine a proper temperature for the annealing step because efficiency and specificity are strongly affected by the annealing temperature. This temperature must be low enough to allow for hybridization of the primer to the strand, but high enough for the hybridization to be specific, i.e., the primer should bind *only* to a perfectly complementary part of the strand, and nowhere else. If the temperature is too low, the primer may bind imperfectly. If it is too high, the primer may not bind at all. A typical annealing temperature is about 3–5 °C below the T_m of the primers used. Stable hydrogen bonds between complementary bases are formed only when the primer sequence very closely matches the template sequence. During this step, the polymerase binds to the primer-template hybrid and begins DNA formation.

- *Extension/elongation*: The temperature at this step depends on the DNA polymerase used; the optimum activity temperature for the thermostable DNA polymerase of *Taq* polymerase is approximately 75–80 °C (167–176 °F), though a temperature of 72 °C (162 °F) is commonly used with this enzyme. In this step, the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding free dNTPs from the reaction mixture that is complementary to the template in the 5′-to-3′ direction, condensing the 5′-phosphate group of the dNTPs with the 3′-hydroxy group at the end of the nascent (elongating) DNA strand.

The precise time required for elongation depends both on the DNA polymerase used and on the length of the DNA target region to amplify. As a rule of thumb, at their optimal temperature, most DNA polymerases polymerize a thousand bases per minute. Under optimal conditions (i.e., if there are no limitations due to limiting substrates or reagents), at each extension/elongation step, the number of DNA target sequences is doubled. With each successive cycle, the original template strands plus all newly generated strands become template strands for the next round of elongation, leading to exponential (geometric) amplification of the specific DNA target region.

- The processes of denaturation, annealing and elongation constitute a single cycle. Multiple cycles are required to amplify the DNA target to millions of copies. The formula used to calculate the number of DNA copies formed after a given number of cycles is 2^n, where *n* is the number of cycles. Thus, a reaction set for 30 cycles results in 2^{30}, or 1,073,741,824, copies of the original double-stranded DNA target region.
- *Final elongation*: This single step is optional, but is performed at a temperature of 70–74 °C (158–165 °F) (the temperature range required for optimal activity of most polymerases used in PCR) for 5–15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully elongated.
- *Final hold*: The final step cools the reaction chamber to 4–15 °C (39–59 °F) for an indefinite time, and may be employed for short-term storage of the PCR products.

Optimization

In practice, PCR can fail for various reasons, in part due to its sensitivity to contamination causing amplification of spurious DNA products. Because of this, a number of techniques and procedures have been developed for optimizing PCR conditions.Contamination with extraneous DNA is addressed with lab protocols and procedures that separate pre-PCR mixtures from potential DNA contaminants. This usually involves spatial separation of PCR-setup areas from areas for analysis or purification of PCR products, use of disposable plasticware, and thoroughly cleaning the work surface between reaction setups. Primer-design techniques are important in improving PCR product yield and in avoiding the formation of spurious products, and the usage of alternate buffer components or polymerase enzymes can help with amplification of long or otherwise problematic regions of DNA. Addition of reagents, such as formamide, in buffer systems may increase the specificity and yield of PCR. Computer simulations of theoretical PCR results (Electronic PCR) may be performed to assist in primer design

Effort By Amaan Khan

LESSON 71

Applications Of PCR

Selective DNA isolation

PCR allows isolation of DNA fragments from genomic DNA by selective amplification of a specific region of DNA. This use of PCR augments many ways, such as generating hybridization probes for Southern or northern hybridization and DNA cloning, which require larger amounts of DNA, representing a specific DNA region. PCR supplies these techniques with high amounts of pure DNA, enabling analysis of DNA samples even from very small amounts of starting material.

Selective DNA isolation

Other applications of PCR include DNA sequencing to determine unknown PCR-amplified sequences in which one of the amplification primers may be used in Sanger sequencing, isolation of a DNA sequence to expedite recombinant DNA technologies involving the insertion of a DNA sequence into a plasmid, phage, or cosmid (depending on size) or the genetic material of another organism. Bacterial colonies *(such as E. coli)* can be rapidly screened by PCR for correct DNA vector constructs. PCR may also be used for genetic fingerprinting; a forensic technique used to identify a person or organism by comparing experimental DNAs through different PCR-based methods.

Selective DNA isolation

Some PCR fingerprint methods have high discriminative power and can be used to identify genetic relationships between individuals, such as parent-child or between siblings, and are used in paternity testing This technique may also be used to determine evolutionary relationships among organisms when certain molecular clocks are used (i.e. the 16S rRNA and recA genes of microorganisms

Amplification and quantification of DNA

PCR amplifies the regions of DNA that it targets, PCR can be used to analyze extremely small amounts of sample. This is often critical for forensic analysis, when only a trace amount of DNA is available as evidence. PCR may also be used in the analysis of ancient DNA that is tens of thousands of years old. These PCR-based techniques have been successfully used on animals, such as a forty-thousand-year-old mammoth, and also on human DNA, in applications ranging from the analysis of Egyptian mummies to the identification of a Russian tsar and the body of English king Richard

Quantitative PCR or Real Time PCR (qPCR, not to be confused with RT-PCR) methods allow the estimation of the amount of a given sequence present in a sample—a technique often applied to quantitatively determine levels of gene expression. Quantitative PCR is an established tool for DNA quantification that measures the accumulation of DNA product after each round of PCR amplification.

Medical and diagnostic applications

Prospective parents can be tested for being genetic carriers, or their children might be tested for actually being affected by a disease. DNA samples for prenatal testing can be obtained by amniocentesis, chorionic villus sampling, or even by the analysis of rare fetal cells circulating in the mother's bloodstream. PCR analysis is also essential to preimplantation genetic diagnosis, where individual cells of a developing embryo are tested for mutations.

- PCR can also be used as part of a sensitive test for *tissue typing*, vital to organ transplantation. As of 2008, there is even a proposal to replace the traditional antibody-based tests for blood type with PCR-based tests.
- Many forms of cancer involve alterations to *oncogenes*. By using PCR-based tests to study these mutations, therapy regimens can sometimes be individually customized to a patient. PCR permits early diagnosis of malignant diseases such as leukemia and lymphomas, which is currently the highest-developed in cancer research and is already being used routinely. PCR assays can be performed directly on genomic DNA samples to detect translocation-specific malignant cells at a sensitivity that is at least 10,000 fold higher than that of other methods. PCR is very useful in the medical field since it allows for the isolation and amplification of tumor suppressors. Quantitative PCR for example, can be used to quantify and analyze single cells, as well as recognize DNA, mRNA and protein confirmations and combinations.

Research applications

- PCR has been applied to many areas of research in molecular genetics:
- PCR allows rapid production of short pieces of DNA, even when not more than the sequence of the two primers is known. This ability of PCR augments many methods, such as generating *hybridization probes* for Southern or northern blot hybridization. PCR supplies these techniques with large amounts of pure DNA, sometimes as a single strand, enabling analysis even from very small amounts of starting material.
- The task of *DNA sequencing* can also be assisted by PCR. Known segments of DNA can easily be produced from a patient with a genetic disease mutation. Modifications to the amplification technique can extract segments from a completely unknown genome, or can generate just a single strand of an area of interest.

sources

- https://en.wikipedia.org/wiki/Polymerase_chain_reaction#:~

Lesson 72

Types of polymerase chain reaction-PCR

Types

- Real-Time **PCR** (quantitative **PCR** or qPCR)
- Reverse-Transcriptase (RT-**PCR**)
- Multiplex **PCR**.
- Nested **PCR**.
- High Fidelity **PCR**.
- Fast **PCR**.
- Hot Start **PCR**.
- GC-Rich **PCR**

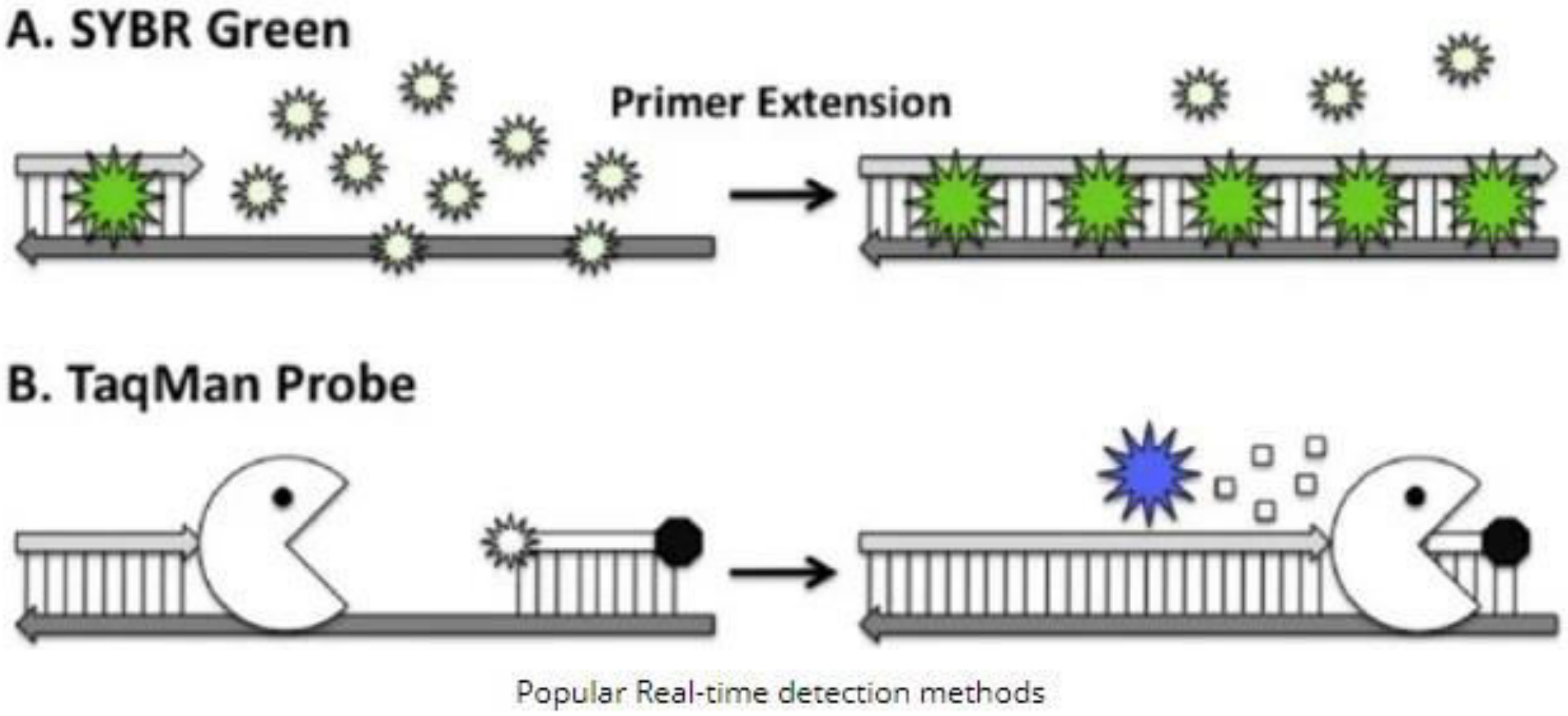
Real-time PCR

Real-time PCR also called quantitative PCR (qPCR) is a variant of standard polymerase chain reaction in which amplification and simultaneous quantitation of a target DNA is done in the same PCR machine, using commercially available fluorescence-detecting thermocyclers. Fluorescent dyes specifically label DNA of interest and the amount of fluorescence generated is proportional to the quantity of DNA present.

Real-time PCR

Depending on the available excitation source and detection filters, a variety of fluorescent dyes may be used in qPCR. Two most commonly use real-time PCR methods uses **SYBR green** (a dye that binds to double-stranded DNA but not to single-stranded DNA, and, when so bound, fluoresces) and **TaqMan probes** respectively.

Real-time PCR

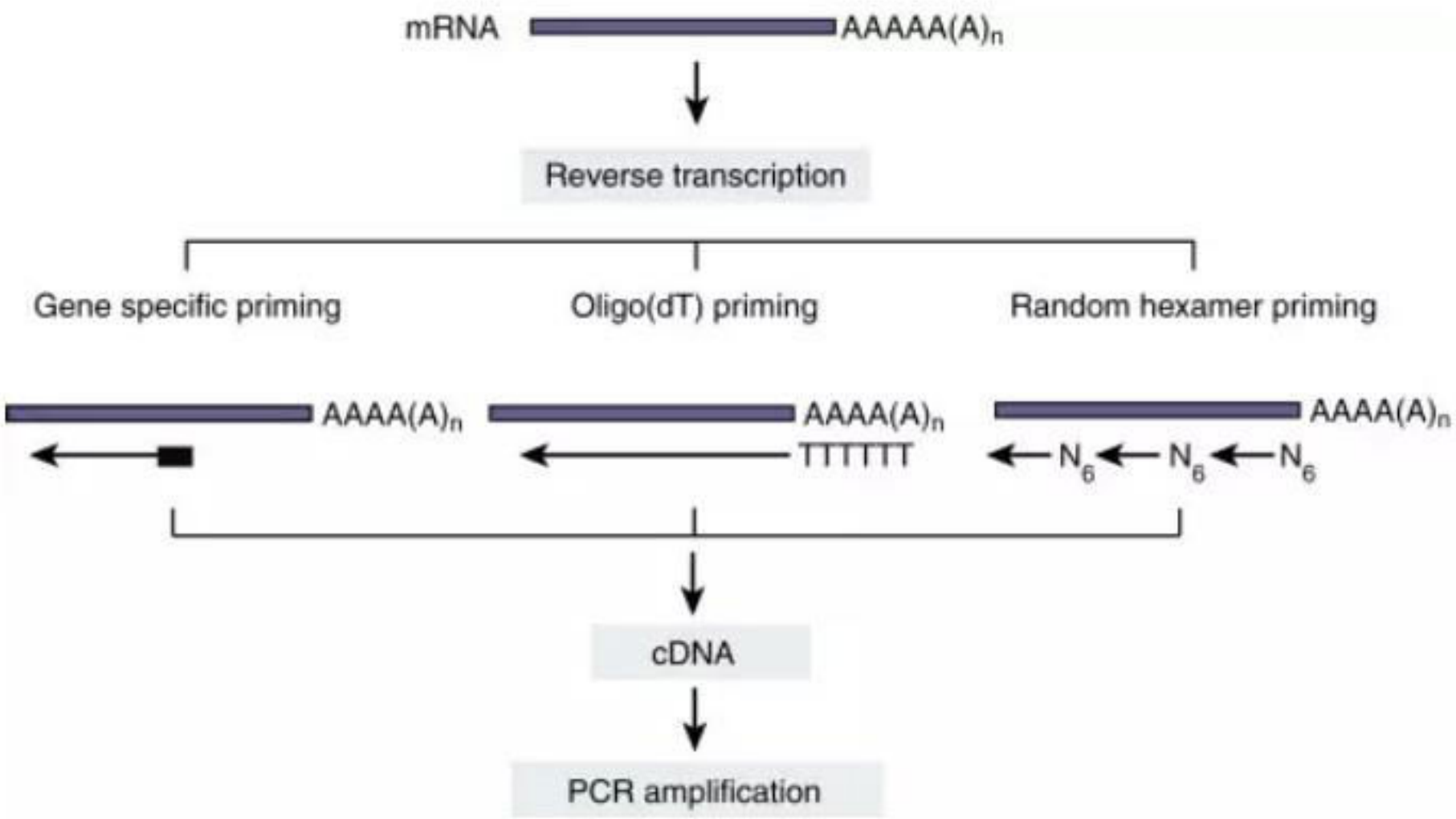


Popular Real-time detection methods

Real-time PCR Principle

Real-time PCR is accomplished in the same manner as conventional PCR-based assays (denaturation of double-stranded DNA followed by primer annealing and extension). However, it is the detection process that discriminates real-time PCR from conventional PCR assays. In real-time PCR assays, accumulation of amplicon is monitored as it is generated using labelling of primers with dyes capable of fluorescence. These labels produce a change in fluorescent signal that is measured by the instrument following their direct interaction with or hybridization to the amplicon. This signal is related to the amount of amplified product present during each cycle and increases as the number of specific amplicon increases.

Reverse-Transcriptase (RT-**PCR**)



Reverse-Transcriptase (RT-**PCR**)

- Reverse Transcriptase PCR (RT-PCR) is a variation of the polymerase chain reaction that amplifies target RNA. Addition of reverse transcriptase (RT) enzyme prior to PCR makes it possible to amplify and detect RNA targets.
- Reverse transcriptase enzyme transcribes the template RNA and forms complementary DNA (cDNA). Single-stranded cDNA is converted into double-stranded DNA using DNA polymerase. These DNA molecules can now be used as templates for a PCR reaction.

Principle of RT-PCR

- Reverse transcription and PCR amplification can be performed as a two-step process in a **single tube** or with **two separate reactions**. In both cases, RNA is first reverse-transcribed into cDNA, which is then used as the template for PCR amplification.
- The primers used for cDNA synthesis can be either non–sequence-specific primers (a mixture of random hexamers or oligo-dT primers) or sequence-specific primers.

Principle of RT-PCR

- **Non-sequence-specific primers:**
  - **Random hexamers** are a mixture of all possible combinations of six nucleotide sequences that can attach randomly to mRNA and initiate reverse transcription of the entire RNA pool.
  - **Oligo-dT primers** are complementary to the poly-A tail of mRNA molecules and allow synthesis of cDNA only from mRNA molecules.
- **Sequence-specific primers:**
  - Sequence-specific primers are the most restricted because they are designed to bind selectively to mRNA molecules of interest, which makes reverse transcription a target-specific process.
-

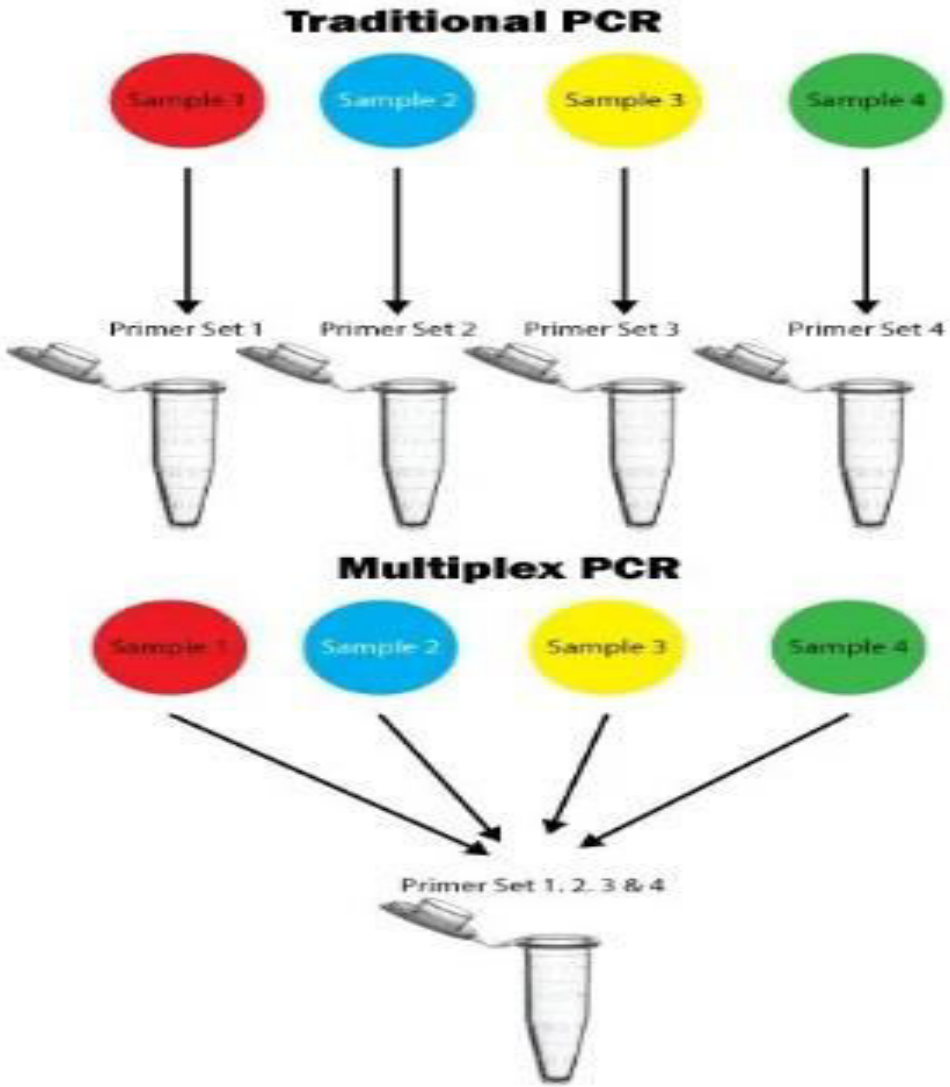
Multiplex PCR

Multiplex PCR is a variant of **PCR method** in which more than one target sequence is amplified using multiple sets of primers within a single PCR mixture. This enables amplification of several gene segments at the same time, instead of specific test runs for each. This technology was first used by Chamberlain et al. for the diagnosis of Duchenne muscular dystrophy (1988).

Multiplex PCR

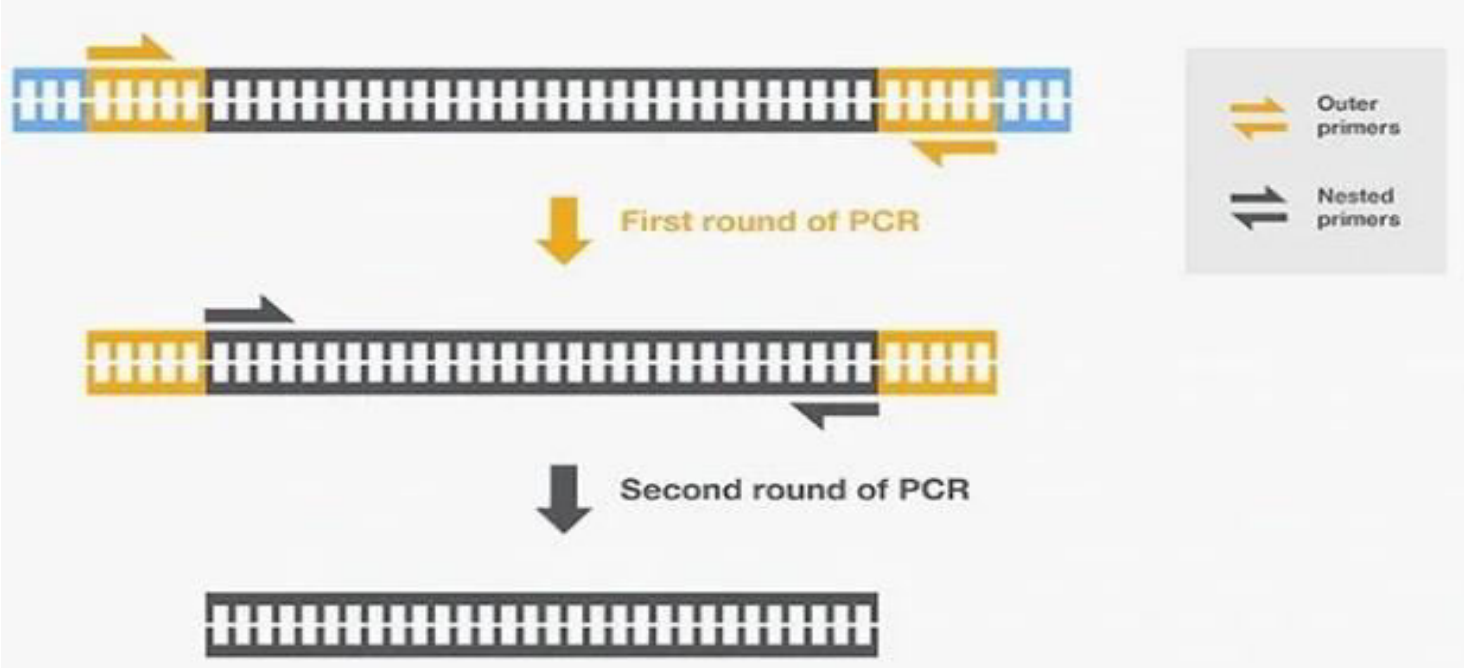
Multiplex PCR is a space, time and cost-effective method for the genetic analyses that need to be repeated many times (e.g. sequencing). It requires a small amount of DNA (10–200 ng) as the starting template and can be performed on specimens with a suboptimal DNA quality. Though multiplex PCR has many benefits, optimization of it is equally challenging. While using multiple primer pairs, primers from one pair can interact with primers from another one. As each primer pair could have different requirements, there is not a single optimum melting temperature (Tm) and ΔG.

Multiplex PCR



Nested PCR

- Nested PCR is a modification of PCR designed to increase the sensitivity and specificity of the assay reaction. It involves the use of two primer sets directed against the same target and two successive PCR reactions.



High Fidelity **PCR**

The fidelity of a polymerase refers to its ability to insert the correct base during PCR. Conversely, the rate of misincorporation is known as a polymerase's error rate. High-fidelity PCR, utilizes a DNA polymerase with a low error rate and results in a high degree of accuracy in the replication of the DNA of interest. NEB scientists were the first to identify and commercialize a high-fidelity DNA polymerase suitable for PCR, namely Vent® DNA Polymerase. Since its discovery, several other high fidelity polymerases have been discovered and engineered – all with differing processivity, speed and levels of accuracy.

Fast PCR

Enzymes capable of **fast PCR** can help in achieving faster turnaround times from sample to result and give improved throughput on existing thermocyclers. **PCR** Biosystems' range of **PCR** enzymes have extension rates that are typically 3-6kb/min compared to approximately 1kb/min for wild type Taq DNA polymerase.

Hot Start PCR

Hot Start PCR is a technique that reduces non-specific amplification and offers the convenience of reaction set up at room temperature. The polymerases used in Hot Start PCR are unreactive at ambient temperatures. Polymerase activity can be inhibited at these temperatures through different mechanisms, including antibody interaction, chemical modification and aptamer technology. At permissive reaction temperatures reached during PCR cycling, the polymerase dissociates from its inhibitor and commences polymerization. Use of hot start DNA polymerases is most often recommended for high-throughput applications, experiments requiring a high degree of specificity, or even routine PCR where the added security offered by a hot start enzyme is desired.

GC-Rich PCR

- The **GC-RICH PCR** System is a blend of Taq DNA Polymerase and a proofreading polymerase for amplifying longer nucleic acid fragments. The **GC-Rich** Solution provides an solution to enable amplification of all kind of difficult **PCR** products.

Sources:

- https://microbeonline.com/real-time-pcr-principles-and-applications/
- https://microbeonline.com/rt-pcr-principles-applications/
- https://microbeonline.com/multiplex-pcr-principle-applications-and-limitations/
- https://microbeonline.com/nested-pcr-principle-applications/

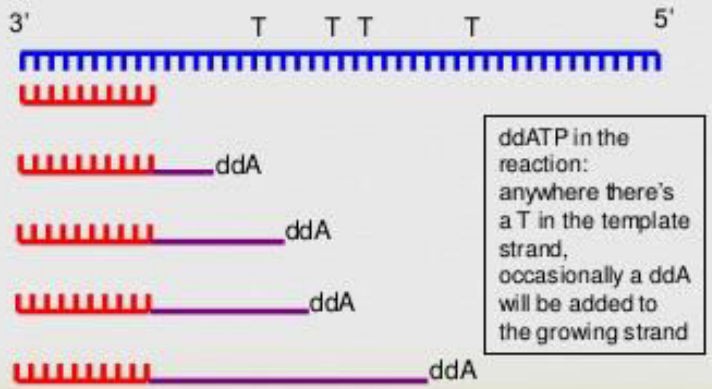
Lesson 73
DNA sequencing (Sanger)

PRINCIPLE

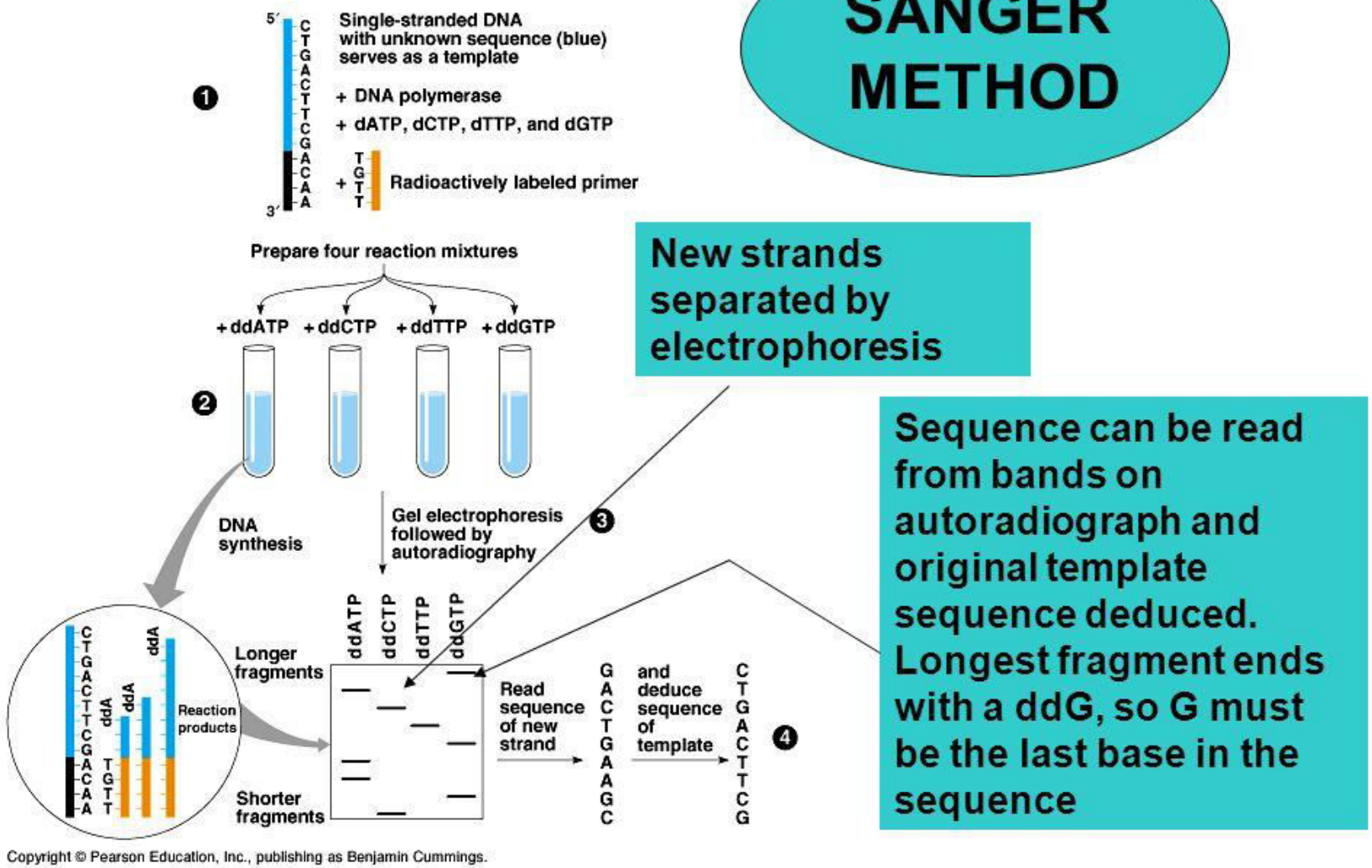
- The 5’ carbon of an “incoming” deoxynucleotide (dNTP) is joined to the 3’ carbon at the end of the chain. Hydroxyl groups in each position form ester linkages with a central phosphate. , the nucleotide chain elongates.
- The key to Sanger’s sequencing method is the peculiar chemistry of dideoxynucleotides (ddNTP). Like a deoxynucleotide, a ddNTP is incorporated into a chain by forming a phosphodiester linkage at its 5’ end.
- However, the ddNTP lacks a 3’ hydroxyl group (OH) necessary to form the linkage with an incoming nucleotide.

BASIC PRINCIPLE

- This method generally is an In-Vitro synthesis of DNA strand and by using terminators (di-deoxynucleotide) the growing strand terminates at specific site.
- Upon termination the strands are overlap to got original sequence of unknown DNA Strand.



SANGER METHOD



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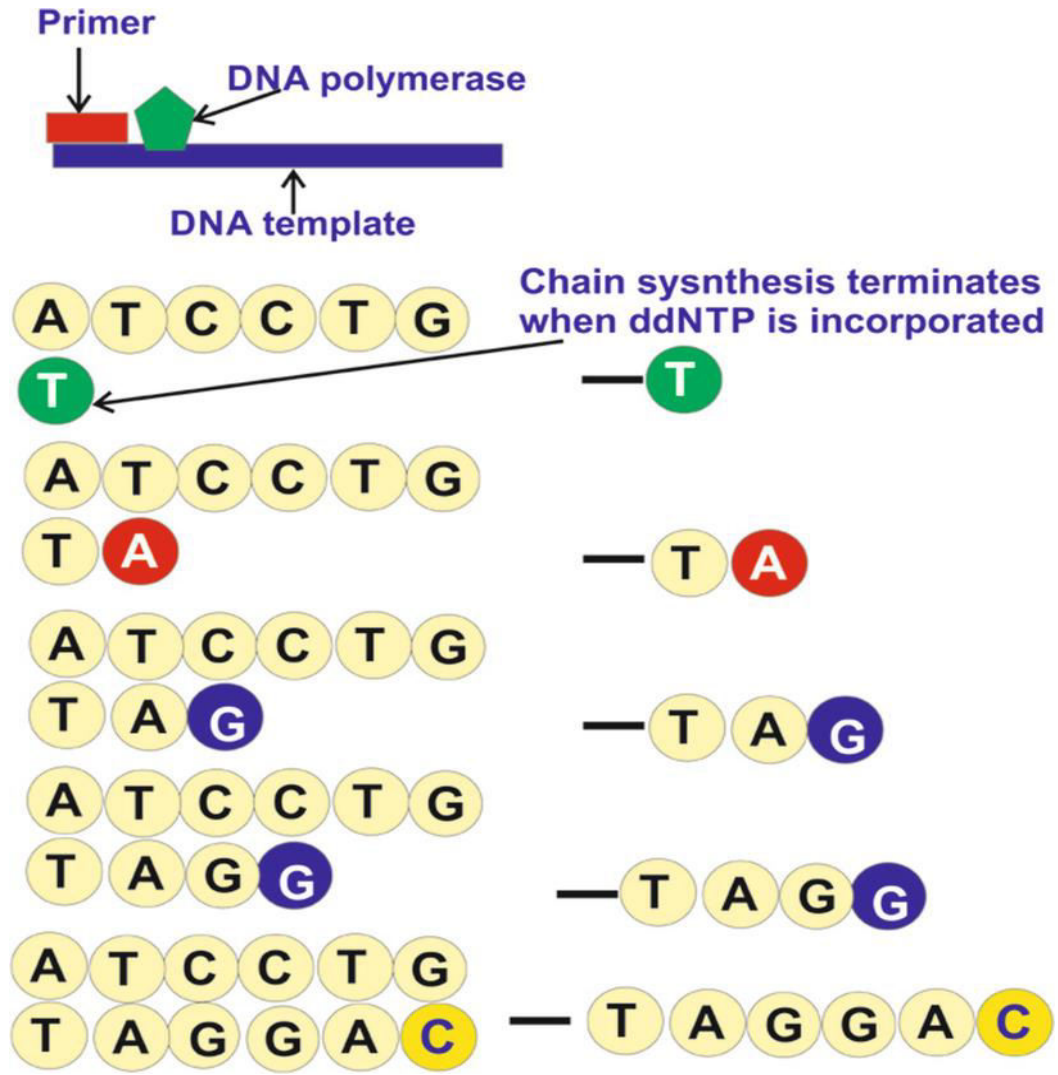
New strands separated by electrophoresis

Sequence can be read from bands on autoradiograph and original template sequence deduced. Longest fragment ends with a ddG, so G must be the last base in the sequence

METHOD

- To sequence DNA, four separate reactions are necessary – one to provide sequence information about each of the nucleotides. Each reaction contains: template DNA, a short primer (about 20 nucleotides), DNA polymerase, and the four dNTPs (one radioactively labeled). • One type of ddNTP – A, T, C, or G – is added to each.

- Each reaction is then loaded into a separate lane of a polyacrylamide gel containing urea, which prevents the DNA strands from renaturing during electrophoesis.
- Ionized phosphates give the DNA molecule a negative charge, so DNA migrate toward the positive pole of an electric field. The movement of DNA molecules through the polyacrylamide matrix is size dependent.
- Over the course of electrophoresis, shorter DNA molecules will move further down the gel than larger ones. Millions of terminated molecules of the same size will migrate to the same place and “band” in the gel.


Primer
DNA polymerase
DNA template
Chain sysnthesis terminates when ddNTP is incorporated
A T C C T G
T
T
A T C C T G
T A
T A
A T C C T G
T A G
T A G
A T C C T G
T A G G
T A G G
A T C C T G
T A G G A C
T A G G A C

- THANK YOU

Lesson 74
DNA sequencing (Sanger)

What is sequencing?

- DNA sequencing is the process of determining the sequence of nucleotide bases(As, Ts, Cs, and Gs)in a piece of DNA.Today, with the right equipment and materials, sequencing a short piece of DNA is relatively straightforward.

Sanger sequencing: The chain termination method

- **INTRODUCTION**
- The DNA sequencing method developed by Fred Sanger from the basis of automated “cycle” sequencing reaction today.
- In the 1980s two key developments allowed researchers to believe that sequencing the entire genome could be possible.
- The first was a technique called polymerase chain reaction (PCR)that enabled many copies of DNA sequence to be quickly and accurately produced.

- Sanger sequencing method of DNA sequencing was first commercialized by Applied Biosystems.
- It was the most widely used sequencing method for approximately 40 years.
- Regions of DNA up to about 900 base pairs in length are routinely sequenced using this method.
- In the Human Genome Project, Sanger sequencing was used to determine the sequences of many relatively small fragments of human DNA.
- The fragments were aligned based on overlapping portions to assemble the sequences of larger regions of DNA and, eventually, entire chromosomes.

REQUIREMENTS

- Requirements for Sanger sequencing Fred Sanger’s method of DNA sequencing was based on Arthur Kornberg’s earlier work on DNA Replication. A new DNA strand is synthesized using an existing strand as a template.Sanger sequencing involves making many copies of a target DNA region. They include:
- A DNA polymerase enzyme
- A primer, which is a short piece of single-stranded DNA that binds to the template DNA and acts as a "starter" for the polymerase •

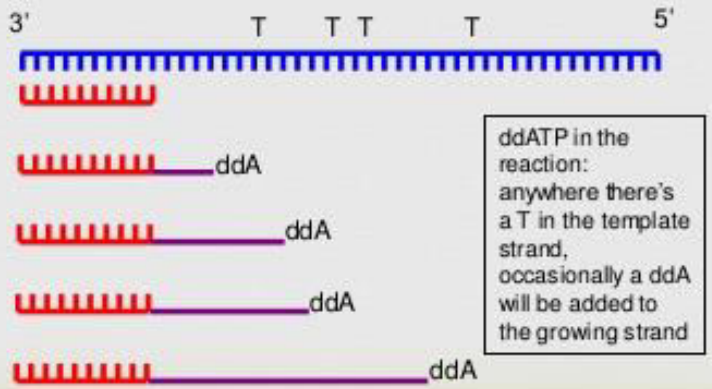
- The four DNA nucleotides (dATP, dTTP, dCTP, dGTP)
- The template DNA to be sequenced However, a Sanger sequencing reaction also contains a unique ingredient:
- Dideoxy, or chain-terminating, versions of all four nucleotides (ddATP, ddTTP, ddCTP, ddGTP), each labeled with a different color of dye.

PRINCIPLE

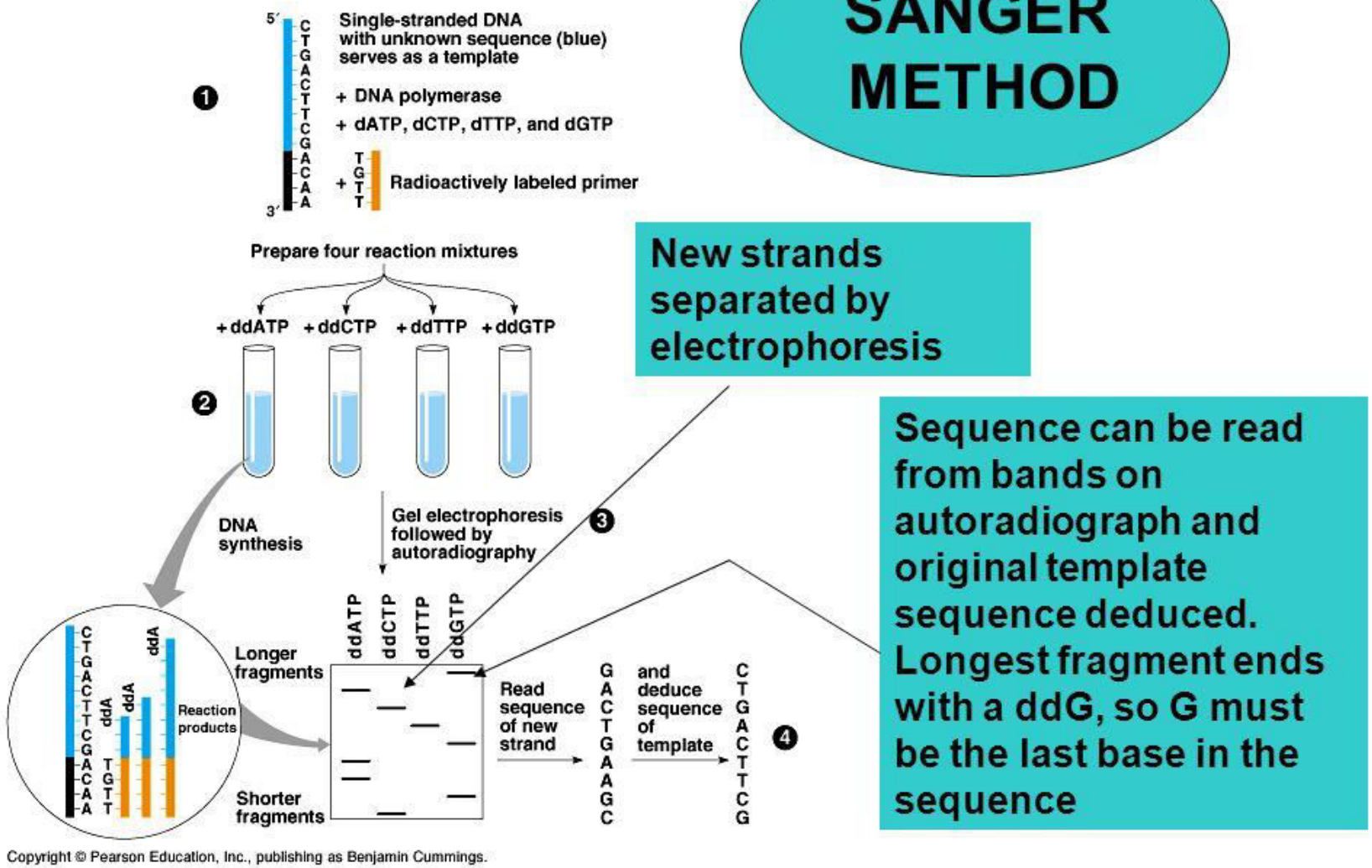
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SANGER METHOD



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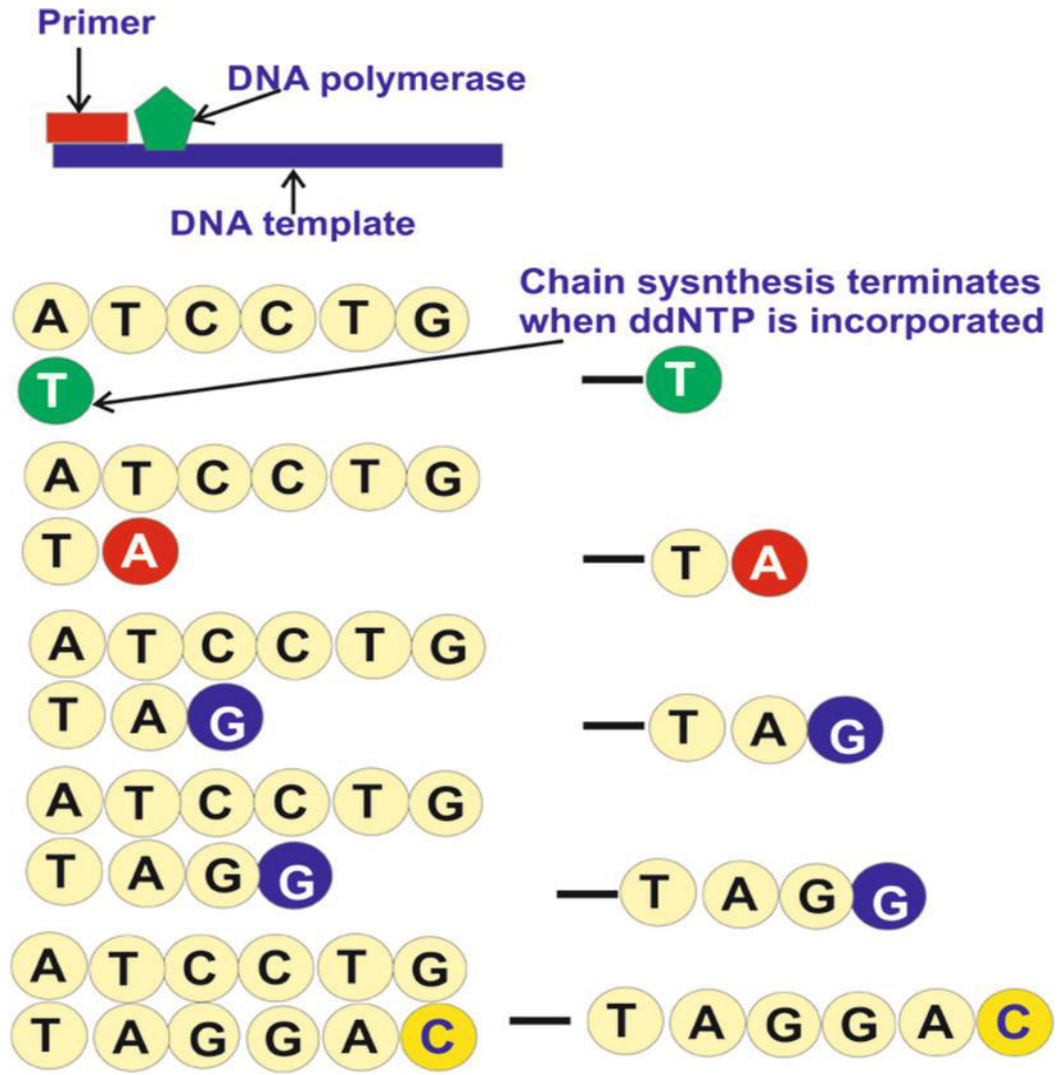
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DNA polymerase
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T A
A T C C T G
T A G
T A G
A T C C T G
T A G G
T A G G
A T C C T G
T A G G A C
T A G G A C

- THANK YOU

Lesson 75
DNA Sequencing(Maxam Gilbert)

Method

- Maxam-Gilbert method of DNA Sequencing
- It is a method by which the sequence of a DNA fragment is identified by using chemicals, thats cut DNA at specific points.
- Also called Chemical degradation method of DNA sequencing.

Procedure

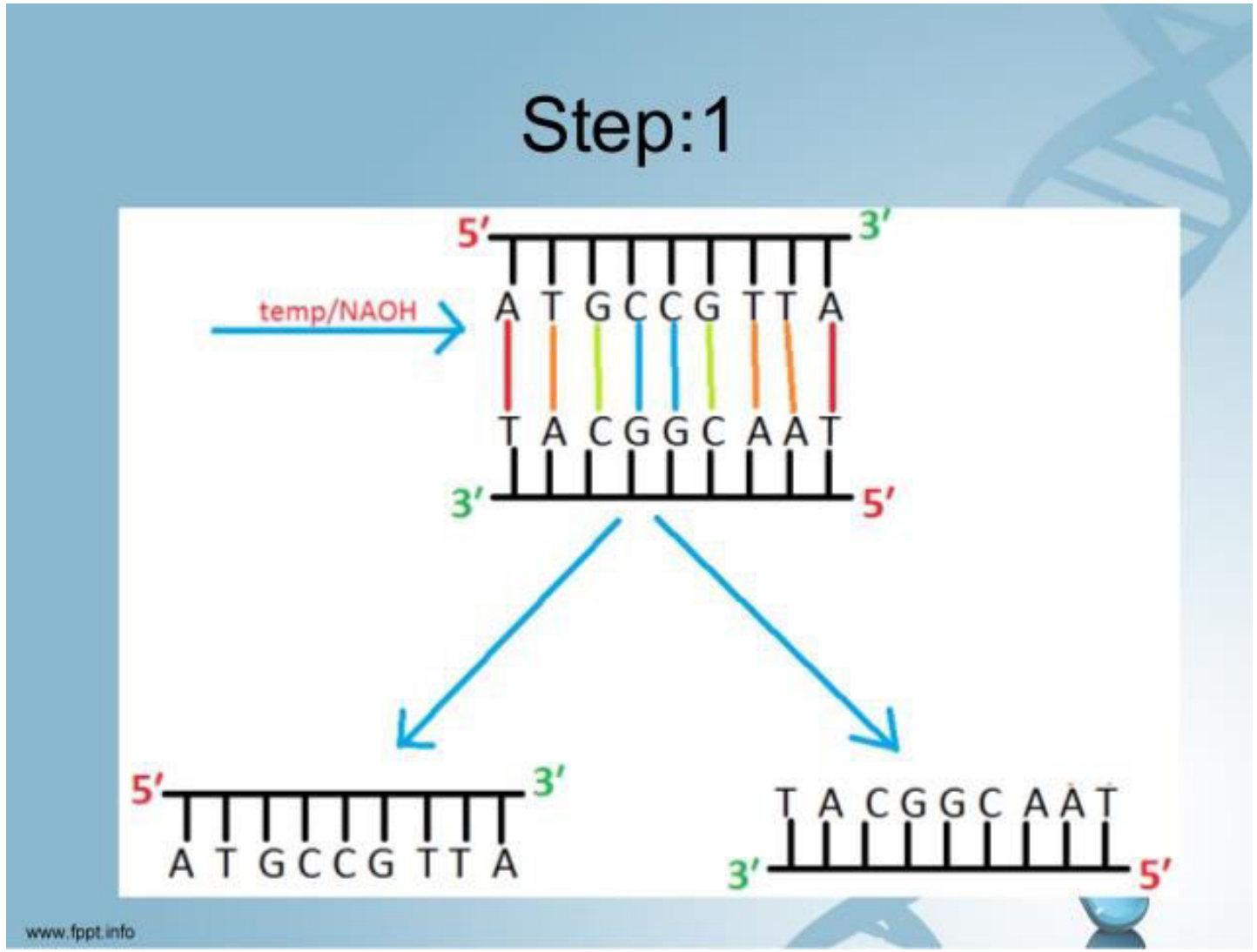
it is a fragment of double stranded DNA, and you do not its sequence:

Step :1

• As the sequence of both strands are unknown, but if we find out the sequence of one strand, we would get to know sequence of other one also.

At first, the double stranded fragment is separeted into two single strands by applying high Temperature or high PH.

Step:1



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Step 2:

- Run the single stranded fragments on gel. As lighter fragment band will move further than the heavy fragment band.
- How will we know which one is the lighter band?
- The band having larger number of purines(A,G) would be heavier.
- Single stranded DNA fragments in Gel Electrophoresis gel.

Step: 3

- Take one of fragment band from the gel.

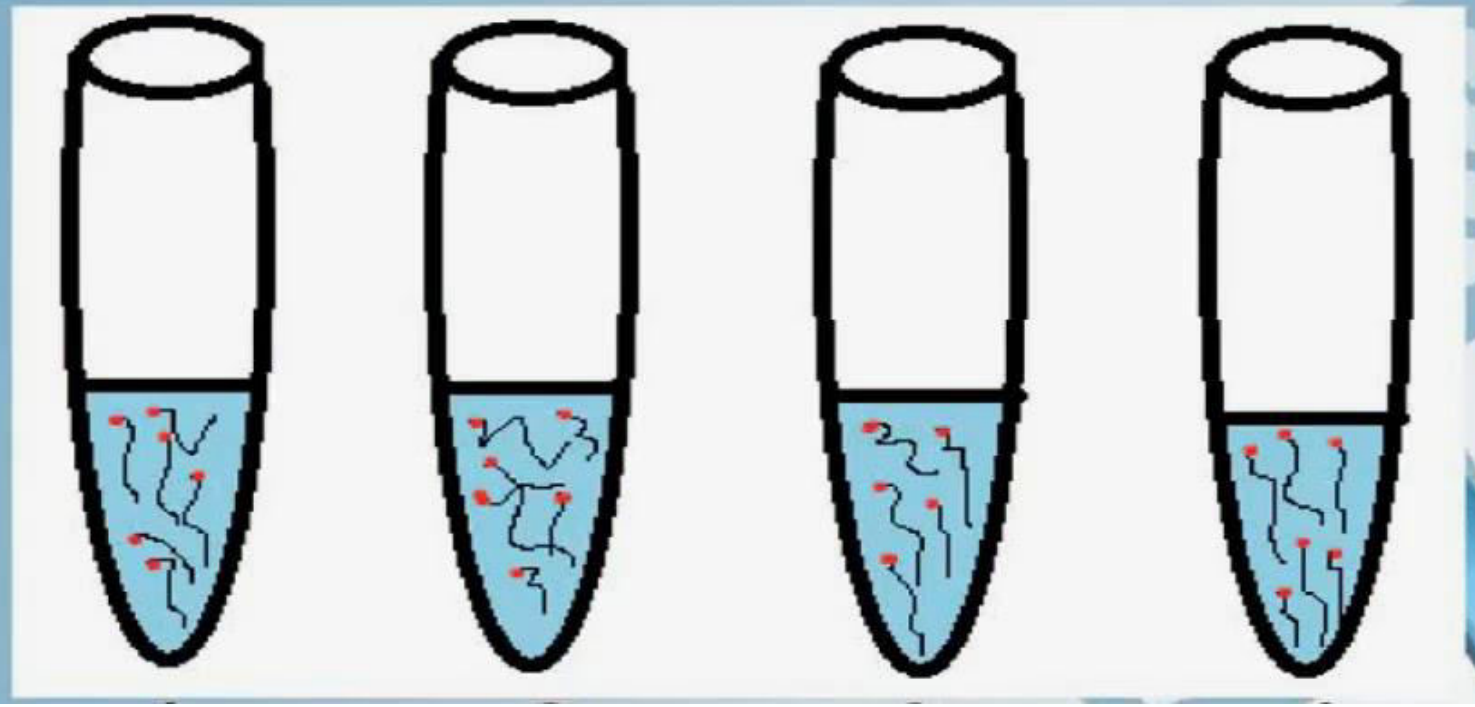
- Remove the Phosphate at 5’ end and incorporate Radioactive Phosphate 32- PO4 enztmatically.

Step: 4

- Radioactive Labelling
- Now put all the radioactively labelled fragments in four tubes. 1 2 3 4

Step 4: Radioactive Labelling

- Now put all the radioactively labelled fragments in four tubes.



Step: 5

- Chemical Degradation Tube
- Tube:1 : Increase Temperature and PH(by adding NAOH), that would cause fragments to break down. Dimethyl sulfate will be added that would make cuts at Adenine and Guanine positions.
- Tube 2: Dimethyl sulfate and dilute HCL will be added that would cuts the fragment at Adenine position
- Tube 3: Reagents Hydrazine and Piperidine are added that would cuts the fragment at position Cytocine and Thyamine.
- Tube 4: In the last tube, Hydrazine, Piperdine and NACL is added that would cuts the fragment at Cytocine position. Dimethylsulphate + High Temp + NAOH Dimethylsulphate + dil HCL Hydrazine + Piperidine Hydrazine + NACL + Piperidine

Step:6

- Gel electrophoresis
- All of the fragments from each four tubes are pour in Gel.
- Four wells will be make on Gel, in 1st well, fragments from 1st tube is pour, in 2nd well fragments from 2nd tubes and so on.
- Fragments would separate on Gel according to size.
- Smaller fragments would move farther than larger fragments.
- After placing radioactive film on top of gel, radioactive labelled fragments would emit a spot at their position.

Advantages

- Directly read purified DNA.
- Used sequence heterogenous DNA as well as Homopolymeric sequences.
- used to analyze DNA-Protein interaction.
- Used to analyze Epigenic modification and nucleic acid structure.

Disadvantages

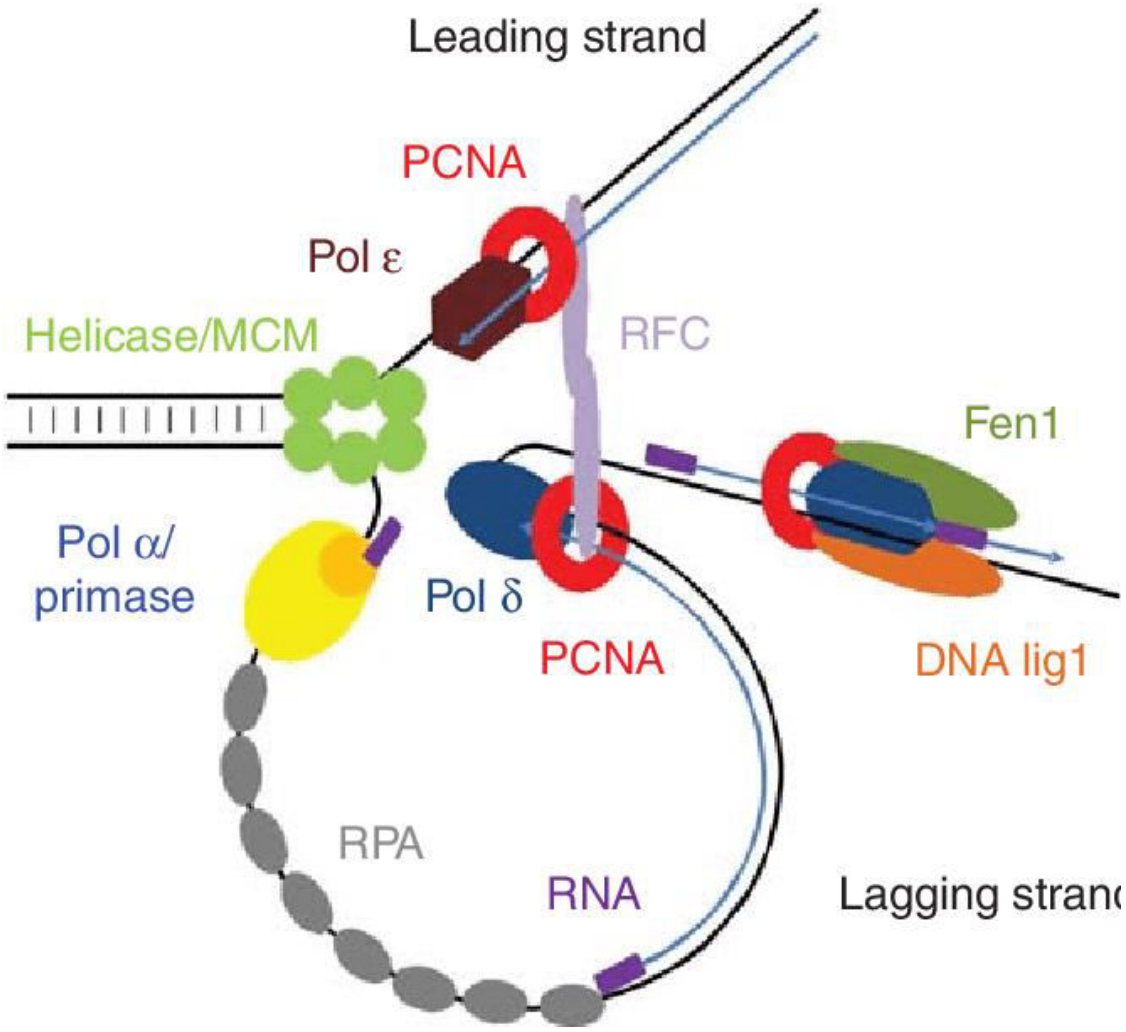
- Use of toxic chemicals and extensive use of radioactive isotopes. highly poisonous and unstable.
- Cannot read more than 500bp.
- Setup is quite complex.
- It is difficult to make Maxam-Gilbert DNA sequencing kit.
- Read size decrease with incomplete cleavage reactions.

Lesson 76

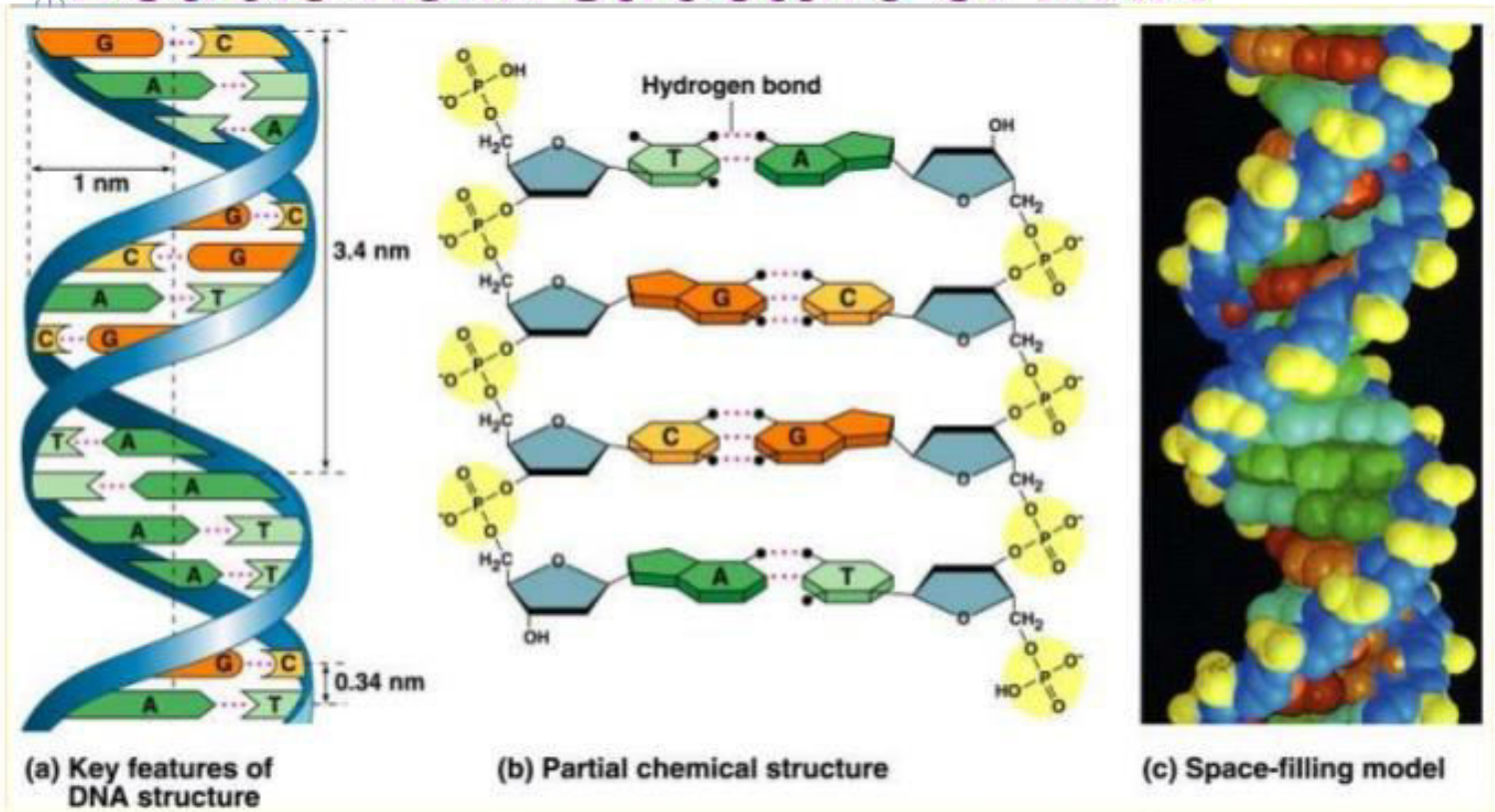
DNA SYNTHESIS

- DNA synthesis occurs by the process of replication. During replication, each of the two parental strands of DNA serves as a template for the synthesis of a Complementary strand
- Each molecule generated by the replication process contains one intact parental strand and one newly synthesized strand

In eukaryotes, DNA replication occurs during the S phase of the cell cycle The cell divides during the next phase (M), and each daughter cell receives an exact copy of the DNA of the parent cells.



Double helix structure of DNA

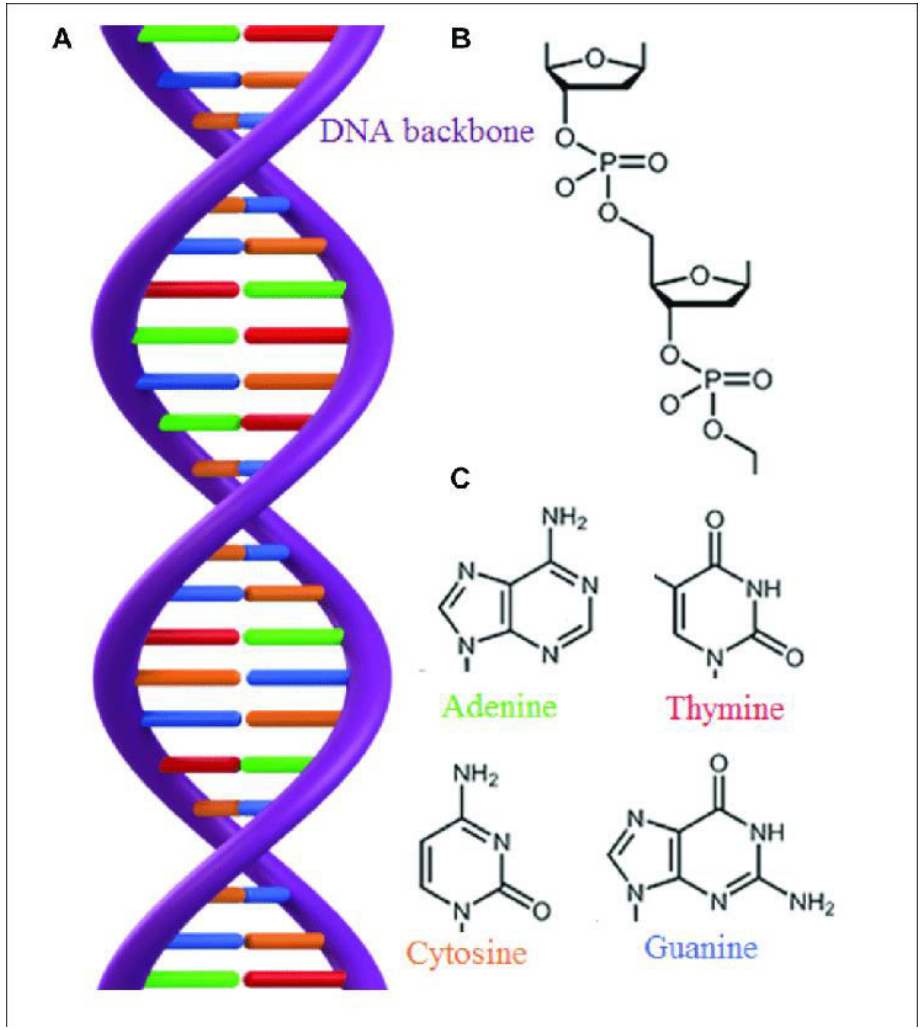


"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."

Watson & Crick

The DNA backbone

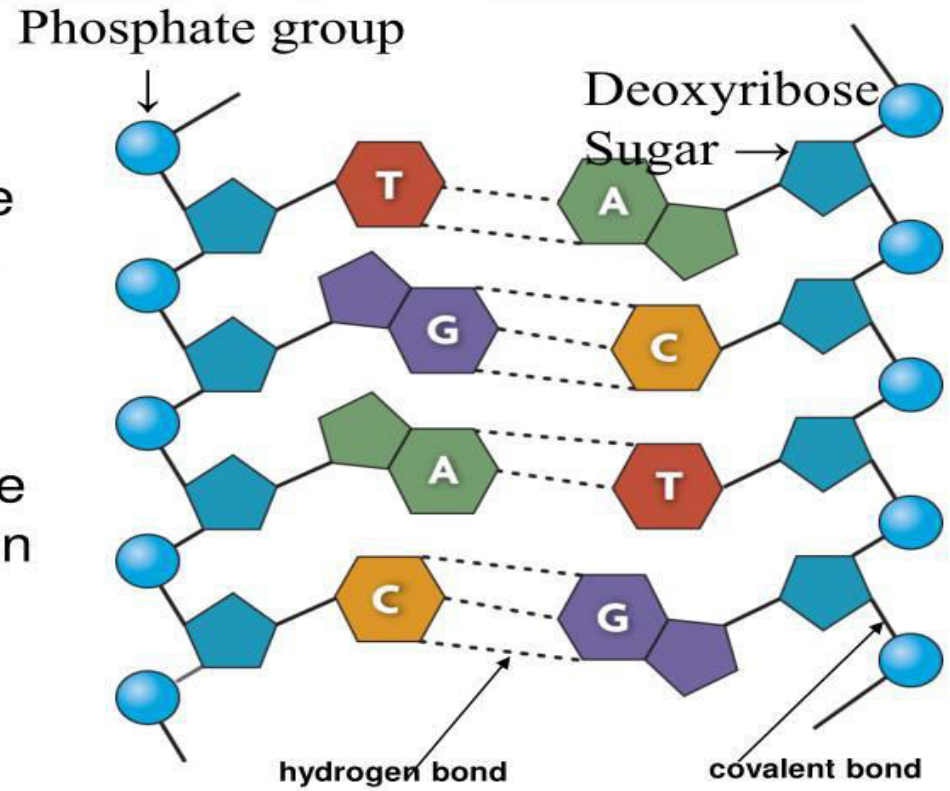
Putting the DNA backbone together



8.2 Structure of DNA

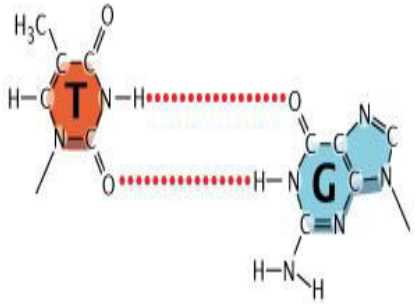
Bonding in DNA

- The **DNA backbone:** alternating deoxyribose (sugar) and phosphate group is connected by strong covalent bonds.
- The nitrogen bases are connected by hydrogen bonds.

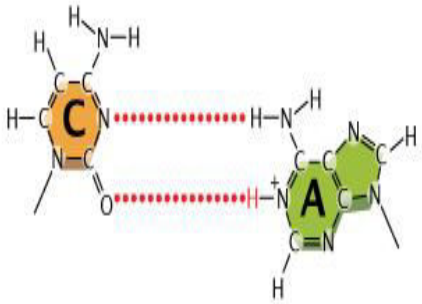


Pairing DNA

- Base pairing in DNA Purines
- adenine (A)
- guanine (G)
- Pyrimidines thymine (T)
- cytosine (C)
- Pairing A : T(2 bonds)
- C : G (3 bonds)



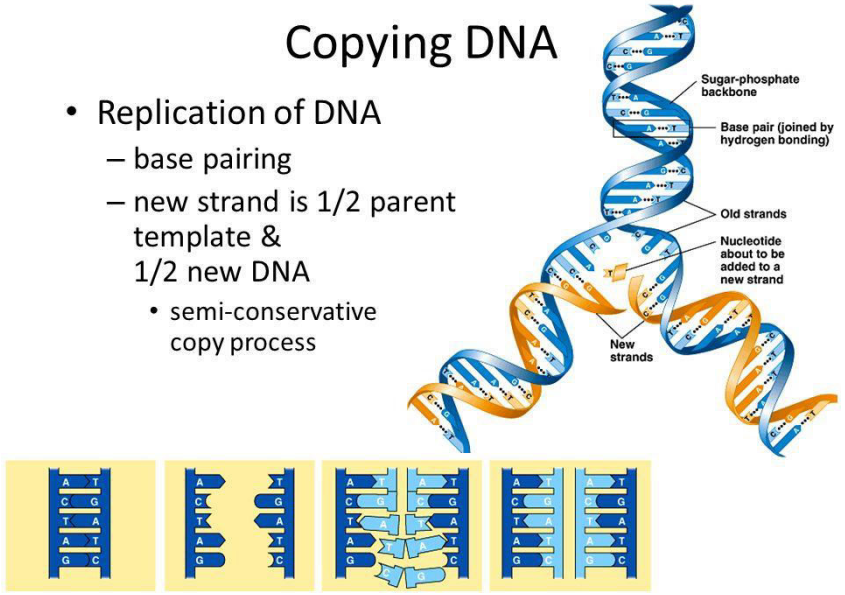
Thymine–guanine wobble



Cytosine–adenine protonated wobble

Copying DNA

- Replication of DNA
- base pairing allows each strand to serve as a template for a new strand new strand is 1/2 parent template & 1/2 new DNA



THANK YOU

Lesson 77
DNA Microarray

DNA **microarray**

- A DNA microarray (also commonly known as DNA chip or biochip) is a collection of microscopic DNA spots attached to a solid surface. Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome. Each DNA spot contains picomoles (10^{-12} moles) of a specific DNA sequence, known as *probes* (or *reporters* or *oligos*). These can be a short section of a gene or other DNA element that are used to hybridize a cDNA or cRNA (also called anti-sense RNA) sample (called *target*) under high-stringency conditions. Probe-target hybridization is usually detected and quantified by detection of fluorophore-, silver-, or chemiluminescence-labeled targets to determine relative abundance of nucleic acid sequences in the target.

Principle

- The core principle behind microarrays is hybridization between two DNA strands, the property of complementary nucleic acid sequences to specifically pair with each other by forming hydrogen bonds between complementary nucleotide base pairs. A high number of complementary base pairs in a nucleotide sequence means tighter non-covalent bonding between the two strands. After washing off non-specific bonding sequences, only strongly paired strands will remain hybridized

Principle

- Fluorescently labeled target sequences that bind to a probe sequence generate a signal that depends on the hybridization conditions (such as temperature), and washing after hybridization. Total strength of the signal, from a spot (feature), depends upon the amount of target sample binding to the probes present on that spot. Microarrays use relative quantitation in which the intensity of a feature is compared to the intensity of the same feature under a different condition, and the identity of the feature is known by its position

Uses and types

- Many types of arrays exist and the broadest distinction is whether they are spatially arranged on a surface or on coded beads:
- The traditional solid-phase array is a collection of orderly microscopic "spots", called features, each with thousands of identical and specific probes attached to a solid surface, such as glass, plastic or silicon biochip (commonly known as a *genome chip*, *DNA chip* or *gene array*). Thousands of these features can be placed in known locations on a single DNA microarray.
- The alternative bead array is a collection of microscopic polystyrene beads, each with a specific probe and a ratio of two or more dyes, which do not interfere with the fluorescent dyes used on the target sequence.

- DNA microarrays can be used to detect DNA (as in comparative genomic hybridization), or detect RNA (most commonly as cDNA after reverse transcription) that may or may not be translated into proteins. The process of measuring gene expression via cDNA is called expression analysis or expression profiling.

Sources:

- https://en.wikipedia.org/wiki/DNA_microarray

LESSON 78

Molecular Hybridization

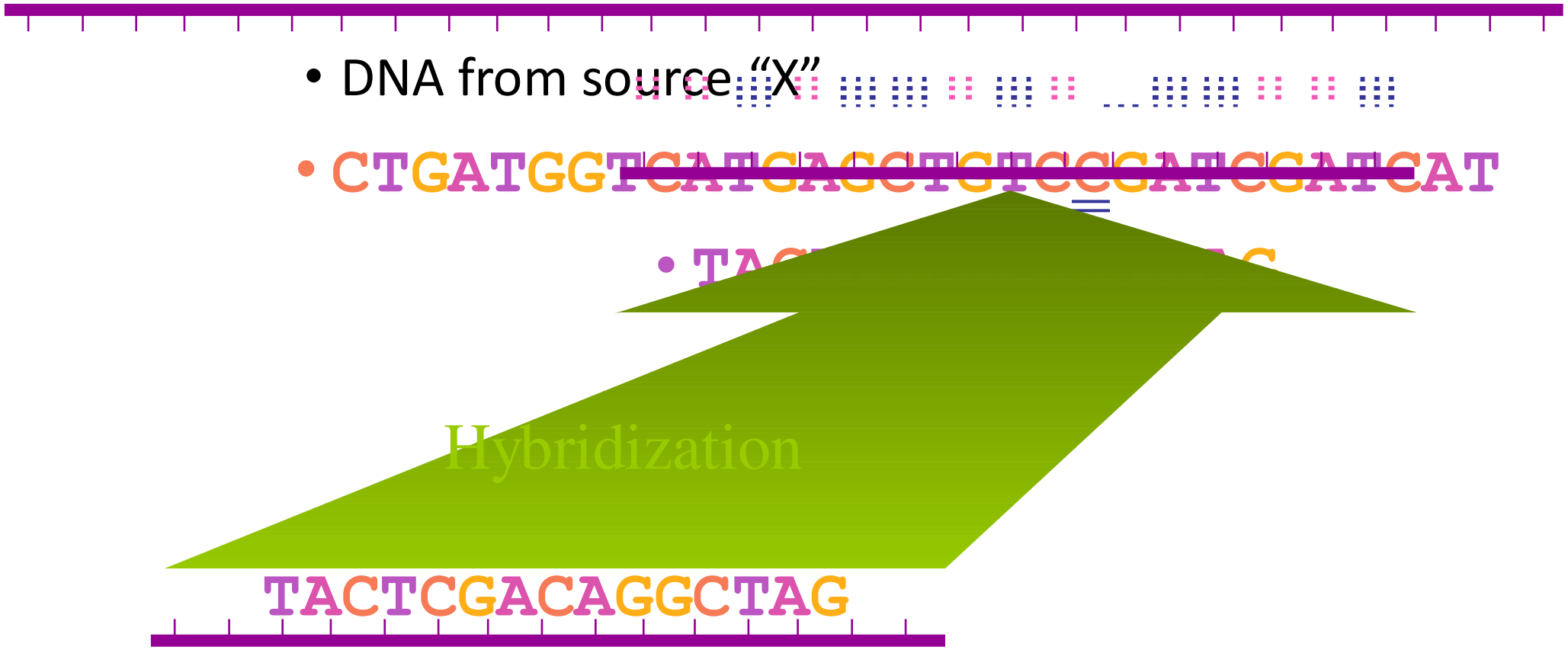
Molecular hybridization of nucleic acids is the process in which two single-stranded nucleic acid molecules with complementary base sequences form a double-stranded nucleic acid molecule. Nucleic acid hybridization technology is a fundamental tool in molecular biology, and has been applied in various fields such as detection of gene expression, screening specific clone from cDNA or genomic library, determining the location

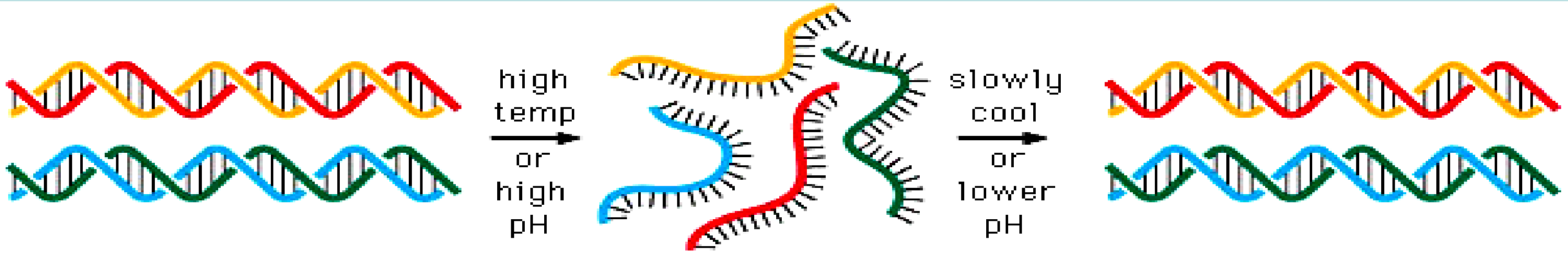
of a gene in chromosome and diagnosis of diseases.

PRINCIPLE

The technique of nucleic acid hybridization is established and developed on the basis of the denaturation and renaturation of nucleic acids. Hydrogen bonds in double-stranded nucleic acids can be disrupted by some physicochemical elements, and two strands of nucleic acids are separated into single strand.

Hybridization





DNA double helices

denaturation to single strands (nucleotide pairs broken)

renaturation restores DNA double helices (nucleotide pairs re-formed)

If different single-stranded DNA molecules, or DNA and RNA molecules, or RNA molecules are mixed together in a solution, and the renaturation is allowed to occur under proper conditions, single- stranded DNA or RNA will bind with each other to form a local or whole molecule of double-stranded structure as long as the single-stranded molecules are complementary, no matter what kind of sources they come from.

Nucleic acid hybridization as a technique involves using a labeled nucleic acid probe, which is a known DNA or RNA fragment, to bind with the target nucleic acids, which is usually a poorly understood, heterogeneous population of nucleic acids. A probe labeled with detectable tracer is the prerequisite for determining a specific DNA sequence or gene in a sample or genomic DNA by nucleic acid hybridization.

The target nucleic acids to be analyzed are usually denatured, and then mixed with the labeled probe in the hybridization system. The probe will bind to the segment of nucleic acid with complementary sequence under proper conditions. The hybridization can be identified by the detection of the tracer labeling the probe. Thus the existence or the expression of specific gene can be determined.

Lesson 79

Molecular Hybridization

Preparation of probes

Probes may be single-stranded or double- stranded molecules, but the working probe must be single-stranded molecules. The probes used in hybridization of nucleic acids include oligonucleotide(15-50 nucleotides), genomic DNA fragment, cDNA fragment and RNA

Oligonucleotide probes are short single- stranded DNA fragments designed with a specific sequence complementary to the given region of the target DNA. They are usually synthesized in vitro

- Genomic DNA probes can be prepared from the cloned DNA fragment in plasmid.
- cDNA probes can be prepared from the cloned cDNA in plasmid, or amplified directly from mRNA by RT-PCR.
- RNA probes are usually transcribed in vitro from a cloned cDNA in a proper vector. The size of genomic DNA probes, cDNA probes and RNA probes may be 0.1 kb to 1 kb.

Labeling of probes

Probe is usually labeled with a detectable tracer, which is either isotopic or non-isotopic. The purified oligonucleotide is labeled in vitro by using a suitable enzyme to add the labeled nucleotide to the end of the oligonucleotide.

Genomic DNA probes and cDNA probes

Genomic DNA probes and cDNA probes are usually labeled in the process of DNA synthesis in vitro. In the reaction of DNA synthesis with a DNA probe as template, if a labeled-dNTP, which can be incorporated into newly-synthesized DNA chain, is added as a substrate, the labeled DNA probe will be formed.

There are different, sensitive detecting methods for each of the labels used in nucleic acid hybridization. After hybridization, the location and the quantity of the hybrid molecules can be determined. The labels in common use include radioactive (^{32}P and ^{35}S) and nonradioactive (digoxigenin, biotin, fluorescein) substances which are used to label dNTP.

Southern blot hybridization

Southern blot hybridization is an assay for sample DNA by DNA-DNA hybridization which detects target DNA fragments that have been size-fractionated by gel electrophoresis (Figure 4-1). In Southern blot hybridization, the target DNA is digested with restriction endonucleases, size-fractionated by agarose gel electrophoresis, denatured and transferred to a nitrocellulose or nylon membrane for hybridization.

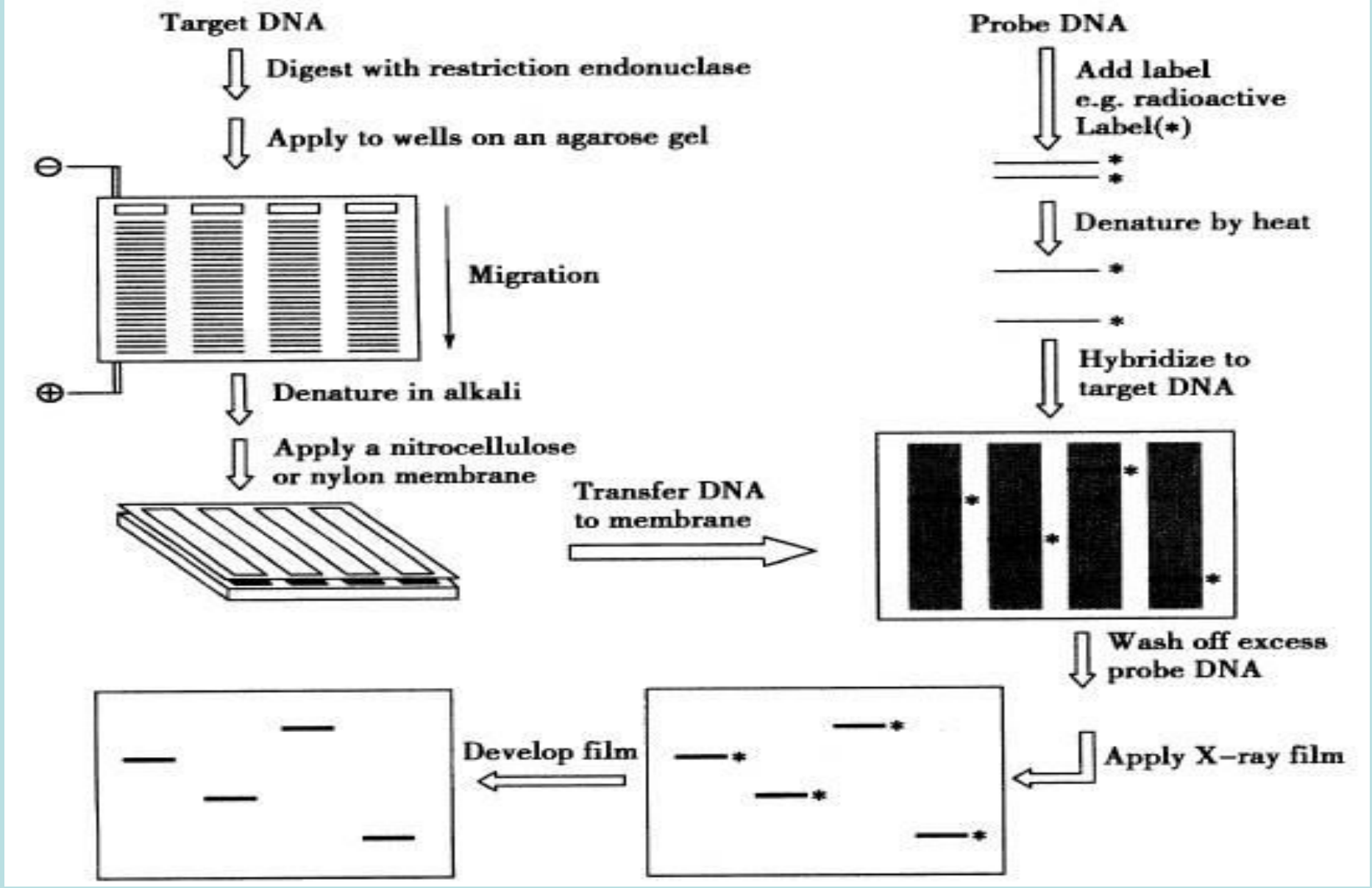
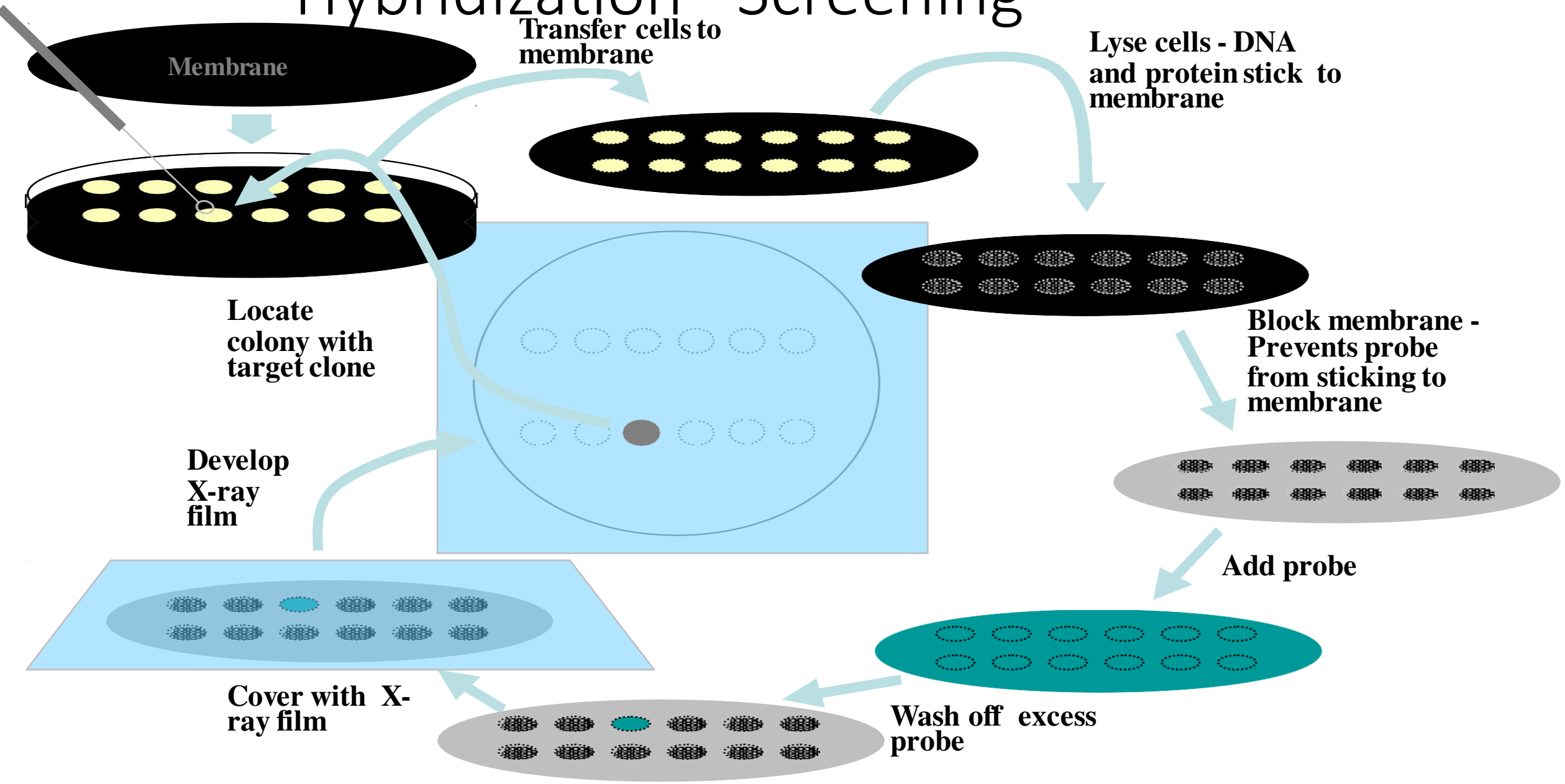


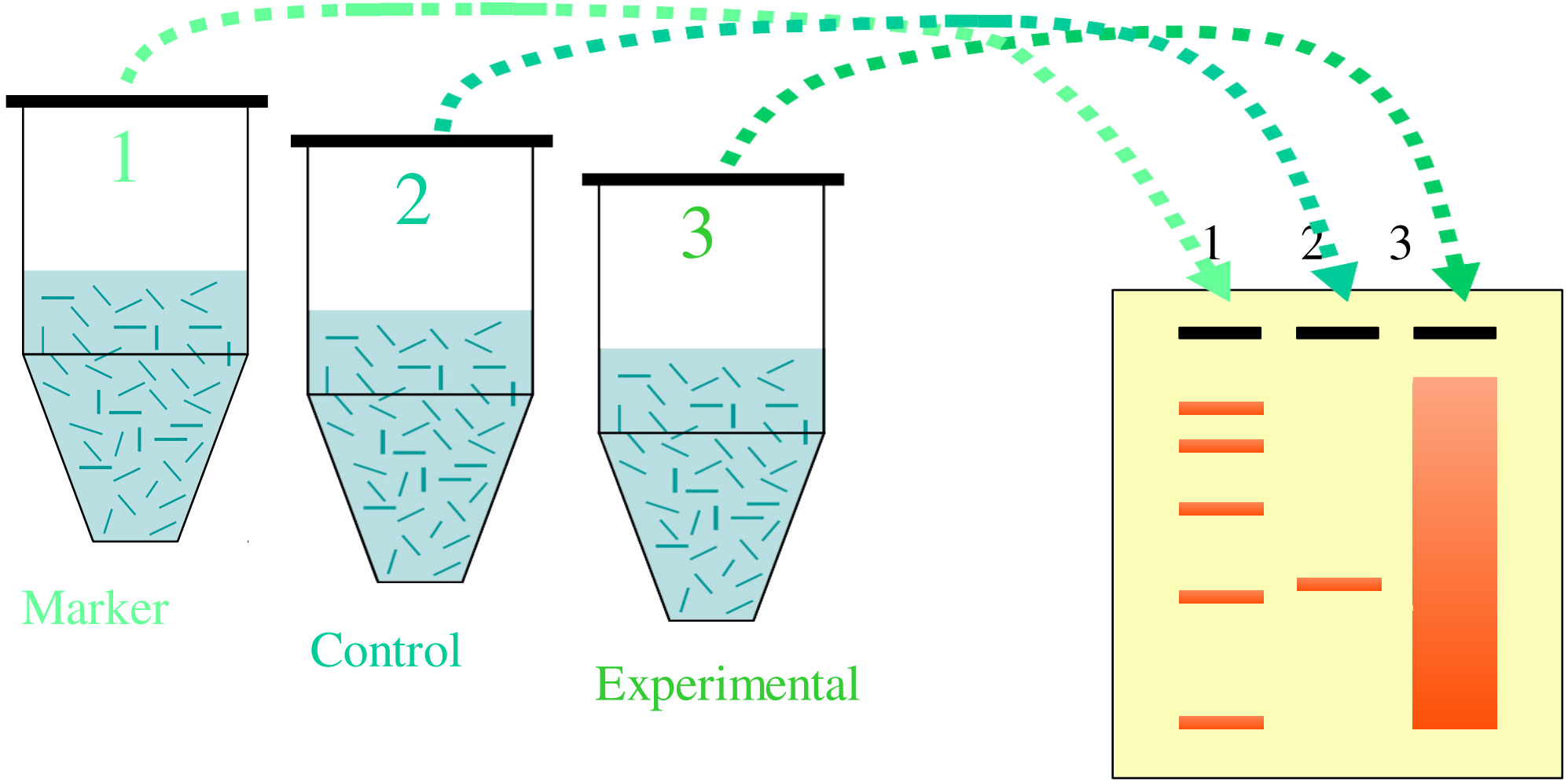
Figure 4-1 Southern blot hybridization detects target DNA fragments that have been size-fractionated by gel electrophoresis

Southern blot hybridization technique is widely applied in researches since its invention. It could be applied for analysis of gene expression, screening of recombinant plasmids, analysis of gene mutation, and identification of the existence of a given DNA such as DNA from pathogenic microorganism. It could also be used to detect deletion of gene by restrictions mapping.

Hybridization Screening



Making A Southern Blot 1 + 2 Digestion and Electrophoresis



LESSON 80

Southern Blot Hybridization

Southern blot hybridization

Southern blot hybridization is an assay for sample DNA by DNA-DNA hybridization which detects target DNA fragments that have been size-fractionated by gel electrophoresis (Figure 4-1). In Southern blot hybridization, the target DNA is digested with restriction endonucleases, size-fractionated by agarose gel electrophoresis, denatured and transferred to a nitrocellulose or nylon membrane for hybridization.

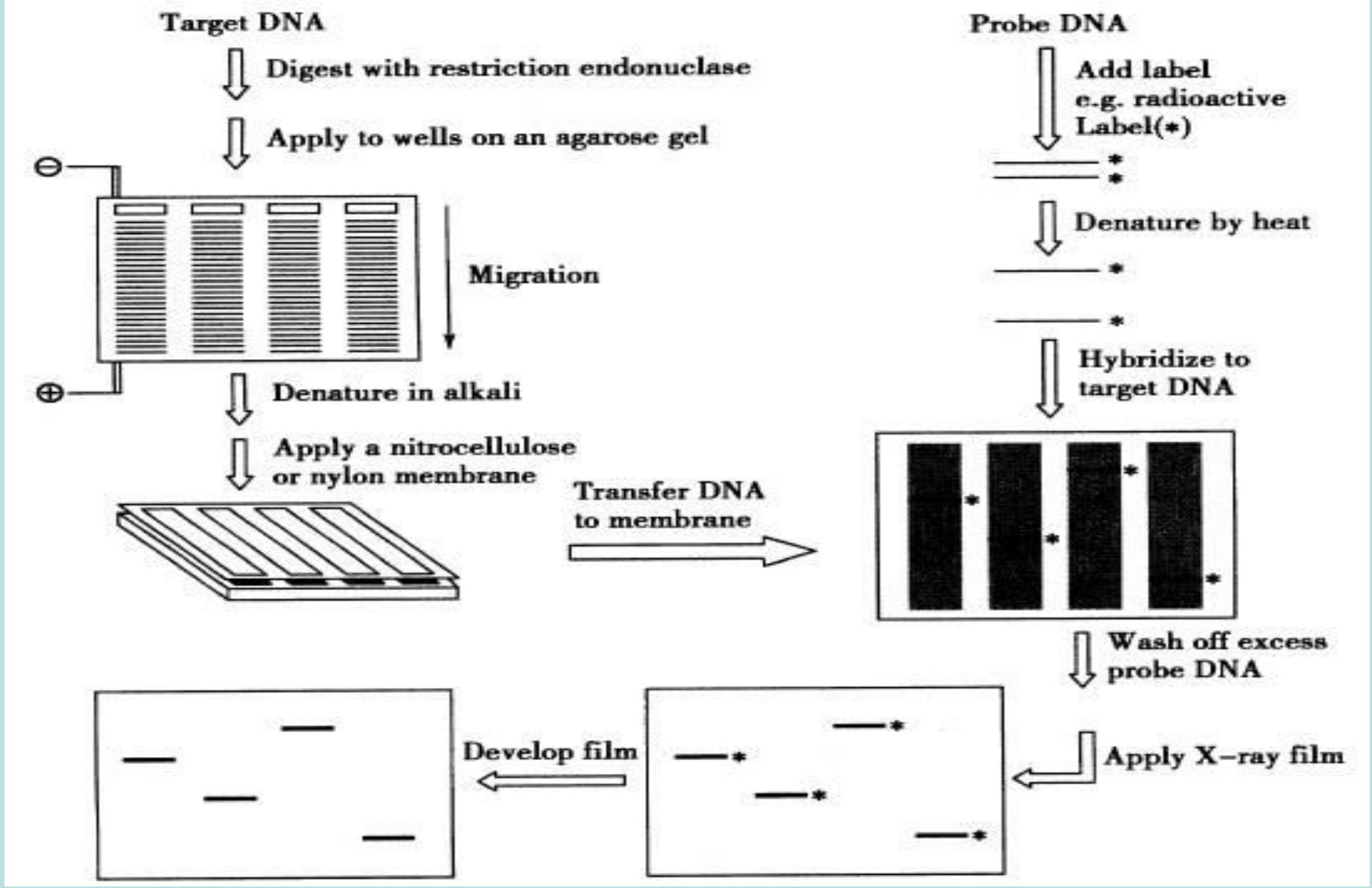
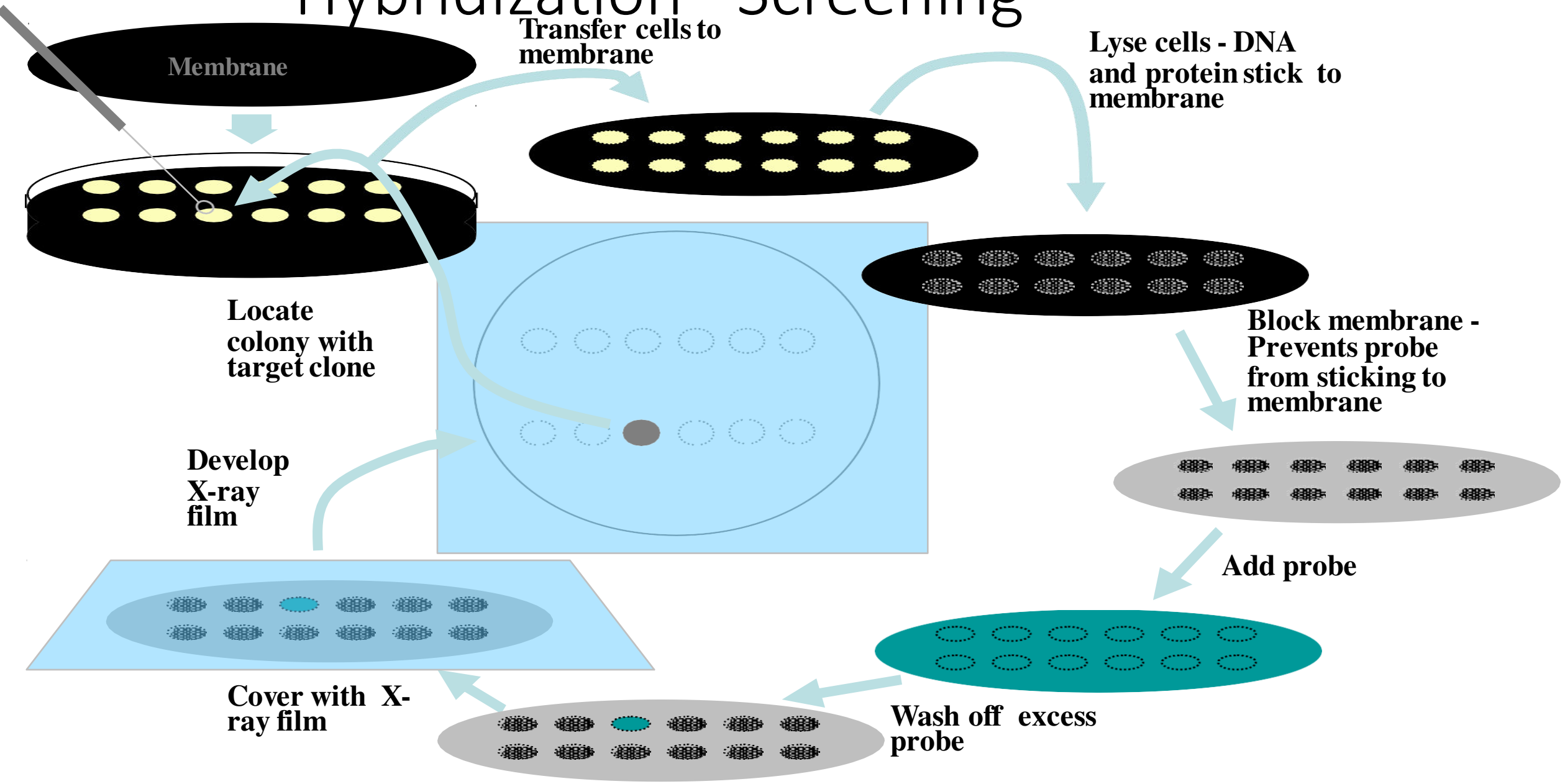


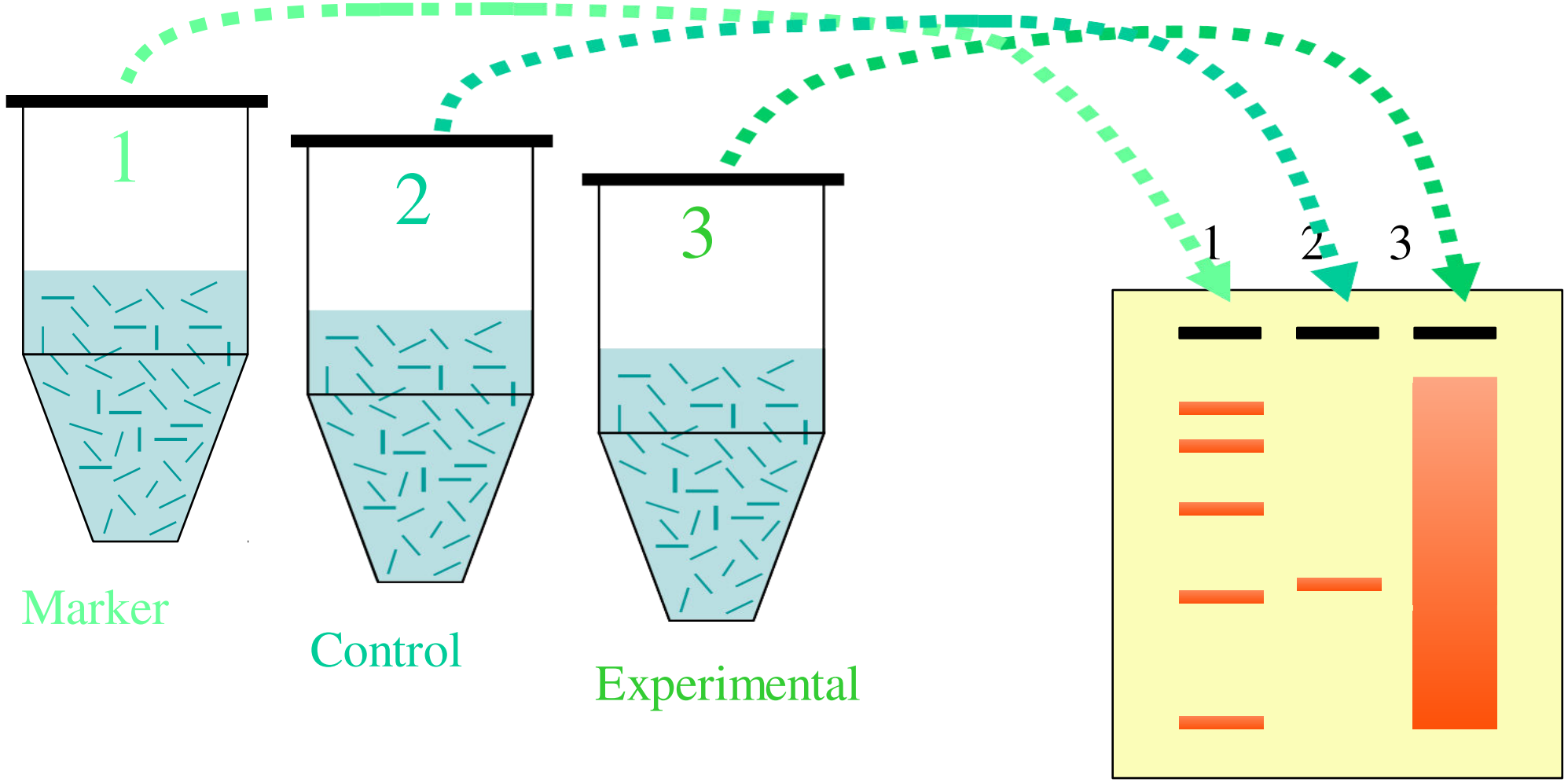
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Hybridization Screening



Making A Southern Blot 1 + 2 Digestion and Electrophoresis



Lesson 81

Ligation

Restriction enzyme = Restriction Endonuclease

Is an enzyme that cuts DNA at or near specific recognition nucleotides sequences known as Restriction site.

These enzymes are found in bacteria and provide a defense mechanism againstinvading viruses.

To cut DNA, at restriction enzymes make two incisions, once through each sugar-phosphate backbone (i.e. each strand) of the DNA double helix.

Restriction enzyme = Restriction Endonuclease

Over 3000 restriction enzymes have been studied in detail, and more than 600 of these are available commercially.

Restriction site:

Restriction enzymes recognize a specific sequence of nucleotides and produce a double-stranded cut in the DNA.

The recognition sequences usually vary between 4 and 8 nucleotides, and many of them are **palindromic,** meaning the base sequence reads the same backwards and forwards

Restriction site:

In theory, there are two types of palindromic sequences that can be possible in DNA.

The mirror-like palindrome is similar to those found in ordinary text, in which a sequence reads the same forward and backwards on a single strand of DNA strand.

Restriction site:

- EcoRI digestion produces "sticky" ends:

- whereas SmaI restriction enzyme cleavage produces "blunt" ends:

Ligation (Ligases & Phosphatases) Ligation

It is an essential laboratory procedure in the molecular cloning by which DNA fragments are joined together to create recombinant DNA molecules, such as when a foreign DNA fragment is inserted into a plasmid.

The ends of DNA fragments are joined together by the formation of phosphodiester bonds between the 3'- hydroxyl of one DNA termini with the 5'-phosphoryl of another.

Blunt-end ligation

Blunt end may be ligated to another blunt end, Blunt ends may be generated by restriction enzymes such as *Sma*I and *Eco*RV.

However a major advantage of blunt-end cloning is that the desired insert does not require any restriction sites in its sequence as blunt-ends are usually generated in a PCR, and the PCR generated blunt-ended DNA fragment may then be ligated into ablunt-ended vector generated from restriction digest.

Disadvantages of blunt-end ligation:

1ligation is much less efficient than sticky end ligation, typically the reaction is 100Xslower than sticky-end ligation.

2The concentration of ligase used is higher than sticky end ligation (10xormore).

3The concentration of DNA used in blunt-end ligation is also higher to increase the likelihood of collisions between ends.

4Longer incubation time may also be used for blunt- end ligations.

Method of Ligation

Add 1 µl of the PD rive vector of the ligation in o.2 µl or 0.5 µl PCR tubes,

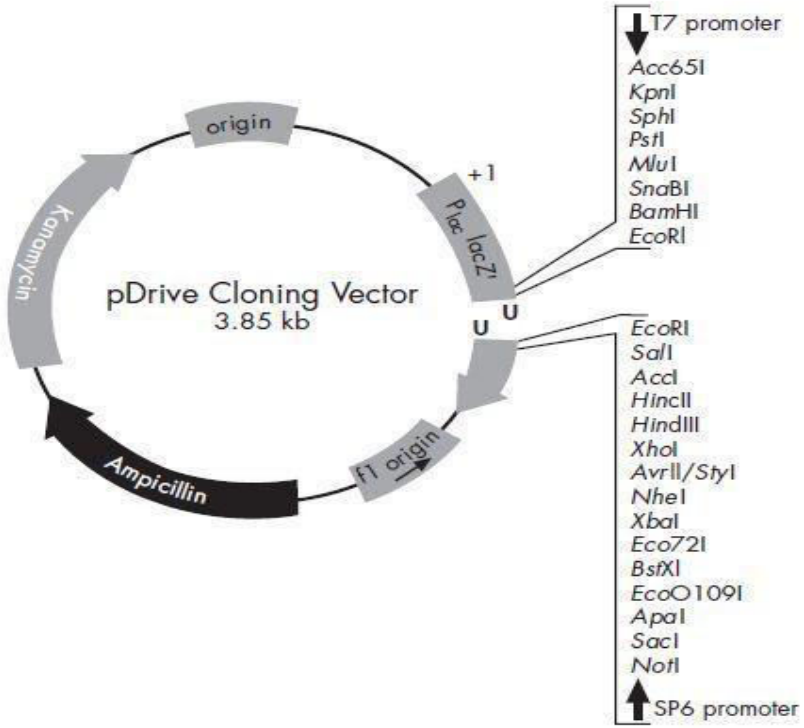
Add 1:4 µl of the purified PCR product, Add 5µl of ligation master mix &

Variable µl of water then put in the thermo cycler for four hours at 16°C.

pDrive Cloning Vector

Ampicillin and kanamycin selection

- Blue/white screening
- Unique restriction endonuclease recognition sites around the cloning site
- T7 and SP6 promoter on either side of the cloning site allowing *in vitro* transcription of cloned PCR products as well as sequence analysis using standard sequencing primers.



Trouble-shooting

Sometimes ligation fail to produce the desired ligated products, and some of the possible reasons may be:

1Damaged DNA - over-exposure to UV radiation during preparation of DNA for ligation can damage the DNA.

2 Excessive amount of DNA used.

3Incomplete DNA digest, The vector DNA that is incompletely digested will give rise to a high background. Insert that is not completely digested will also not ligate properly and circularize.

4Incomplete ligation. Blunt-ends DNA and some sticky-ends DNA that have low-melting temperature require more ligase and longer incubation time.

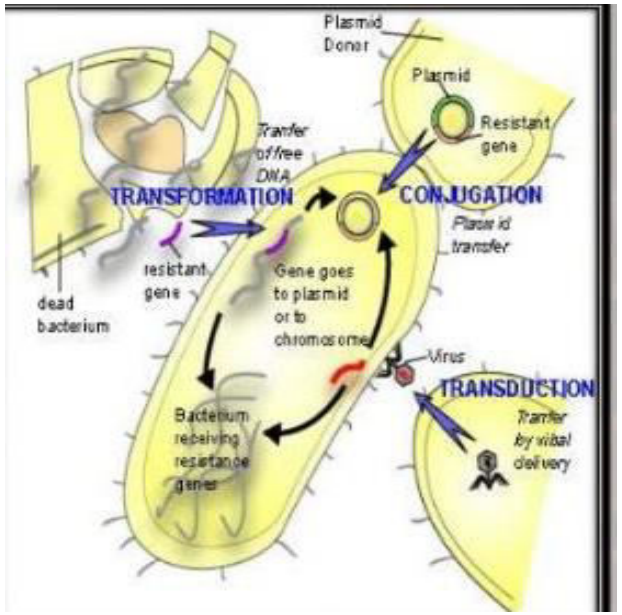
Lesson 82

Transformation

What is transformation?

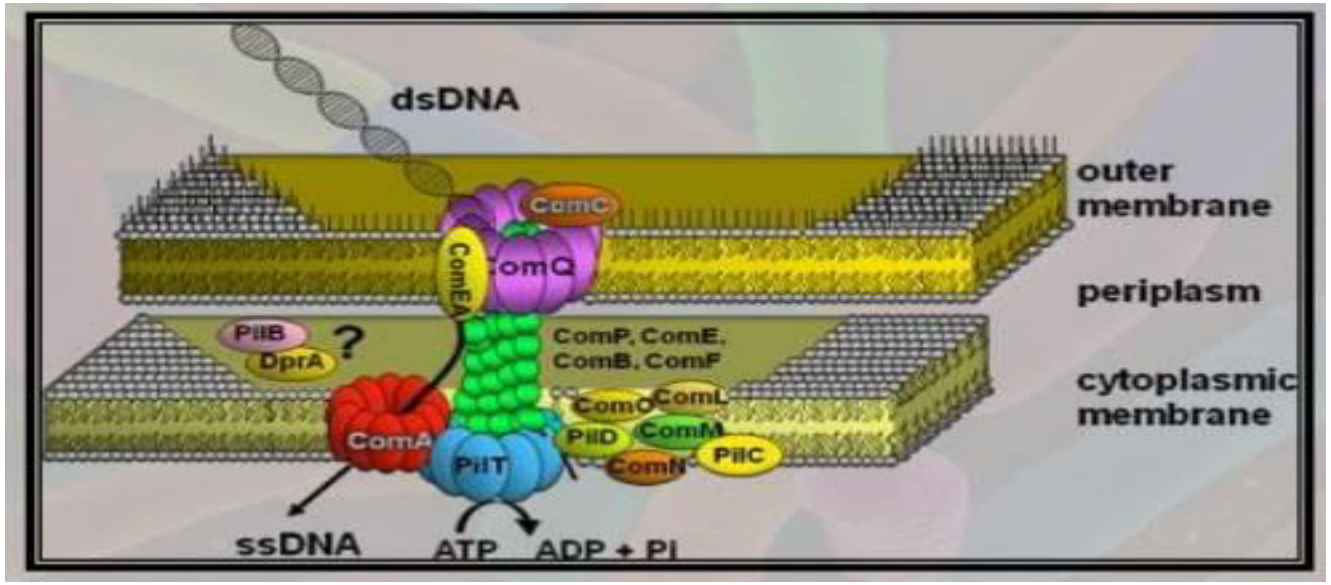
Genetic transformation is the incorporation of naked DNA from the extracellular environment.

It is one of three possible mechanisms of HGT (Horizontal gene transfer)

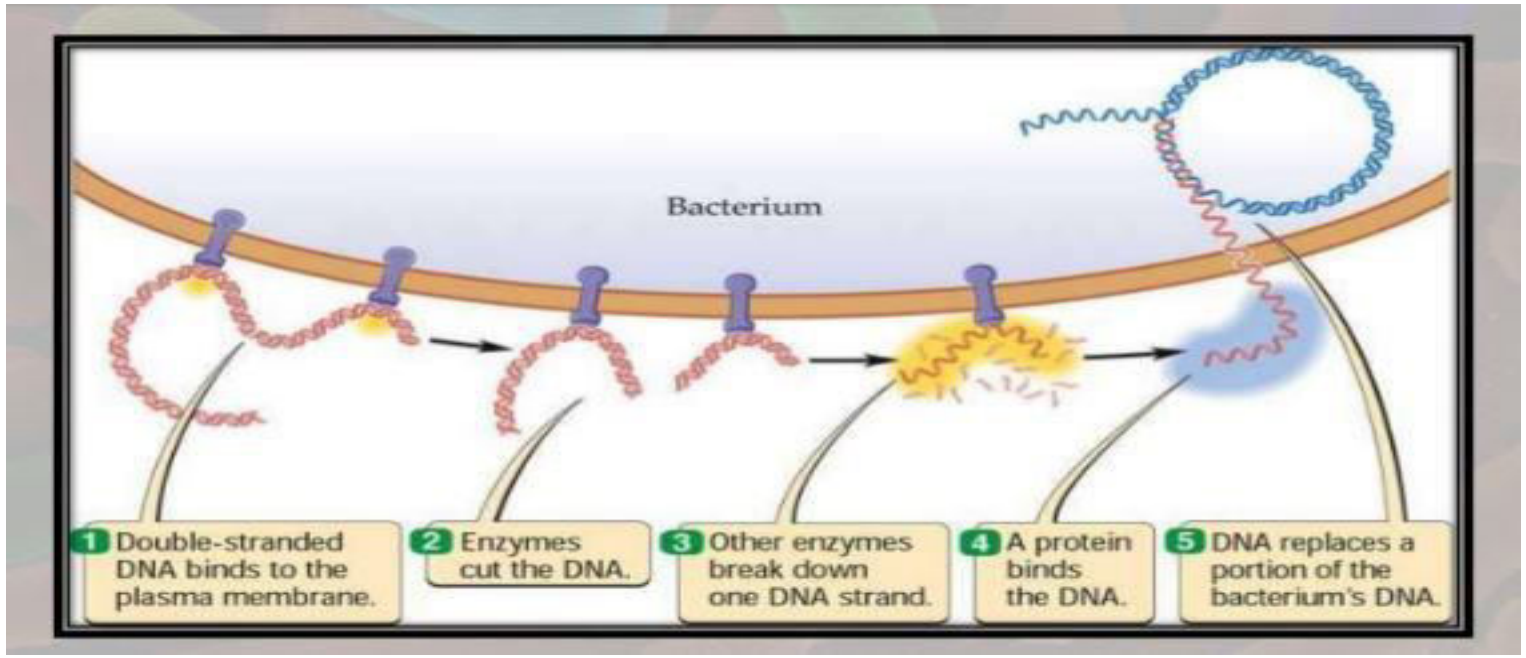


Types of transformation

First Natural Transformation

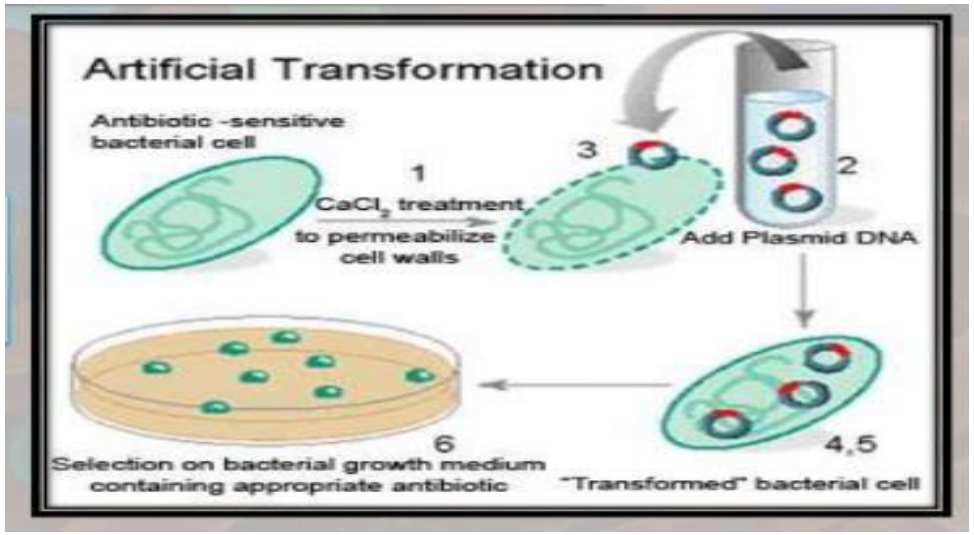


First Natural Transformation



Second: Artificial transformation

- Chemical Treatment
- Physical Treatment
- Enzymatic Treatment



Types of Component Cell fro transforamtion

- **<u>Chemically Competent Cells</u>**
  - calcium chloride-treated = attachment of the plasmid DNA
  - heat-shocked in a water bath = opens the pores of the cell membrane
  - entry of plasmid DNA from the buffer.

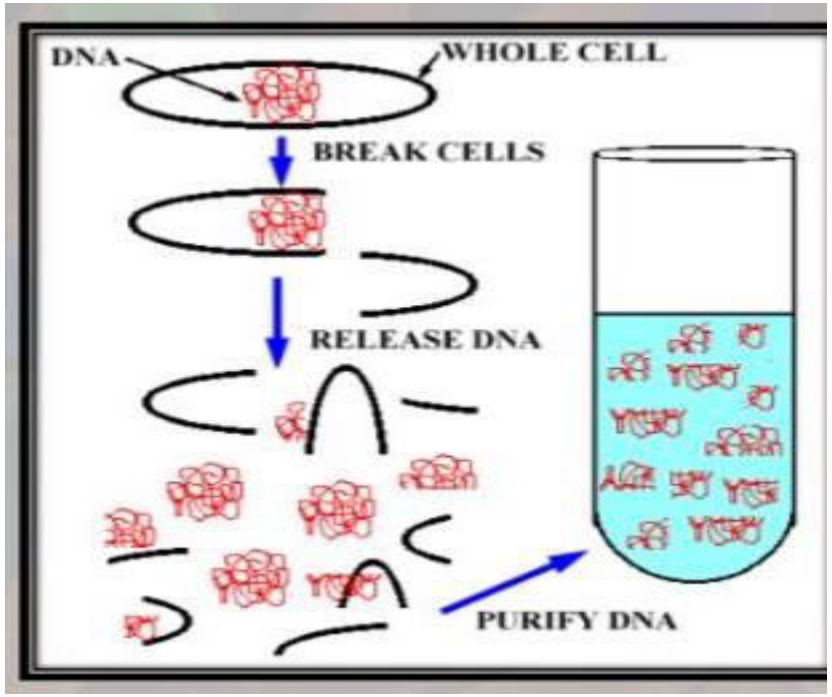
Types of Component Cell fro transforamtion

- Electro competent Cells
  - using electroporation
  - create pores
  - genetic material enters the cells.

What is the basic procedure transformation?

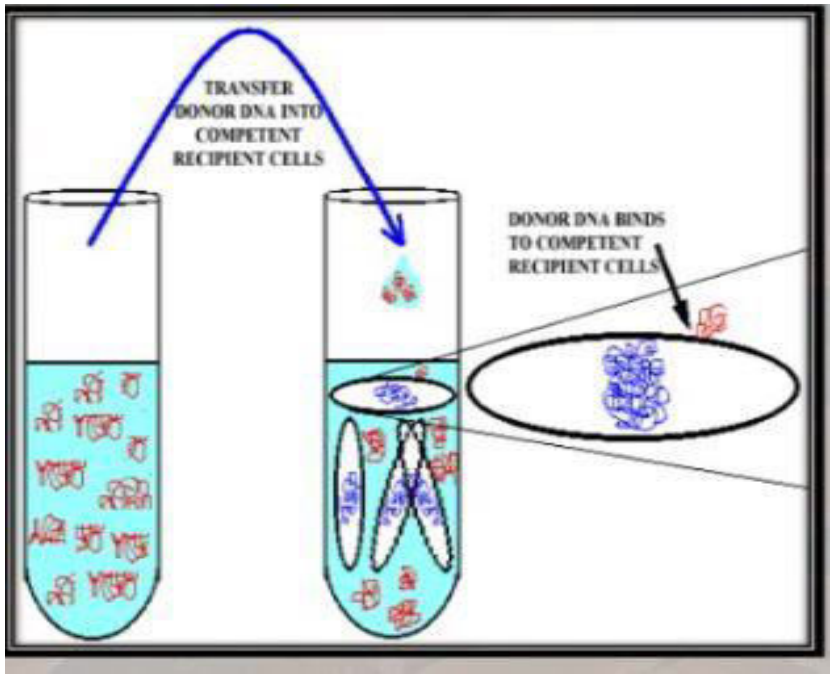
First:

Isolation of CELL-FREE or NAKED DNA. The cells are broken and the DNA released. The cell-free DNA is subsequently isolated and collected.



Second:

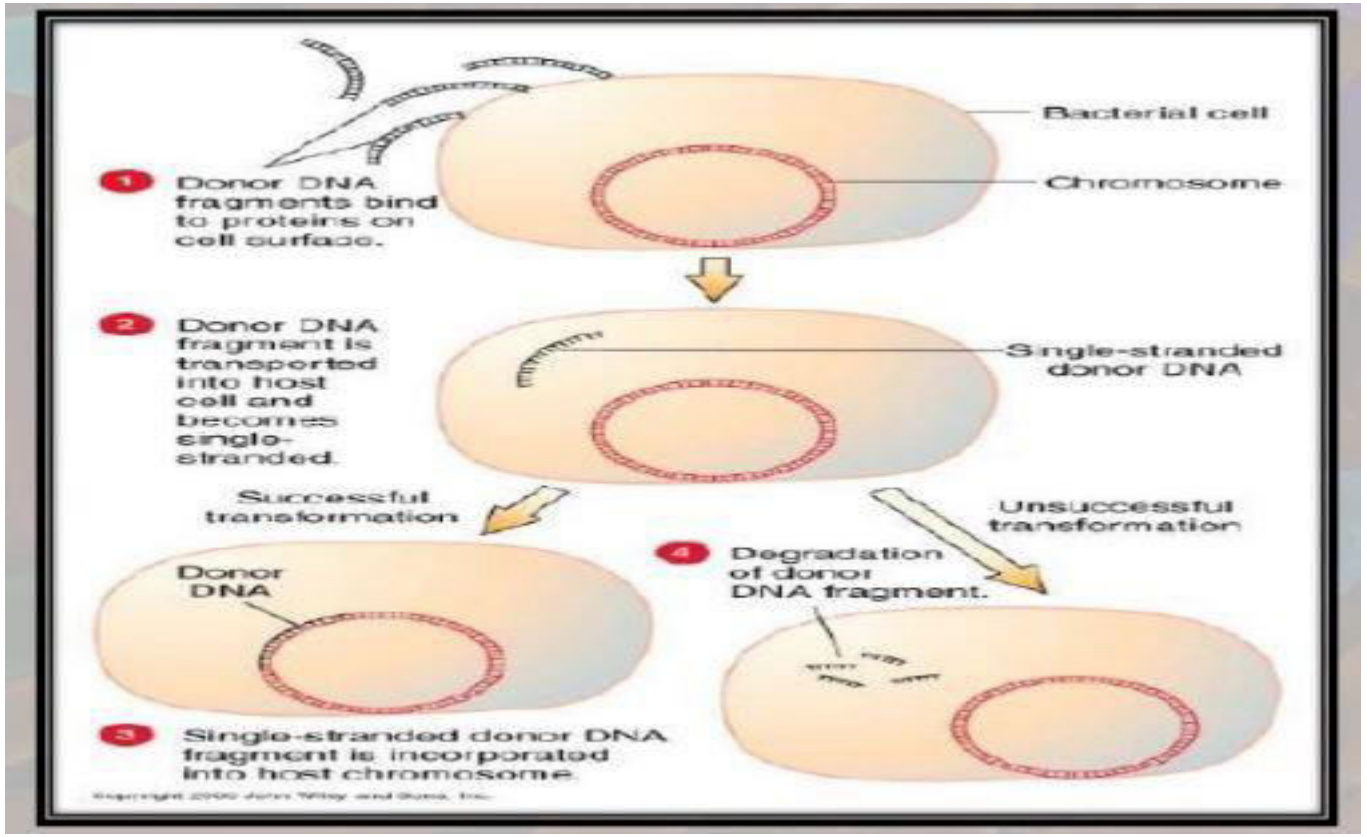
Mixing of Donor DNA with Recipient Competent Cells. The naked donor DNA is incubated with the competent recipient cells to which it binds.



Third

Uptake and Recombination of Donor DNA. Donor DNA binds to competent recipient cells, following which it enters the recipient celis. Portions of the donor DNA align, at random, with genes on the recipient DNA and segments of the two DNAs ore exchanged.

What is the basic procedure transformation?



LESSON 85

- **Flow Cytometry**
  - **Protocol/Procedure/Process/Steps of Flow Cytometry**

Protocol/Procedure/Process/Steps of Flow Cytometry

1. Sample Preparation

Before running in the flow cytometers, the cells under analysis must be in a single-cell suspension.

Clumped cultured cells or cells present in solid organs should first be converted into a single cell suspension before the analysis by using enzymatic digestion or mechanical dissociation of the tissue, respectively.

Protocol/Procedure/Process/Steps of Flow Cytometry

It is then followed by mechanical filtration should to avoid unwanted instrument clogs and obtain higher quality flow data.

The resulting cells are then incubated in test tubes or microtiter plates with unlabeled or fluorescently conjugated antibodies and analyzed through the flow cytometer machine.

Protocol/Procedure/Process/Steps of Flow Cytometry

2. Antibody Staining

Once the sample is prepared, the cells are coated with fluorochrome-conjugated antibodies specific for the surface markers present on different cells. This can be done either by direct, indirect, or intracellular staining.

Indirect staining, cells are incubated with an antibody directly conjugated to a fluorophore.

In indirect staining, the fluorophore-conjugated secondary antibody detects the primary antibody

Protocol/Procedure/Process/Steps of Flow Cytometry

The intracellular staining procedure allows direct measurement of antigens presents inside the cell cytoplasm or nucleus. For this, the cells are first made permeable and then are stained with antibodies in the permeabilization buffer.

3. Running Samples

At first, control samples are run to adjust the voltages in the detectors.

The flow rates in the cytometer are set and the sample is run.

Types of Flow Cytometry

1. **Traditional flow cytometers**

The traditional cytometers are the common cytometer using sheath fluid for focusing the sample stream.

The most common lasers used in traditional flow cytometers are 488 nm (blue), 405nm (violet), 532nm (green), 552nm (green), 561 nm (green-yellow), 640 nm (red) and 355 nm (ultraviolet).

Types of Flow Cytometry

2. Acoustic Focusing Cytometers

In these cytometers, ultrasonic waves are used to focus the cells for analysis.

This prevents sample clogging and also allows higher sample inputs.

3. Cell sorters

Cell sorters are a category of traditional flow cytometers which allows the user to collect samples after processing.

The cells that are positive for the desired parameter can be separated from those that are negative for the parameters.

Types of Flow Cytometry

4. Imaging flow cytometer

Imaging cytometers are traditional cytometers combined with fluorescence microscopy.

Imaging cytometer allows for rapid analysis of a sample for morphology and multi-parameter fluorescence at both a single cell and population level.

THANK YOU

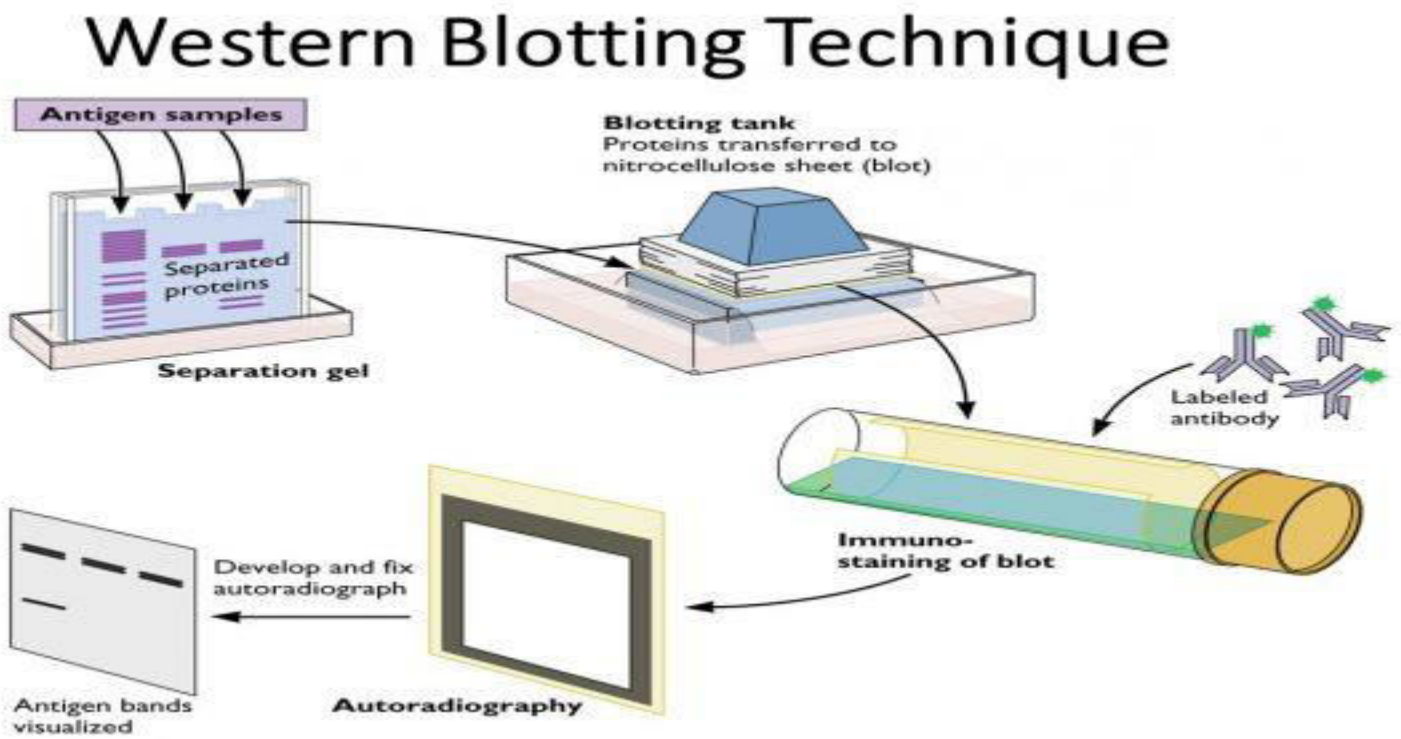
Lesson 86

Western Blotting

Western Blotting Technique

- Western blot (Immunoblotting) : A technique for detecting specific proteins separated by electrophoresis by use of labeled antibodies

Western Blotting Technique



PROCEDURE

1. Tissue Preparations

- Samples may be taken from whole tissue or from cell culture. In most cases, solid tissues are first broken down mechanically using a blender. It should be noted that bacteria, virus or environmental samples can be the source of protein and thus Western blotting is not restricted to cellular studies only.

PROCEDURE

- Assorted detergents, salts, and buffers may be employed to encourage lysis of cells and to solubilize proteins.
- Tissue preparation is often done at cold temperatures to avoid protein denaturing

PROCEDURE

2. Gel Electrophoresis: The proteins of the sample are separated using gel electrophoresis. Separation of proteins may be by isoelectric point, molecular weight, electric charge, or a combination of these factors.

PROCEDURE

3. Transferring: In order to make the proteins accessible to antibody detection, they are moved from within the gel onto a membrane made of nitrocellulose or polyvinylidene difluoride (PVDF).

The membrane is placed on top of the gel, and a stack of filter papers placed on top of that. The entire stack is placed in a buffer solution which moves up the paper by capillary action, bringing the proteins with it.

Another method for transferring the proteins is called electro blotting and uses an electric current to pull proteins from the gel into the PVDF or nitrocellulose membrane.

PROCEDURE

4. Blocking • The membrane has the ability to bind to proteins. In this case both the target and antibodies are proteins and so there could be some unwanted binding. • Blocking of non-specific binding is achieved by placing the membrane in a dilute solution of protein typically Bovine serum albumin(BSA) with a minute percentage of detergent such as Tween 20.

PROCEDURE

The protein in the dilute solution attaches to the membrane in all places where the target proteins have not attached. • Thus, when the antibody is added, there is no room on the membrane for it to attach other than on the binding sites of the specific target protein.

THANK YOU

Lesson 87

Western Blotting Technique

PROCEDURE

5. Detection : There are two steps for the detection of the protein-Primary antibody

•

After blocking, a dilute solution of primary antibody (generally between 0.5 and 5 ug/ml) is incubated with the membrane under gentle agitation.

• The solution is composed of buffered saline solution with a small percentage of detergent, and sometimes with powdered milk.

PROCEDURE

The antibody solution and the membrane can be sealed and incubated together for anywhere from 30 minutes to overnight. Secondary antibody

• After rinsing the membrane to remove unbound primary antibody, the membrane is exposed to another antibody, directed at a species-specific portion of the primary antibody.

• Several secondary antibodies will bind to one primary antibody and enhance the signal.

PROCEDURE

6. ANALYSIS:

calorimetric detection, The colorimetric detection method depends on incubation of the western blot with a substrate that reacts with the reporter enzyme (such as peroxidase).

Chemiluminescent detection, Chemiluminescent detection methods depend on incubation of the western blot with a substrate that will luminescent when exposed to the reporter on the secondary antibody.

PROCEDURE

Radioactive detection: Radioactive labels do not require enzyme substrates, but rather allow the placement of medical X-ray film directly against the western blot which develops as it is exposed to the label and creates dark regions which correspond to the protein bands of Interest.

Applications

- The confirmatory HIV test employs a western blot to detect anti-HIV antibody in human serum sample. Proteins from known HIV-infected cells are separated and blotted on a membrane as above.

Then, the serum to be tested is applied in the primary antibody incubation step; free antibody is washed away, and a secondary anti-human antibody linked to an enzyme signal is added

Applications

- The stained bands then indicate the proteins to which the patient's serum contains antibody.
- Western blot can also be used as a confirmatory test for Hepatitis B infection.

Disadvantages

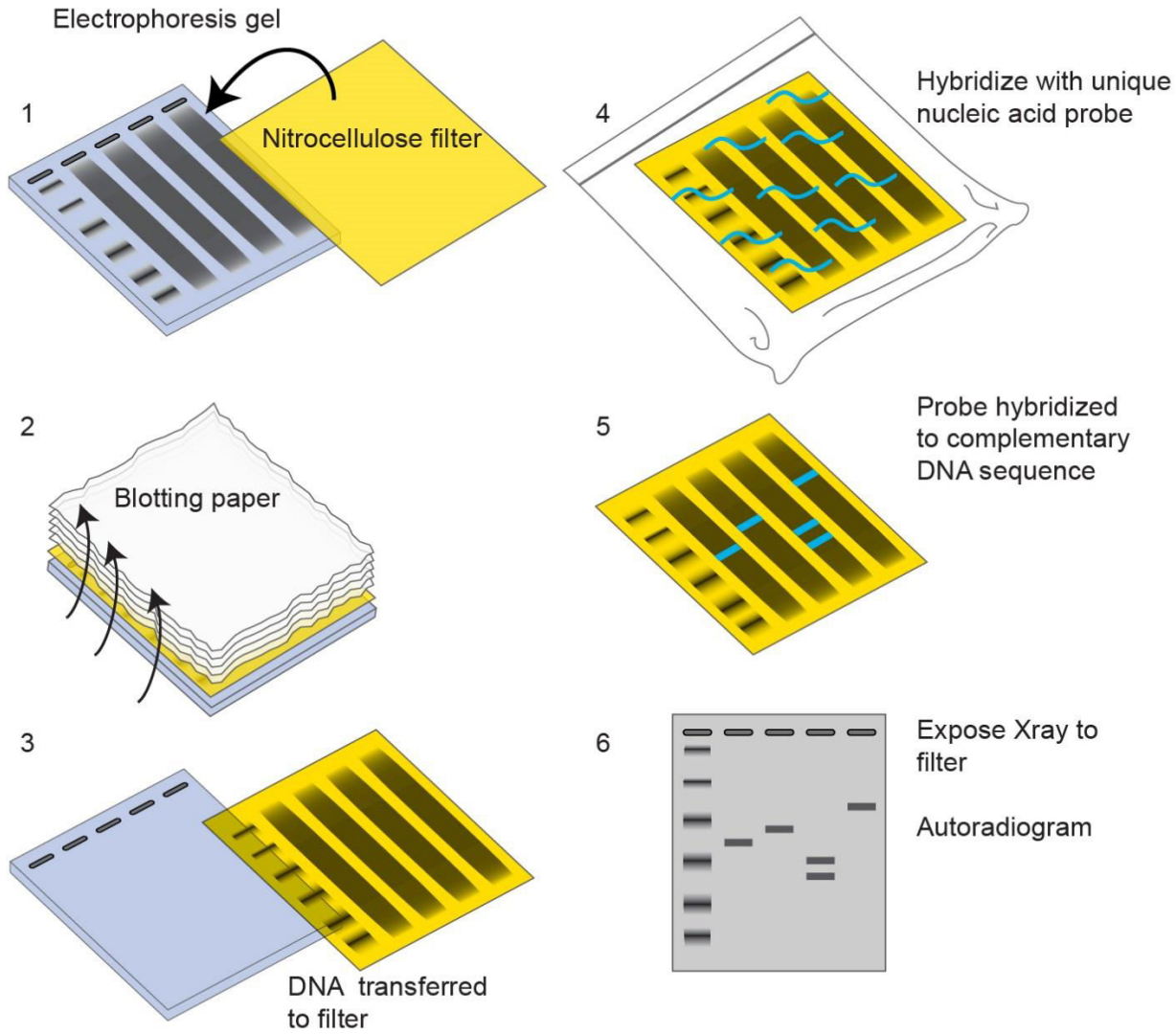
- If a protein is degraded quickly, Western blotting won't detect it well.
- This test takes longer that other existing tests.
- It might also be more costly

THANK YOU

Lesson: 89
Southern blotting

Southern blotting

- Southern blotting is a laboratory technique used to detect a specific DNA sequence in a blood or tissue sample. A restriction enzyme is used to cut a sample of DNA into fragments that are separated using gel electrophoresis. The DNA fragments are transferred out of the gel to the surface of a membrane. The membrane is exposed to a DNA probe labeled with a radioactive or chemical tag. If the probe binds to the membrane, then the probe sequence is present in the sample.


Electrophoresis gel
1
Nitrocellulose filter
2
Blotting paper
3
DNA transferred
to filter
4
Hybridize with unique
nucleic acid probe
5
Probe hybridized
to complementary
DNA sequence
6
Expose Xray to
filter
Autoradiogram

Lesson: 90
Southern blotting

Southern blotting Technique

- **1. Digest the DNA** with an appropriate restriction enzyme.
- **2. Run the digest** on an agarose gel.
- **3. Denature the DNA** (usually while it is still on the gel).
  For example, soak it in about 0.5M NaOH, which would separate double-stranded DNA into single-stranded DNA. Only ssDNA can transfer.

4. Transfer the denatured DNA to the membrane.

- Traditionally, a nitrocellulose membrane is used, although nylon or a positively charged nylon membrane may be used. Nitrocellulose typically has a binding capacity of about 100µg/cm, while nylon has a binding capacity of about 500 µg/cm. Many scientists feel nylon is better since it binds more and is less fragile. Transfer is usually done by capillary action, which takes several hours. Capillary action transfer draws the buffer up by capillary action through the gel an into the membrane, which will bind ssDNA

5. Probe the membrane with labeled ssDNA.

- This is also known as hybridization.
  Whatever you call it, this process relies on the ssDNA hybridizing (annealing) to the DNA on the membrane due to the binding of complementary strands.
  Probing is often done with 32P labeled ATP, biotin/streptavidin or a bioluminescent probe.

6. Visualize

- your radioactively labeled target sequence. If you used a radiolabeled 32P probe, then you would visualize by autoradiograph. Biotin/streptavidin detection is done by colorimetric methods, and bioluminescent visualization uses luminesence.

32P labeled ATP

- Treat the dsDNA fragment that you are using as a probe with a limiting amount of Dnase, which causes double-stranded nicks in DNA. Add 32P, dATP, and other dNTPs to DNA polymerase I, which has 5' to 3' polymerase activity and 5' to 3' exonuclease activity.

References

- https://askabiologist.asu.edu/southern-blotting

Lesson 91

Nothern Blotting

Nothern Blotting

A northern blot is a laboratory method used to detect specific RNA molecules among a mixture of RNA. Northern blotting can be used to analyze a sample of RNA from a particular tissue or cell type in order to measure the RNA expression of particular genes.

The first step in a northern blot is to denature, or separate, the RNA within the sample into single strands, which ensures that the strands are unfolded and that there is no bonding between strands. The RNA molecules are then separated according to their sizes using a method called gel electrophoresis. Following separation, the RNA is transferred from the gel onto a blotting membrane. (Although this step is what gives the technique the name "northern blotting," the term is typically used to describe the entire procedure.)

Once the transfer is complete, the blotting membrane carries all of the RNA bands originally on the gel. Next, the membrane is treated with a small piece of DNA or RNA called a probe, which has been designed to have a sequence that is complementary to a particular RNA sequence in the sample; this allows the probe to hybridize, or bind, to a specific RNA fragment on the membrane. In addition, the probe has a label, which is typically a radioactive atom or a fluorescent dye. Thus, following hybridization, the probe permits the RNA molecule of interest to be detected from among the many different RNA molecules on the membrane.

- https://www.nature.com/scitable/definition/northern-blot-

Lesson 92

Nothern Blotting

Applications of Nothern Blotting

Northern blotting allows one to observe a particular gene's expression pattern between tissues, organs, developmental stages, environmental stress levels, pathogen infection, and over the course of treatment.

The technique has been used to show overexpression of oncogenes and downregulation of tumor-suppressor genes in cancerous cells when compared to 'normal' tissue, as well as the gene expression in the rejection of transplanted organs

If an upregulated gene is observed by an abundance of mRNA on the northern blot the sample can then be sequenced to determine if the gene is known to researchers or if it is a novel finding.

The expression patterns obtained under given conditions can provide insight into the function of that gene. Since the RNA is first separated by size, if only one probe type is used variance in the level of each band on the membrane can provide insight into the size of the product, suggesting alternative splice products of the same gene or repetitive sequence motifs.

Advantages and disadvantages

Analysis of gene expression can be done by several different methods including RT-PCR, RNase protection assays, microarrays, RNA-Seq serial analysis of gene expression (SAGE), as well as northern blotting.

Microarrays are quite commonly used and are usually consistent with data obtained from northern blots; however, at times northern blotting is able to detect small changes in gene expression that microarrays cannot.

The advantage that microarrays have over northern blots is that thousands of genes can be visualized at a time, while northern blotting is usually looking at one or a small number of genes.

A problem in northern blotting is often sample degradation by RNases (both endogenous to the sample and through environmental contamination), which can be avoided by proper sterilization of glassware and the use of RNase inhibitors such as DEPC (diethylpyrocarbonate).

The chemicals used in most northern blots can be a risk to the researcher, since formaldehyde, radioactive material, ethidium bromide, DEPC, and UV light are all harmful under certain exposures.

Compared to RT-PCR, northern blotting has a low sensitivity, but it also has a high specificity, which is important to reduce false positive results.

Microdialysis can produce crystals by salting out, employing high concentrations of salt or other small membrane-permeable compounds that decrease the solubility of the protein. Very occasionally, some proteins can be crystallized by dialysis salting in, by dialyzing against pure water, removing solutes, driving self-association and crystallization.

- https://en.wikipedia.org/wiki/Northern_blot#cite_note-

Lesson 93

Basic principles of Sampling

Introduction

Sample Collection, such as handling, labeling, processing, aliquoting, storage, and transportation, may affect the results of the study

If case sample are handled differently from controls samples, differential misclassification may occur

Information linked to Sample

- Time and date of collection
- Recent diet and supplement use,
- Reproductive information (menstrual cycle)
- Recent smoking
- current medication use
- Recent medical illness
- Storage conditions

Quality Assurance

Systematic Application of optimum procedures to ensure valid, reproducible, and accurate results

Quality Assurance

- Adoption of standardized operation procedures for each aspect of biospecimen handling
- Stored specimens should be tested on a regular basis to detect sample deterioration
- Aliquoting material into multiple small vials

Quality Assurance

- Storing each person's specimen in at least two different physical locations to avoid the likelihood of loss of a large volume of specimen as a result of accidental thawing due to freezer failure or electronic blackout.

Quality Assurance

Sample should be selected from specimens that received the same treatment throughout the storage process or the same variations in handling.

Quality Assurance: Careful Record of disbursement

- Barcode system to check in and out the bio- specimens
- which specimen, how much material remains, documentation on factors such as thawing which could influence future use of the material.



bar code scanner

LESSON 94

Types 0f Sampling

Sampling Types

A Sample is “a smaller (but hopefully representative) collection of units from a population used to determine truths about that population”.

Why sample?

- Resources (time, money) and workload
- Gives results with known accuracy that can be calculated mathematically

Random Sampling

Random sampling is the purest form of probability sampling.

- Each member of the population has an equal and known chance of being selected.

- When there are very large populations, it is often ‘difficult’ to identify every member of the population, so the pool of available subjects becomes biased.

Systematic Sampling

- **Systematic sampling** is often used instead of random sampling. It is also called an Nth name selection technique.

- After the required sample size has been calculated, every Nth record is selected from a list of population members.

- As long as the list does not contain any hidden order, this sampling method is as good as the random sampling method.

- Its only advantage over the random sampling technique is simplicity (and possibly cost effectiveness).

Stratified Sampling

- **Stratified sampling** is commonly used probability method that is superior to random sampling because it reduces sampling error.

- A stratum is a subset of the population that share at least one common characteristic; such as males and females.

    - Identify relevant stratums and their actual representation in the population.

Random sampling is then used to select a *sufficient* number of subjects from each stratum.

Stratified sampling is often used when one or more of the stratums in the population have a low incidence relative to the other stratums.

Cluster Sampling

- Cluster Sample: a probability sample in which each sampling unit is a collection of elements.

- Effective under the following conditions:

  - A good sampling frame is not available or costly, while a frame listing clusters is easily obtained

  - The cost of obtaining observations increases as the distance separating the elements increases

Convenience Sampling

- **Convenience sampling** is used in exploratory research where the researcher is interested in getting an inexpensive approximation.

- The sample is selected because they are convenient.

- It is a nonprobability method.
  - Often used during preliminary research efforts to get an estimate without incurring the cost or time required to select a random sample

Judgment Sampling

- **Judgment sampling** is a common nonprobability method.
- The sample is selected based upon judgment.
  - an extension of convenience sampling
- When using this method, the researcher must be confident that the chosen sample is truly representative of the entire population.

Lesson 95

Air Sampling Techniques

Air Sampling Techniques

To investigate biological particles in the air, they have to be caught by a sampling device with a design format suitable for viewing particles under a microscope or other form of analysis. Although quantitative analytical methods such as immunological and molecular techniques (real-time PCR), which can quantify target DNA accurately, are now available.

Passive traps

- This is the simplest way of collecting airborne biological particles. To trap particles by sedimentation, a microscope slide is made sticky by coating one side of it with petroleum jelly, glycerine jelly or silicone grease. The slide is placed horizontally with the sticky surface upwards. As an alternative, Petri dishes containing selective medium may be left open to sample viable propagules. These are usually exposed for 10-30 minutes but possibly longer or much shorter durations would be appropriate depending on the environment. For outdoor use, a rain-shield may be mounted above the coated-slide or Petri dish.

- A variety of different sampling devices are currently available to acquire air samples of microbial and viral particles. These technologies include filters, impingers, impactors, and wet or dry cyclones.

AIR SAMPLING METHODOLOGIES

- Most air sampling technologies depend on the aerodynamic diameter of the airborne particles, the adhesion properties of airborne particles, Brownian motion, thermal gradients, and the inertia of the particles. Aerosolized particles attach to any surface with which they come into contact. Adhesive forces such as van der Waals forces, electrostatic forces, and surface tension partly explain this adhesion.

- Airborne particles with aerodynamic diameters on the order of 100 nm or less are prone to a particular way of moving, mainly due to the billions of collisions they encounter with gas molecules. This is called Brownian motion, and the smaller the particle, the greater the movement and the more likely that the particle will diffuse, come into contact with a surface, and adhere to it. When this happens, the other suspended particles occupy the space left vacant by the particle that has adhered to the surface. This phenomenon is the basis for the efficient removal of very small particles by filtration, particularly when the distance between two surfaces of the filter is sufficient for the particles to pass through.

- Larger particles with aerodynamic diameters on the order of a micrometer or more are less influenced by Brownian motion but have greater inertia. Gravitational attraction has a significant impact on these particles and causes them to settle on surfaces. These particles are also more easily diverted from a gas streamline, leading to impaction on surfaces, especially at high velocity and when the angle of the airflow is drastically altered. Very small particles have less inertia and will more likely follow the streamline.

- https://mmbr.asm.org/content/72/3/413/Willey.Com

Lesson 96
Soil Sampling Techniques

Sampling from Soil

Soil microbial communities form the foundation of the terrestrial ecosystem and are critical drivers of biogeochemical processes. Microbes play an essential role in nutrient cycling and soil fertility and influence global climate, water quality, and atmospheric composition. A few grams of soil may contain billions of microorganisms, including bacteria, archaea, and fungi.

Sampling Design and Methods

Soil microbial samples are collected as part of a soil sampling about at all terrestrial field sites. There are two main types of sampling bouts: non-coordinated and coordinated. A non-coordinated bout occurs most years at a site and consists primarily of microbial sampling, while a coordinated bout occurs every five years at a site and includes additional soil biogeochemical and isotopic measurements, as well as coordinated plant biogeochemical measurements.

Three soil cores are taken from ten pre-determined soil plots at each field site during each sampling bout, including four plots within the tower airshed and six plots distributed throughout the field site. Plots are selected to reflect the dominant vegetation types at each site. Sampling bouts occur up to three times per year at each field site. One sampling event occurs at all sites during the period of "peak greenness.

- For most sites (excluding sites with very short growing seasons), two additional sampling events occur during seasonal transitions. For temperature-driven sites, these samples capture snowmelt/thaw in the spring and plant senescence in the fall. For precipitation-driven sites, these windows capture the onset of the wet or dry season. Timing of sampling bouts is specific to each site. Within each sampling event, samples from all plots within the field site are typically collected within 14 calendar days

Collection

- Soil samples are collected at randomly assigned locations within the soil sampling plots located across each NEON site. Typically, the soil is collected using a 1.5-2.5 in. diameter coring device, with the exact collection device being dictated by the local soil conditions such as rockiness or clay content. Soil sampling is conducted to a maximum depth of 30 cm wherever possible. Cores are taken from undisturbed soil and each sampled location is tracked to prevent future sampling from that exact location.

- This coring method allows soil to be separated by soil horizon (horizontal layers with distinct physical, chemical, and biological characteristics) for microbial and biogeochemical analyses. Organic and mineral horizons are separated prior to analysis, allowing researchers to determine how microbial communities differ at different depths, soil types, and soil horizons.

- Soil cores are separated by horizon and homogenized, and rocks, roots, and organic debris are removed. The homogenized soil is divided into subsamples for separate microbial and chemical analyses. Samples for microbial analysis are prepared in two ways:

- Genetic subsamples, used for microbial genetic and metagenomic analysis, microbe community composition and abundances, or archiving, are made and frozen in the field and then shipped on dry ice to the analytic laboratory or archive facility.

- Non-genetic subsamples, used for microbial biomass analysis, are generated in the lab with a subset of soil from each horizon. Mineral soils are sieved, and organic soils are re-picked to remove remaining rocks, roots, and non-soil material. 5-10 g of this collected soil is transferred to a sterile container, kept field moist, frozen, and shipped on dry ice

Processing and Lab Analysis

- Once microbial samples have been collected in the field, they are frozen and sent to an external laboratory for further analysis. Samples are minimally processed in the field and must be sealed and kept in sterile conditions to prevent cross-contamination from other microbes while in storage or transit. Once shipped to the analytical facility, samples are further prepared for laboratory analysis and DNA is extracted from the homogenized soil subsamples.

Sources

- https://www.neonscience.org/data-collection/soil-microbes

LESSON 97
BLOOD SAMPLING

Blood sampling

It is critical to collect samples not only for main biomarkers of interest in your study, but also to process and store material in a way that allows the new biomarkers to be tested in future.

Types of Biospecimens: Blood

The use of skilled technicians and precise procedures when perform phlebotomy are important because painful, prolonged or repeated attempts at venepuncture can cause patient discomfort or injury and result in less than optimum quality or quantity of sample.

Types of Biospecimens: Blood

- Plasma
- Serum
- Lymphocytes
- Erythrocytes
- Platelets

Blood Sample Collection

- When a large amount of blood sample needed, an evacuated tube system with interchangeable glass tubes can be used to avoid multiple venepunctures.
- Evacuated tubes are commercially prepared with or without additives and with sufficient vacuum to draw a predetermined blood volume per tube.



Sterile Blood needles; Sterile Syringes; Plain Vacutainer; Blood Tubes; Alcohol Prep Pads; Tourniquet



Blood Collection

Blood Collection: Color-code Tubes

- Red-top tubes contain no additives. These tubes are used for tests performed on serum samples and DNA.
- When you use the red-top tubes, the sample an be placed for 1-2 hours so that the serum and blood clots will be separated. Blood clots can be used for DNA analysis.

Blood Collection: Color-code Tubes

- Lavender-top tubes contain EDTA, commonly used clinically for complete blood cell counts.
- This is the way to obtain lymphocytes for DNA extraction, plasma for nutritional analysis, and red blood cells for other assays.

EDTA

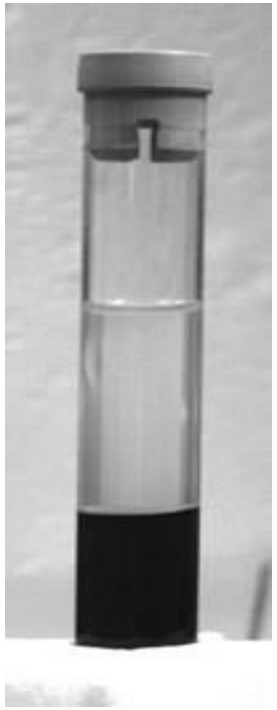
EDTA is a anticoagulant. It works by calcium chelation and is used clinically in heamatology studies. It is well suited to DNA-based assays, but has problems for cytogenetic analysis.



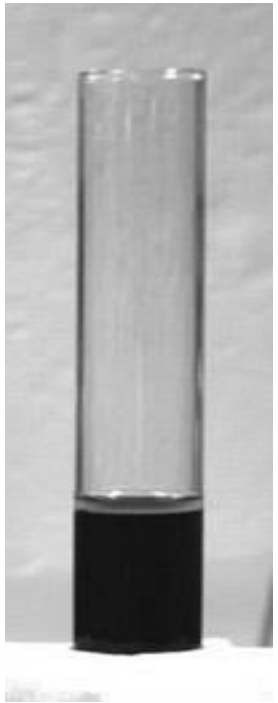
Whole blood in
the collection tube



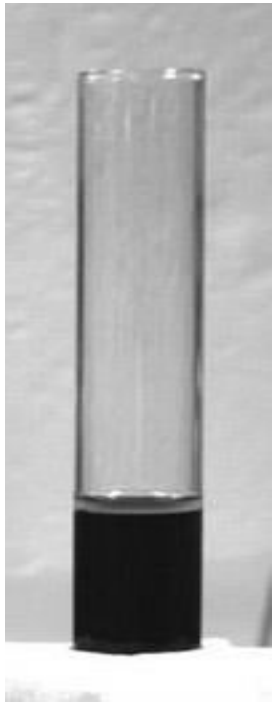
Blood after
centrifugation



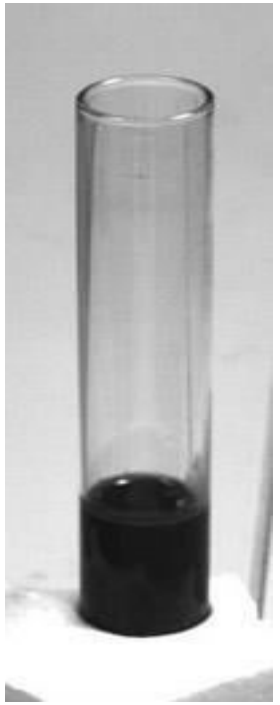
WBCs and RBCs
fter plasma
removal



Top view of the
WBCs (buffy coat)



Top view of sample
after WBC removal



Blood Collection: Color-code Tubes

- Green-top tubes contain heparin
- Blue-top tubes contain sodium citrate and citric acid
- Black-top tubes contain sodium oxalate
- Yellow-top tubes contain acid-citrate-dextrose (ACD) solution.
- Grey-top tubes contain a glycolytic inhibitor.



Heparin

- Heparin is an anticoagulant. There are some reports of occasional problems with heparin in PCR assays, studies generally find that there are no major difference in the use of EDTA or heparin

Citrate

- Citrate also works by calcium chelation and is used in coagulation studies and blood banking. It is optimal for assays conducted on lymphocytes and DNA.

Dried Blood Spot

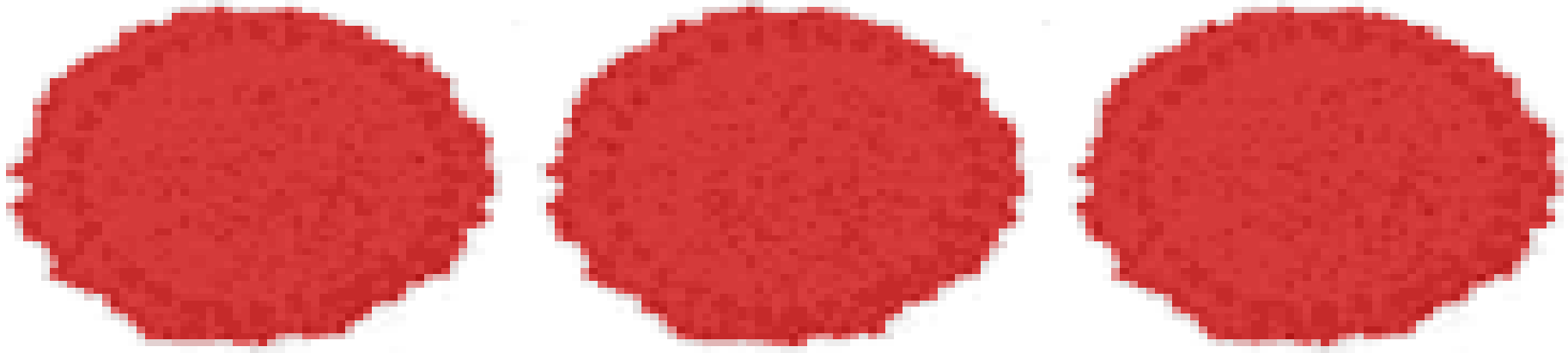
Dried blood spot specimens:

- Small quantities of blood adequate for the characterization of DNA.
- Not require venepuncture or low temperature condition during collection, processing and storage
- Can be from whole blood or antocoagulated with EDTA

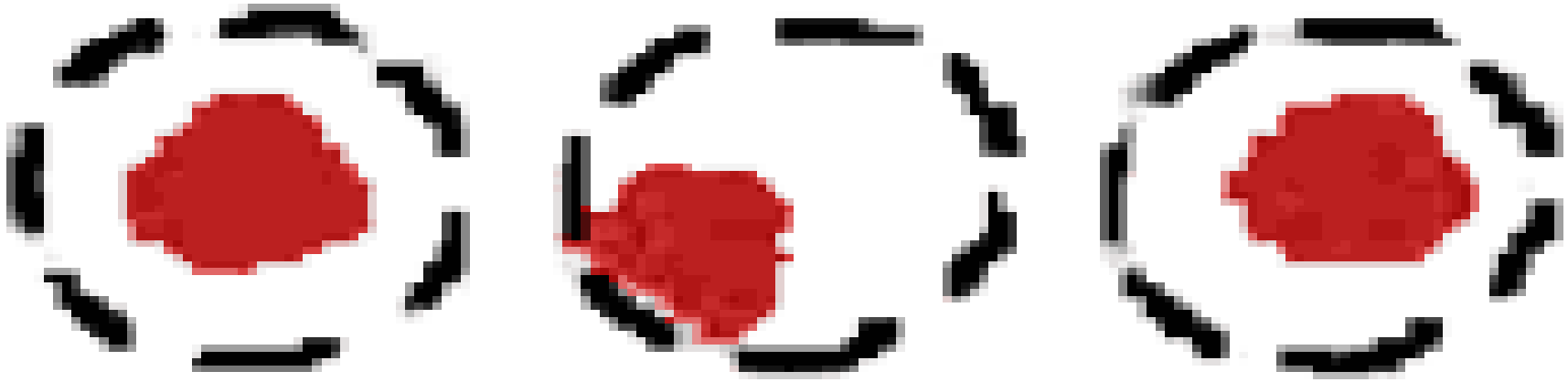
Dried Blood Spot

- Blood specimen is spotted onto clean slides or paper or cotton cloth.
- Transported and stored at room temperature
- Serves as a good source of high-molecular-weight DNA
- A quantity of 50 ul of dried blood can provide 0.5 ug DNA, sufficient for multiple PCR-based assays

Valid Specimen



Invalid Specimen

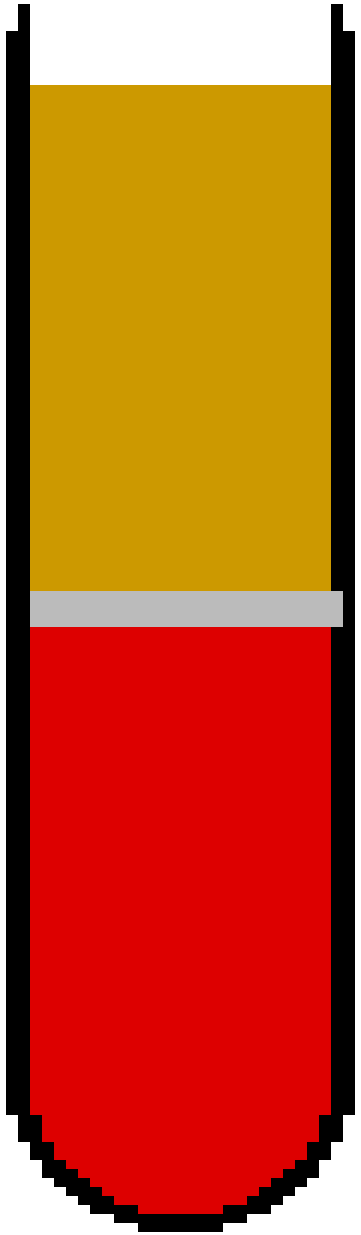


Lesson 98
Blood Sampling

Blood Components

From 10 ml of blood:

- Plasma or serum 6-7 ml
- Lymphocytes and mononuclear cells 10-20 x 10^6 Cells/ml
- Erythrocyte (red blood cells) and other cells – 5 x 10^6 cells/ul; 10-15 mg HB



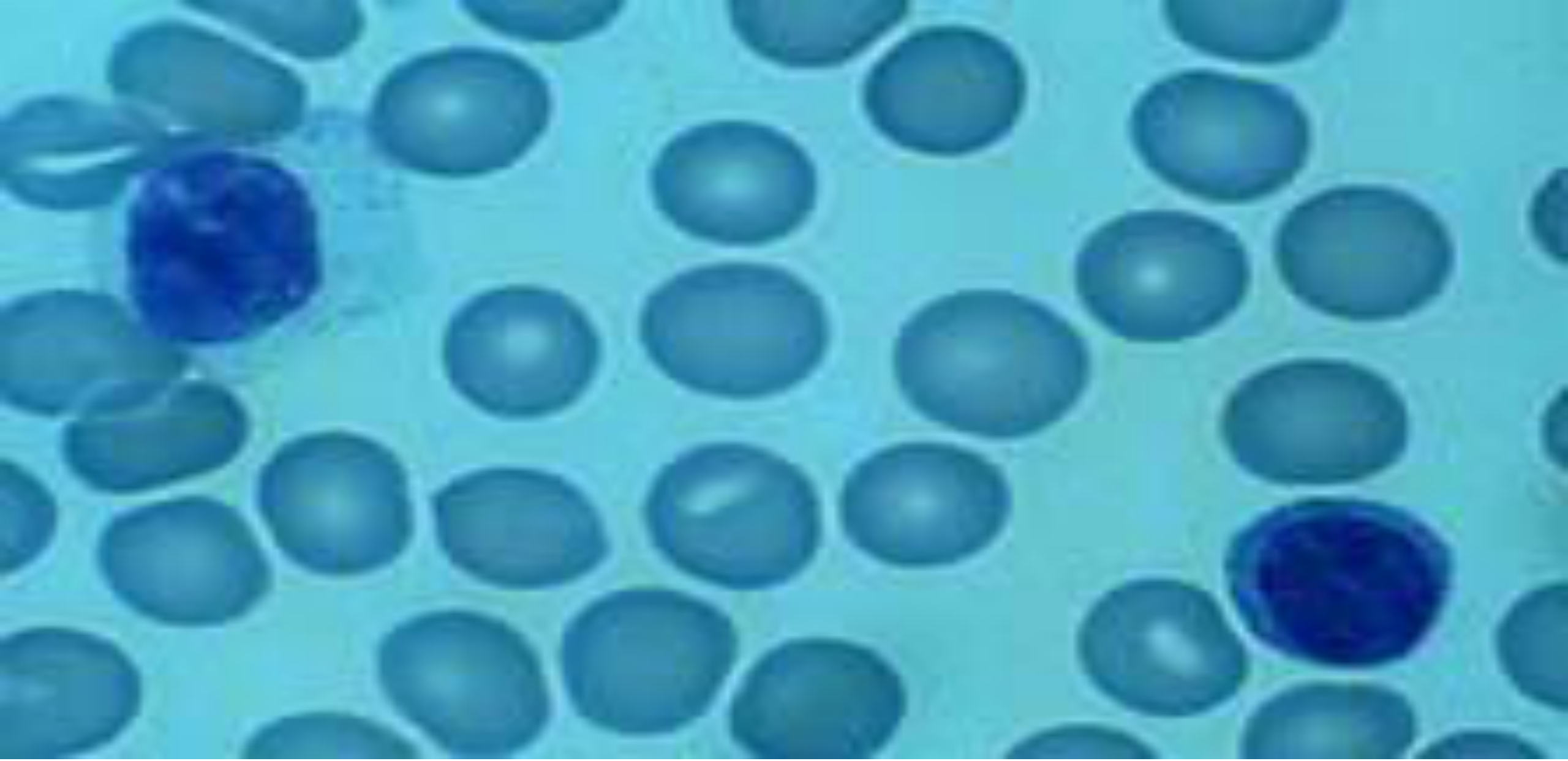
Plasma

White blood cells,
the “buffy coat”

Red blood cells

- Blood is a liquid tissue. Suspended in the watery plasma are seven types of cells and cell fragments.
- Red blood cells (RBCs) or erythrocytes
- Plateletsor thrombocytes
- Five kinds of white blood cells (WBCs) or leukocytes
- Three kinds of granulocytes: Neutrophils; Eosinophils; Basophils
- The number
- Two kinds of leukocytes without granules in their cytoplasm: lymphocytes and monocytes

- White blood cells
- are much less numerous than red (the ratio between the two is around 1:700),
- have nuclei,
- participate in protecting the body from infection,
- consist of lymphocytes and monocytes with relatively clear cytoplasm, and three types of granulocytes, whose cytoplasm is filled with granules.

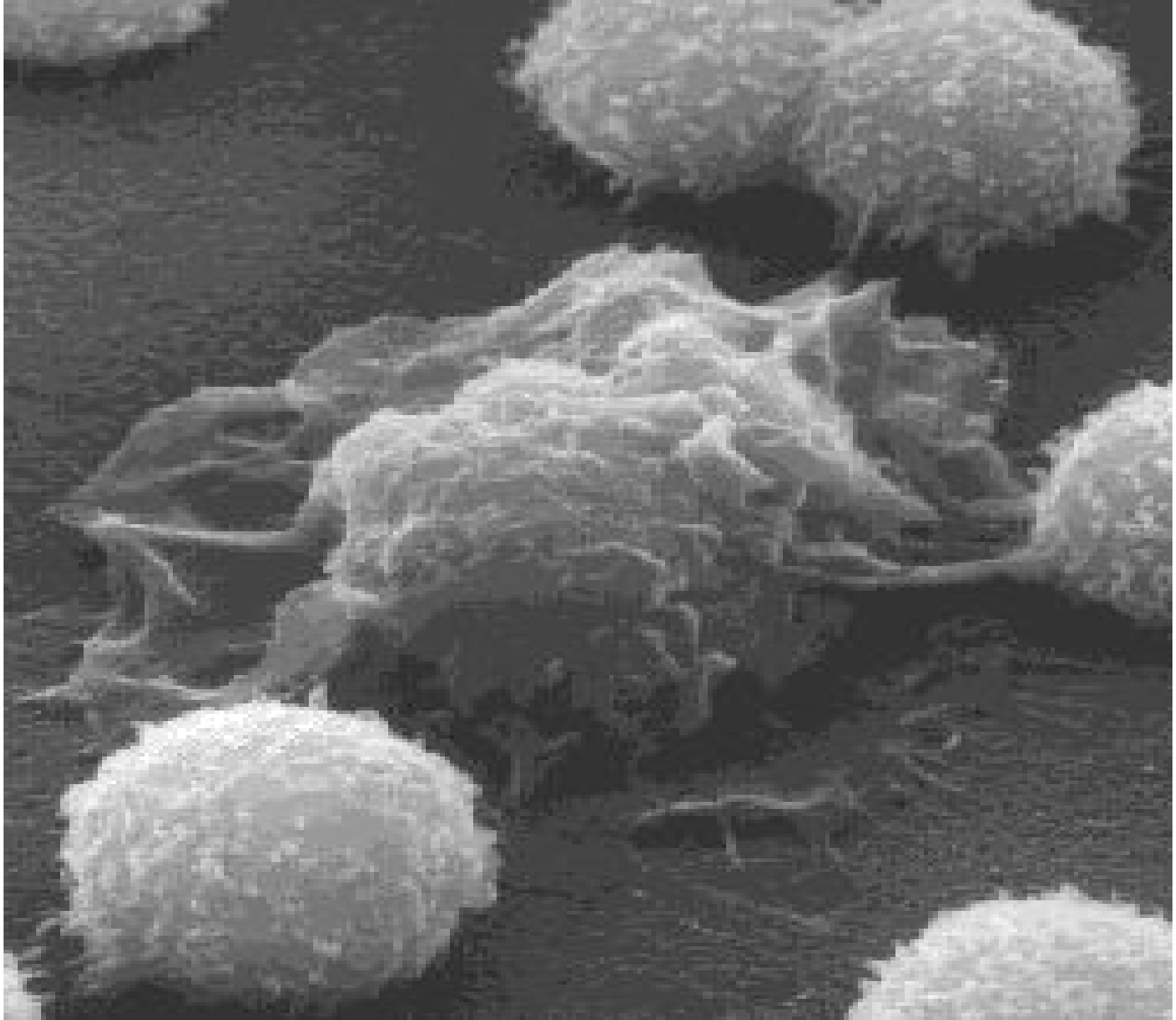


Lymphocytes

Mononuclear leukocytes

- Mononuclear leukocytes are the only cell type in blood capable of growth
- They can be cryopreserved for the establishment of cell lines.
- Cryopreservation permits cell viability and can be the only source to measure RNA

single macrophage (monocyte) surrounded by several lymphocytes



Granulocytes

- Granulocytes can serve as a source of DNA without sacrificing the lymphocytes

Erythrocytes (RBC)

- Stored after washing with physical saline
- Can be useful to study adducts of haemoglobin


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Plasma and Serum

- Can be used to measure microanalytes, diet components, vitamins, xenobiotic exposures and so on.
- Plasma can be obtained from an anticoagulated blood sample through separation from cell components.
- Serum is better for antibody measurements, nutrients, etc.

Lesson 99

Preservation Techniques

Steps for the preservation of specimens for scientific study are as follows:

- **Euthanizing**. Specimens should be euthanized in a way that will leave them undamaged and relaxed.
- **Injection and slitting**. Liquid preservatives must be introduced into the body cavity, limbs and tail, either by hypodermic injection or through slits.
- **Fixing.** While the specimens are still relaxed, they should be arranged in trays so that they will harden in the proper position.

Labeling. Each specimen should be accompanied by certain data, either attached directly or entered in a notebook with a number corresponding to a numbered tag tied to the specimen.

Storage. After specimens have been fixed in the proper position, they should be stored in liquid preservative for at least several days, after which they may be allowed to remain in the liquid, or transferred to plastic bags for temporary storage

Preserving Solutions

Formalin: If at all possible, formalin should be used for injecting and fixing specimens. Formalin is the commercial name of a solution of formaldehyde gas (CH20) in water.

Alcohol: It is usually sold at a strength of 95% (190 proof).

Preparation:

- If specimens are to be made permanently immune to decomposition, it is necessary that liquid preservative be introduced into the body cavity, limbs and tail within as short a time as possible after the animals have been killed. This may be accomplished either by injection (with a hypodermic syringe) or by making deep cuts with a sharp scalpel, razor blade or scissors. The most satisfactory way is by injection. A ten or twenty cc. syringe with a needle lock and several needles (guages 18 to 26) will serve to inject most specimens.

Frogs and Toads:

- Injection should be made through the belly, directly into the body cavity. If the body is puffed with air, it should be deflated by gently squeezing with the fingers. Very small frogs require only a few drops of preservative; frogs two or three inches long only a few cc. Introduce only enough preservative required to make the specimen look natural--it should not look bloated. It is not necessary to inject the legs of any but the largest frogs. If equipment for injection is not available, a single slit may be made in the abdomen, to one side of the midline. The slit should be deep enough to allow free access of the preservative into the body cavity



Salamanders. Most salamanders do not require injection or slitting. If your specimens look "caved in" a small amount of preservative may be injected into the body cavity, or a single slit made in the abdomen to permit preservative.



Tadpoles. Tadpoles and small salamander larvae should always be preserved in 10% formalin, never in alcohol. Simply drop the tadpoles into formalin while they are still alive. Be sure there is enough preservative to cover them and avoid overcrowding. After 24 hours all the liquid should be drained off and replaced with fresh formalin

Lizards

- Injection should be made through the belly directly into the body cavity. Care should be taken not to use too much, or the body will become unnaturally distended. A series of slits should be made in the under side of the tail with a sharp scalpel or razor blade. The slits should be from 1/8 to 1/4 inch long and about 1/4 inch apart, and should extend from the base of the tail to the tip. Very large lizards must be injected or slit in the thicker portions of the limbs and neck. If space does not permit preservation of very large lizards whole, they may be skinned out, except for the head. To skin a large lizard, make a cut down the belly from the neck to the base of the tail.

Work the skin loose from the body, pulling the skin of the arms and legs inside out as far as the wrists and ankles. Do not attempt to skin out the head, hands, feet or tail. Sever the wrists, ankles, neck and base of the tail, and remove the carcass. The skin should then he placed directly into preservative. If possible, one hemipenis of male lizards should be everted. This can be accomplished hy injecting preservative into the base of the tail (before slitting) and at the same time applying pressure with the thumb just behind the anus.



Snakes

Make a series of injections an inch or two apart through the belly into the body cavity. Begin just behind the head and continue the injections to the anus. If a syringe is not available, a series of slits must be made in the belly. For most snakes the slits should be about an inch apart and an inch long; smaller slits closer together for very small snakes. Just as in lizards, a series of slits must be made in the under side of the tail and one hemipenis everted in males (Fig. 1B). Very large snakes may be skinned out, leaving the head and tail attached. To skin a snake make a single, long cut in the belly, just to one side of the midline, beginning about an inch behind the head and continuing to about an inch in front of the anus.

Snakes

Do not cut through the anal plate. Work the skin loose from the body, but do not attempt to remove the skin from the head or tail. Sever the body an inch behind the head and an inch in front of the anus, and (after recording the stomach contents, number of eggs, embryos, etc.) discard the carcass. Put a strip of cloth on the inner side of the skin and roll it up, beginning at the head. Tie the roll with a piece of string and put it directly into preservative.

Alligators and Crocodiles. Small individuals may be preserved just as lizards. Larger individuals should be skinned out with the head attached, rolled up and placed directly into preservative.

Turtles. Preservative should be injected into the body cavity just in front of each of the four limbs, between the carapace and plastron. Use a long needle and continue injections until the head and · limbs are forced out of the shell. If a syringe is not available, make deep cuts into the body cavity just in front of each leg. Limbs, neck and tail should be injected or slit, as in large lizards.



Storage:

After the specimens have been injected or slit, tagged, and fixed, they should be put directly into preservative. If they are to be transferred later to plastic bags for temporary storage or to be shipped they should first be allowed to remain completely immersed in preservative for at least 48 hours if formalin is used, or a week if alcohol is used. The longer they are allowed to stay in preservative, the better. They should be loose and completely covered with plenty of liquid. Specimens which have been hardened in trays should also be allowed to soak in preservative for a day or two before being shipped or placed in plastic bags for storage.

Storage:

- If space is no problem, preserved specimens are best kept in glass containers. Bail-top jars with a glass top and rubber gasket are best. Fruit jars with a metal screw top lid may be used but should be carefully watched for rust and evaporation. Glass jars with polyethylene lids and liners are more commonly used in collections, since the lids form a tight seal and are easier to obtain than the traditional bail-top jars. Metal containers should be used only for temporary storage unless coated on the inside with paraffin, "Bakvar", or some other rustproof material.

Reference

- https://lsa.umich.edu/ummz/herps/collections/preservation-techniques.html

Lesson 100

Fluid Preservation

Fluid Preserved Specimens

- There are three components to a fluid-preserved specimen
- The *fixed specimen*: The specimen is prepared by "fixing" it, achieved by injecting it with chemicals that stop the deterioration and decay process (known as "autolysis"). The most common fixative is formaldehyde, or a formaldehyde and water solution known as formalin. Some specimens may not be fixed before being submersed in the fluid preserve.

- The *fluid preserve*: The preserve is commonly alcohol, either ethanol or isopropyl alcohol.
- The *container*: Containers are typically glass jars or bottles sealed with a closure. Types of closures may vary within a collection and often include lids with gaskets. Large specimens may require the use of open glass tanks.

- Although the fixative and fluid preservation process causes a chemical alteration of the specimen and can lead to discoloration, shrinking, or swelling of the specimen, these collections are able to last for hundreds of years.

- Formalin
  - A strength of 10% formalin is best for most purposes. If the original strength is 40%, it should be mixed at a ratio as nine parts water to one part formalin. The advantages of formalin over other preservatives are: it is inexpensive, it is generally available, a small bulk as concentrated stock solution may be diluted as needed, and specimens almost never decay in it. Its principal disadvantages are: it has a very irritating odor, it is very poisonous and may cause skin irritation or rash, it has a tendency to make specimens become brittle if the solution is too strong, and tends to fade out certain colors rapidly, and it must be stored in rustproof containers. (Buffering of the 10% solution is recommended as formalin is slightly acidic. One buffering system that may be used is a mixture of monobasic and dibasic Sodium Phosphate, at 13 gm/gallon [Monobasic] and 24 gm/gallon [Dibasic]).

Alcohol

- Alcohol which has been stored in open containers loses its strength rapidly due to evaporation. Strength may be tested with an alcoholometer. Specimens which have been fixed in alcohol should be carefully watched for signs of rotting. Alcoholic beverages, shaving lotions and Bay Rum contain ethyl alcohol. They should be used only in an emergency and without dilution. Liquor which is 100 proof is only 50% ethyl alcohol.

Ethanol

- Under most circumstances, ethanol is the preservative of choice for stoneflies. In concentrations ranging from 70% to 80%, it preserves color well and produces supple specimens. In addition, it is pleasant to use and stores safely, and its disposal is much less problematic than for other preservatives/fixatives. A few specimens may be live sorted into 70–80% ethanol and yield well-preserved color patterns without appreciable shrinkage of the body.

Lesson 101

Preservation of dry specimen

Dry preservation

Killing and handling prior to dry mounting

- Insects that are intended to be pinned and stored dry are best killed either in a killing bottle or tube containing a volatile poison, or in a freezer. Freezing avoids the use of chemical killing agents but it is important to place the insects into a small, airtight container to prevent drying out and to freeze them for at least 12–24 h. Frozen insects must be handled carefully and properly thawed before being pinned, otherwise the brittle appendages may break off. The safest and most readily available liquid killing agent is ethyl acetate, which although flammable, is not especially dangerous unless directly inhaled. It should not be used in an enclosed room. More poisonous substances, such as cyanide and chloroform, should be avoided by all except the most experienced entomologists.

- Ethyl acetate killing containers are made by pouring a thick mixture of plaster of Paris and water into the bottom of a tube or wide-mouthed bottle or jar to a depth of 15–20 mm; the plaster must be completely dried before use. To “charge” a killing bottle, a small amount of ethyl acetate is poured onto and absorbed by the plaster, which can then be covered with tissue or cellulose wadding. With frequent use, particularly in hot weather, the container will need to be recharged regularly by adding more ethyl acetate. Crumpled tissue placed in the container will prevent insects from contacting and damaging each other. Killing bottles should be kept clean and dry, and insects should be removed as soon as they die to avoid color loss. Moths and butterflies should be killed separately to avoid them contaminating other insects with their scales.

- Dead insects exhibit *rigor mortis* (stiffening of the muscles), which makes their appendages difficult to handle, and it is usually better to keep them in the killing bottle or in a hydrated atmosphere for 8–24 h (depending on size and species) until they have relaxed, rather than pin them immediately after death. It should be noted that some large insects, especially weevils, may take many hours to die in ethyl acetate vapors and a few insects do not freeze easily and thus may not be killed quickly in a normal household freezer.

- It is important to eviscerate (remove the gut and other internal organs of) large insects or gravid females (especially cockroaches, grasshoppers, katydids, mantids, stick-insects, and very large moths), otherwise the abdomens may rot and the surface of the specimens go greasy. Evisceration, also called gutting, is best carried out by making a slit along the side of the abdomen (in the membrane between the terga and sterna) using fine, sharp scissors and removing the body contents with a pair of fine forceps. A mixture of 3 parts talcum powder and 1 part boracic acid can be dusted into the body cavity, which in larger insects may be stuffed carefully with cotton wool.

- The best preparations are made by mounting insects while they are fresh, and insects that have dried out must be relaxed before they can be mounted. Relaxing involves placing the dry specimens in a water-saturated atmosphere, preferably with a mold deterrent, for one to several days depending on the size of the insects. A suitable relaxing box can be made by placing a wet sponge or damp sand in the bottom of a plastic container or a wide jar and closing the lid firmly. Most smaller insects will be relaxed within 24 h, but larger specimens will take longer, during which time they should be checked regularly to ensure they do not become too wet.

- **Pinning, staging, pointing, carding, spreading, and setting**
- Specimens should be mounted only when they are fully relaxed, i.e. when their legs and wings are freely movable, rather than stiff or dry and brittle. All dry-mounting methods use entomological macropins — these are stainless steel pins, mostly 32–40 mm long, and come in a range of thicknesses and with either a solid or a nylon head. *Never use dressmakers' pins for mounting insects*; they are too short and too thick. There are three widely used methods for mounting insects and the choice of the appropriate method depends on the kind of insect and its size, as well as the purpose of mounting. For scientific and professional collections, insects are either pinned directly with a macropin, micropinned, or pointed, as follows.

- *Direct pinning*
- This involves inserting a macropin, of appropriate thickness for the insect's size, directly through the insect's body; the correct position for the pin varies among insect orders and it is important to place the pin in the suggested place to avoid damaging structures that may be useful in identification. Specimens should be positioned about three-quarters of the way up the pin with at least 7 mm protruding above the insect to allow the mount to be gripped below the pin head using entomological forceps (which have a broad, truncate end).

- Specimens then are held in the desired positions on a piece of polyethylene foam or a cork board until they dry, which may take up to three weeks for large specimens. A desiccator or other artificial drying methods are re- commended in humid climates, but oven temperature should not rise above 35°C.

- *Pointing*
- This is used for small insects that would be damaged by pinning (but *not* for small moths because the glue does not adhere well to scales, nor flies because important structures are obscured), for very sclerotized, small to medium-sized insects (especially weevils and ants) whose cuticle is too hard to pierce with a micropin, or for mounting small specimens that are already dried. Points are made from small triangular pieces of white cardboard which either can be cut out with scissors or punched out using a special point punch. Each point is mounted on a stout macropin that is inserted centrally near the base of the triangle and the insect is then glued to the tip of the point using a minute quantity of water-soluble glue, for example based on gum arabic.

- The head of the insect should be to the right when the apex of the point is directed away from the person mounting. For most very small insects, the tip of the point should contact the insect on the vertical side of the thorax below the wings. Ants are glued to the upper apex of the point, and two or three points, each with an ant from the same nest, can be placed on one macropin. For small insects with a sloping lateral thorax, such as beetles and bugs, the tip of the point can be bent downwards slightly before applying the glue to the upper apex of the point.

Reference

- http://www.entomologa.ru/outline/277.htm

Lesson 102

Preservation of dry specimen

- *Spreading and setting*
- It is important to display the wings, legs, and antennae of many insects during mounting because features used for identification are often on the appendages. Specimens with open wings and neatly arranged legs and antennae also are more attractive in a collection. Spreading involves holding the appendages away from the body while the specimens are drying.
- Legs and antennae can be held in semi-natural positions with pins and the wings can be opened and held out horizontally on a setting board using pieces of tracing paper, cellophane, greaseproof paper, etc.

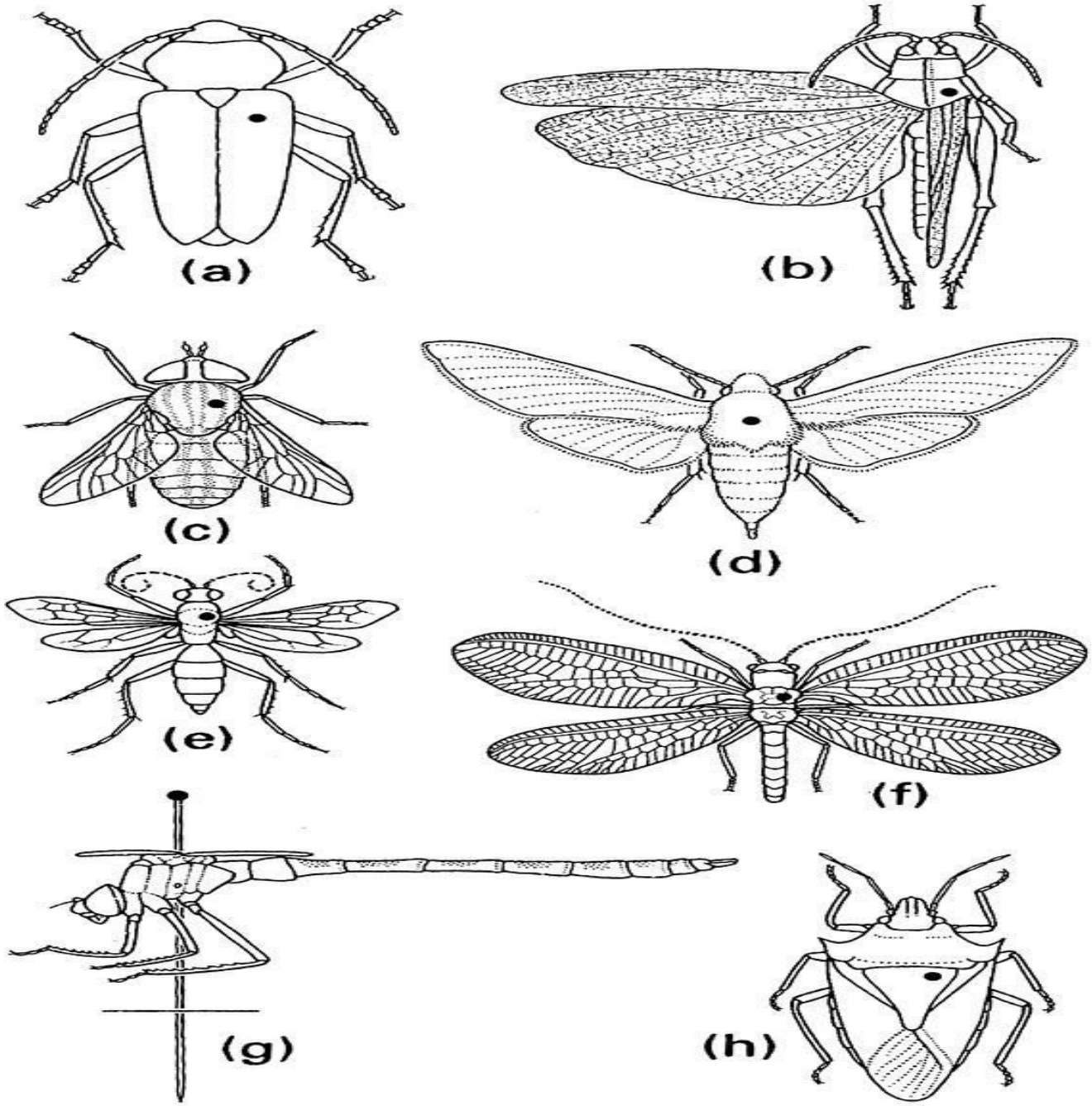
- **Setting boards** can be constructed from pieces of polyethylene foam or soft cork glued to sheets of plywood or masonite; several boards with a range of groove and board widths are needed to hold insects of different body sizes and wingspans. Insects must be left to dry thoroughly before removing the pins and/or setting paper, but it is essential to keep the collection data associated correctly with each specimen during drying. A permanent data label must be placed on each macropin below the mounted insect (or its point or stage) after the specimen is removed from the drying or setting board. Sometimes two labels are used — an upper one for the collection data and a second, lower label for the taxonomic identification.

- *Micropinning (staging or double mounting)*
- This is used for many small insects and involves pinning the insect with a micropin to a stage that is mounted on a macropin; micropins are very fine, headless, stainless steel pins, from 10 to 15 mm long, and stages are small square or rectangular strips of white polyporus pith or synthetic equivalent. The micropins are inserted through the insect's body in the same positions as used in macropinning. Small wasps and moths are mounted with their bodies parallel to the stage with the head facing away from the macropin, whereas small beetles, bugs, and flies are pinned with their bodies at right angles to the stage and to the left of the macropin.

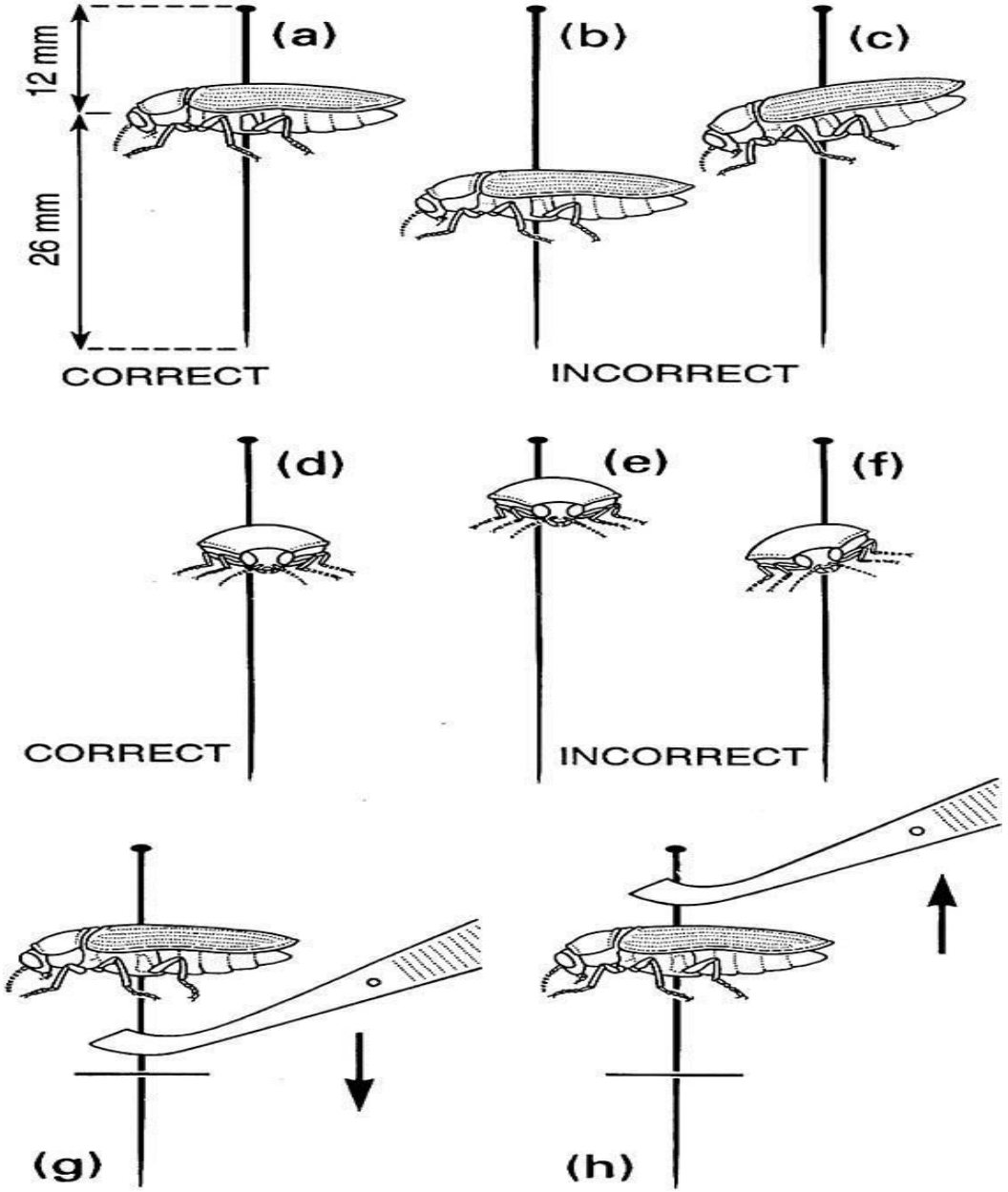
- Some very small and delicate insects that are difficult to pin, such as mosquitoes and other small flies, are pinned to cube mounts; a cube of pith is mounted on a macropin and a micropin is inserted horizontally through the pith so that most of its length protrudes, and the insect then is impaled ventrally or laterally.

Carding

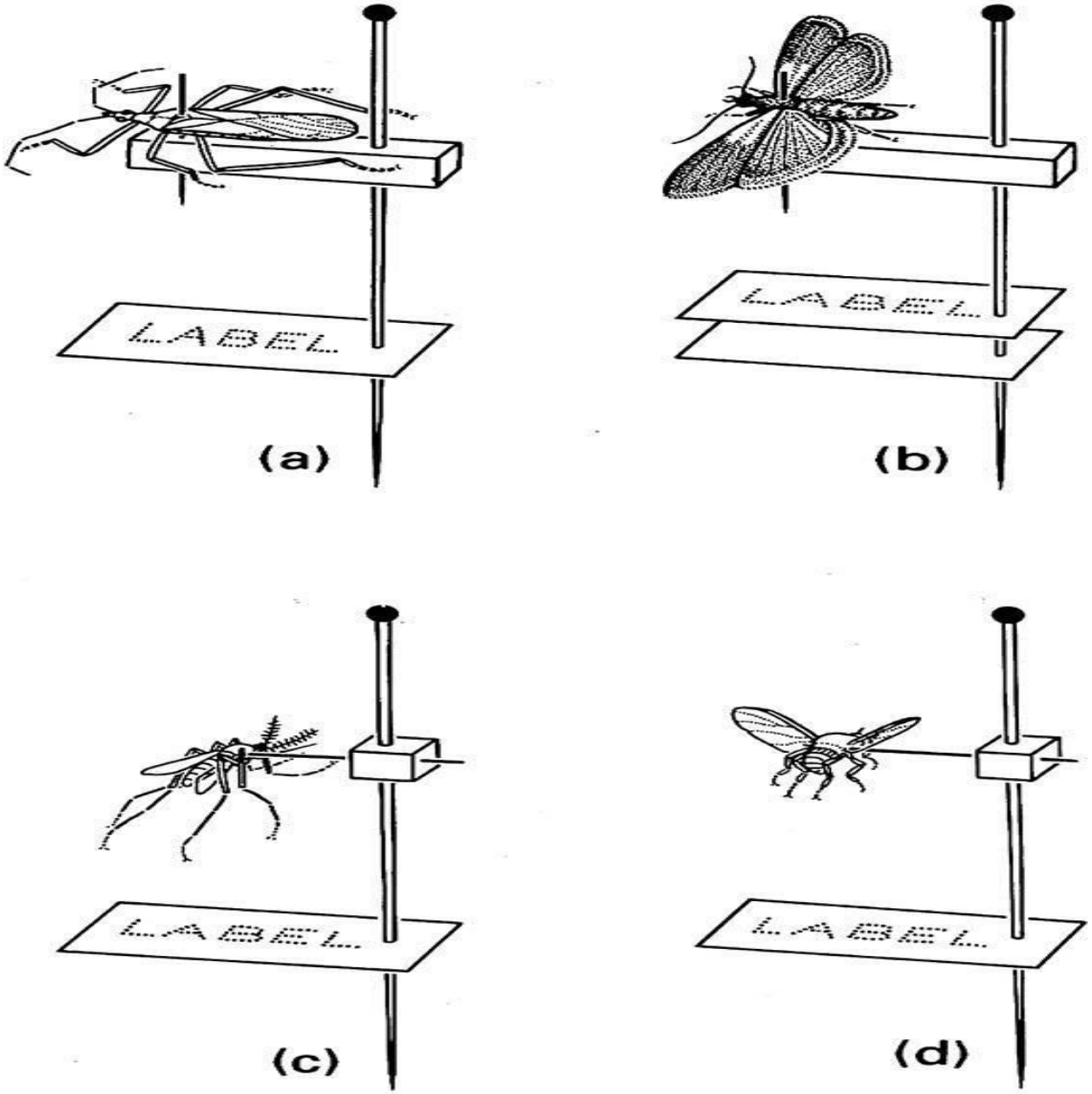
- For hobby collections or display purposes, insects (especially beetles) are sometimes **carded**, which involves gluing each specimen, usually by its venter, to a rectangular piece of card through which a macropin passes. Carding is not recommended for adult insects because structures on the underside are obscured by being glued to the card; however, carding may be suitable for mounting exuviae, pupal cases, puparia, or scale covers.



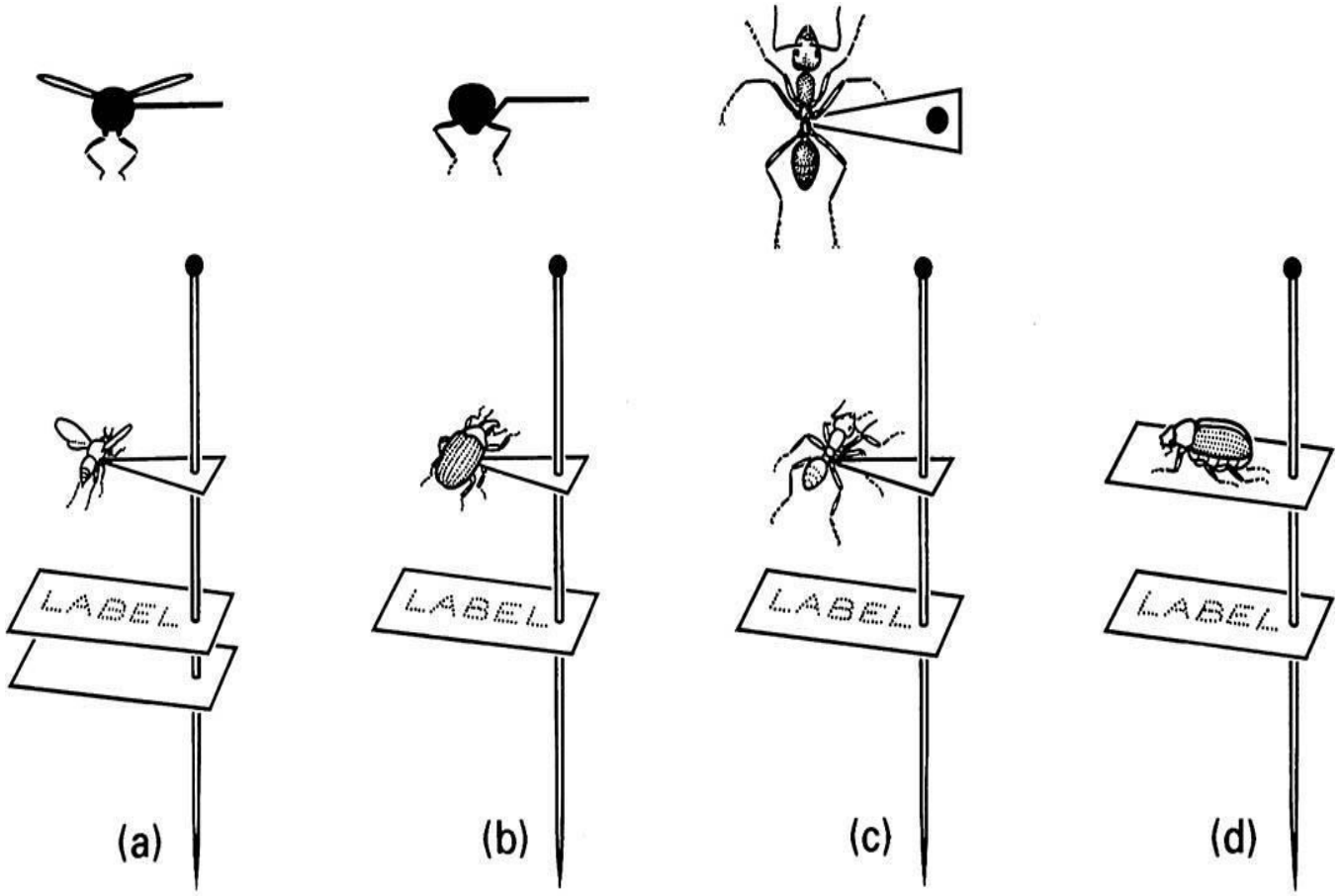
Figures. Pin positions for representative insects:(a) larger beetles (Coleoptera); (b) grasshoppers, katydids, and crickets (Orthoptera); (c) larger flies (Diptera); (d) moths and butterflies (Lepidoptera); (e) wasps and sawflies (Hymenoptera); (f) lacewings (Neuroptera); (g) dragonflies and damselflies (Odonata), lateral view; (h) bugs, cicadas, and leaf- and planthoppers (Hemiptera: Heteroptera, Cicadomorpha, and Fulgoromorpha).



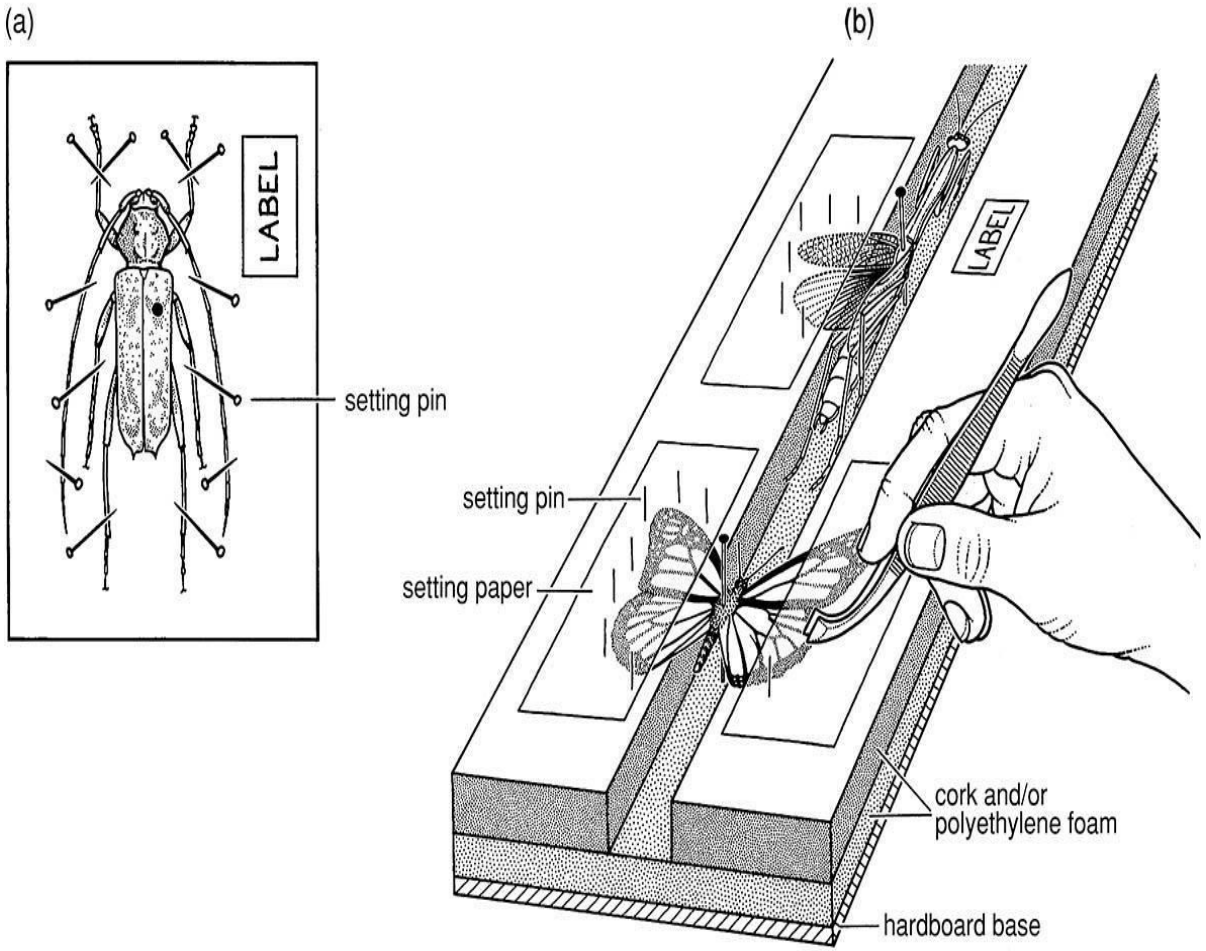
- **Figures. Correct and incorrect pinning:**(a) insect in lateral view, correctly positioned; (b) too low on pin; (c) tilted on long axis, instead of horizontal; (d) insect in front view, correctly positioned; (e) too high on pin; (f) body tilted laterally and pin position incorrect. Handling insect specimens with entomological forceps: (g) placing specimen mount into foam or cork; (h) removing mount from foam or cork. ((g, h) After Upton 1991)



- **Figures. Micropinning with stage and cube mounts:**(a) a small bug (Hemiptera) on a stage mount, with position of pin in thorax as shown in Fig. 17.2h; (b) moth (Lepidoptera) on a stage mount, with position of pin in thorax; (c) mosquito (Diptera: Culicidae) on a cube mount, with thorax impaled laterally; (d) black fly (Diptera: Simuliidae) on a cube mount, with thorax impaled laterally. (After Upton 1991)
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- **Figures. Point mounts:**(a) a small wasp; (b) a weevil; (c) an ant. Carding: (d) a beetle glued to a card mount. (After Upton 1991)



- **Figures Spreading of appendages prior to drying of specimens:** (a) a beetle pinned to a foam sheet showing the spread antennae and legs held with pins; (b) setting board with mantid and butterfly showing spread wings held in place by pinned setting paper. ((b) After Upton 1991)

Reference

- http://www.entomologa.ru/outline/277.htm

Lesson 103

Lyophilization , equipment

INTRODUCTION

- Lyophilization or freeze drying is a process in which water is removed from a product after it is frozen and placed under a vacuum, allowing the ice to change directly from solid to vapor without passing through a liquid phase. The process consists of three separate, unique, and interdependent processes; freezing, primary drying (sublimation), and secondary drying (desorption).
- The advantages of lyophilization include:
- Ease of processing a liquid, which simplifies aseptic handling
- Enhanced stability of a dry powder
- Removal of water without excessive heating of the product
- Enhanced product stability in a dry state

- Removal of water without excessive heating of the product
- Enhanced product stability in a dry state
- Rapid and easy dissolution of reconstituted product
- Disadvantages of lyophilization include:
- Increased handling and processing time
- Need for sterile diluent upon reconstitution
- Cost and complexity of equipment

- The lyophilization process generally includes the following steps:
- Dissolving the drug and excipients in a suitable solvent, generally water for injection (WFI).
- Sterilizing the bulk solution by passing it through a 0.22 micron bacteria-retentive filter.
- Filling into individual sterile containers and partially stoppering the containers under aseptic conditions.
- Transporting the partially stoppered containers to the lyophilizer and loading into the chamber under aseptic conditions.

- Freezing the solution by placing the partially stoppered containers on cooled shelves in a freeze-drying chamber or pre-freezing in another chamber.
- Applying a vacuum to the chamber and heating the shelves in order to evaporate the water from the frozen state.
- Complete stoppering of the vials usually by hydraulic or screw rod stoppering mechanisms installed in the lyophilizers.
- There are many new parenteral products, including anti-infectives, biotechnology derived products, and in-vitro diagnostics which are manufactured as lyophilized products. Additionally, inspections have disclosed potency, sterility and stability problems associated with the manufacture and control of lyophilized products. In order to provide guidance and information to investigators, some industry procedures and deficiencies associated with lyophilized products are identified in this Inspection Guide.

- It is recognized that there is complex technology associated with the manufacture and control of a lyophilized pharmaceutical dosage form. Some of the important aspects of these operations include: the formulation of solutions; filling of vials and validation of the filling operation; sterilization and engineering aspects of the lyophilizer; scale-up and validation of the lyophilization cycle; and testing of the end product. This discussion will address some of the problems associated with the manufacture and control of a lyophilized dosage form.

Equipment and types of freeze dryers

- There are many types of freeze-dryers available, however, they usually contain a few essential components. These are a vacuum chamber,[shelves, process condenser, shelf-fluid system, refrigeration system, vacuum system, and control system.
- **Function of essential components**
- **Chamber**
- The chamber is highly polished and contains insulation, internally. It is manufactured with stainless steel and contains multiple shelves for holding the product.
- A hydraulic or electric motor is in place to ensure the door is vacuum-tight when closed.
-

- **Process condenser**
- The process condenser consists of refrigerated coils or plates that can be external or internal to the chamber. During the drying process, the condenser traps water. For increased efficiency, the condenser temperature should be 20 °C (68 °F) less than the product during primary drying and have a defrosting mechanism to ensure that the maximum amount of water vapor in the air is condensed.
- **Shelf fluid**
- The amount of heat energy needed at times of the primary and secondary drying phase is regulated by an external heat exchanger. Usually, silicone oil is circulated around the system with a pump.
- **Refrigeration system**
- This system works to cool shelves and the process condenser by using compressors or liquid nitrogen, which will supply energy necessary for the product to freeze.

- **Vacuum system**
- During the drying process, a vacuum of 50-100 microbar is applied, by the vacuum system, to remove the solvent. A two-stage rotary vacuum pump is used, however, if the chamber is large then multiple pumps are needed. This system compresses non-condensable gases through the condenser.
- **Control system**
- Finally, the control system sets up controlled values for shelf temperature, pressure and time that are dependent on the product and/or the process. The freeze-dryer can run for a few hours or days depending on the product.

- **Contact freeze dryers**
- Contact freeze dryers use contact (conduction) of the food with the heating element to supply the sublimation energy. This type of freeze dryer is a basic model that is simple to set up for sample analysis. One of the major ways contact freeze dryers heat is with shelf-like platforms contacting the samples. The shelves play a major role as they behave like heat exchangers at different times of the freeze-drying process. They are connected to a silicone oil system that will remove heat energy during freezing and provide energy during drying times.
- Additionally, the shelf-fluid system works to provide specific temperatures to the shelves during drying by pumping a fluid (usually silicone oil) at low pressure. The downside to this type of freeze dryer is that the heat is only transferred from the heating element to the side of the sample immediately touching the heater. This problem can be minimized by maximizing the surface area of the sample touching the heating element by using a ribbed tray, slightly compressing the sample between two solid heated plates above and below, or compressing with a heated mesh from above and below.

- **Radiant freeze dryerS**
- Radiant freeze dryers use infrared radiation to heat the sample in the tray. This type of heating allows for simple flat trays to be used as an infrared source can be located above the flat trays to radiate downwards onto the product. Infrared radiation heating allows for a very uniform heating of the surface of the product, but has very little capacity for penetration so it is used mostly with very shallow trays and homogeneous sample matrices.

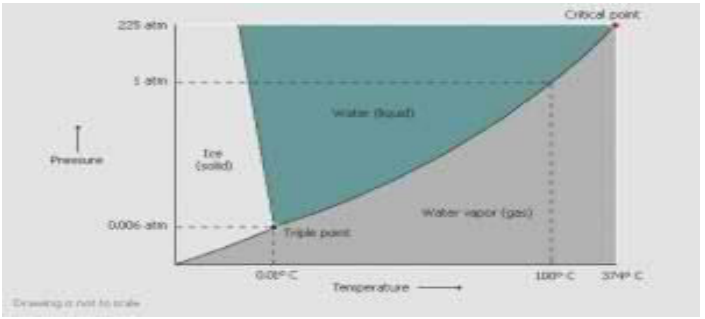
- **Microwave-assisted freeze dryers**
- Microwave-assisted freeze dryers utilize microwaves to allow for deeper penetration into the sample to expedite the sublimation and heating processes in freeze-drying. This method can be very complicated to set up and run as the microwaves can create an electrical field capable of causing gases in the sample chamber to become plasma. This plasma could potentially burn the sample, so maintaining a microwave strength appropriate for the vacuum levels is imperative. The rate of sublimation in a product can affect the microwave impedance, in which power of the microwave must be changed accordingly

Lesson 104

Lyophilization , significance

- Lyophilization vs. Freeze Drying
- What is lyophilization? How does it work?
- Lyophilization and freeze drying are synonymous. Lyophilization is a water removal process typically used to preserve perishable materials, to extend shelf life or make the material more convenient for transport. Lyophilization works by freezing the material, then reducing the pressure and adding heat to allow the frozen water in the material to sublimate.
- Lyophilization's 3 Primary Stages
- Lyophilization occurs in three phases, with the first and most critical being the freezing phase. Proper lyophilization can reduce drying times by 30%.

- **Freezing Phase**
- There are various methods to freezing the product. Freezing can be done in a freezer, a chilled bath (shell freezer) or on a shelf in the freeze dryer. Cooling the material below its triple point ensures that sublimation, rather than melting, will occur. This preserves its physical form.
- Lyophilization is easiest to accomplish using large ice crystals, which can be produced by slow freezing or annealing. However, with biological materials, when crystals are too large they may break the cell walls, and that leads to less-than-ideal freeze drying results. To prevent this, the freezing is done rapidly. For materials that tend to precipitate, annealing can be used. This process involves fast freezing, then raising the product temperature to allow the crystals to grow



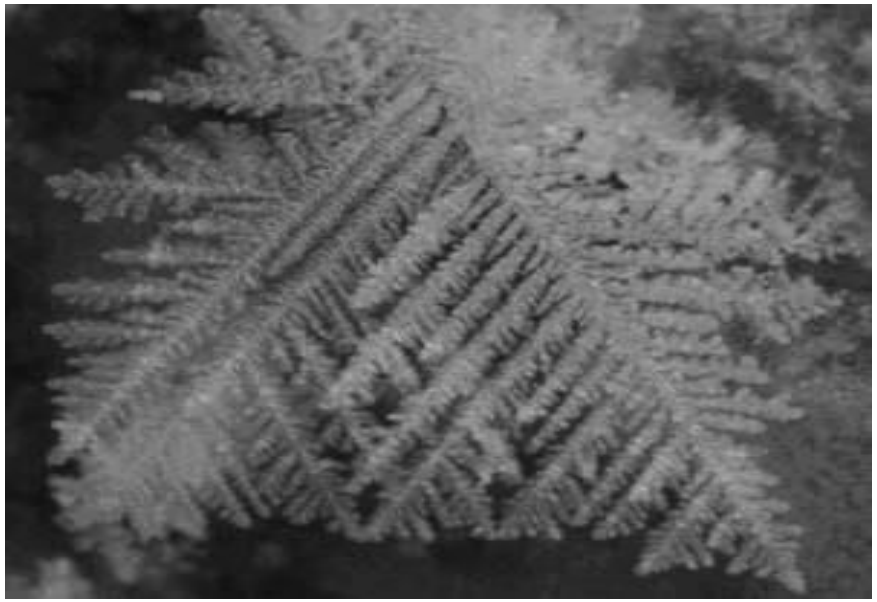
- **Primary Drying (Sublimation) Phase**
- Lyophilization's second phase is primary drying (sublimation), in which the pressure is lowered and heat is added to the material in order for the water to sublimate. The vacuum speeds sublimation. The cold condenser provides a surface for the water vapor to adhere and solidify. The condenser also protects the vacuum pump from the water vapor. About 95% of the water in the material is removed in this phase. Primary drying can be a slow process. Too much heat can alter the structure of the material.
-
- **Secondary Drying (Adsorption) Phase**
- Lyophilization's final phase is secondary drying (adsorption), during which the ionically-bound water molecules are removed. By raising the temperature higher than in the primary drying phase, the bonds are broken between the material and the water molecules. Freeze dried materials retain a porous structure. After the lyophilization process is complete, the vacuum can be broken with an inert gas before the material is sealed. Most materials can be dried to 1-5% residual moisture.

Problems To Avoid During Lyophilization

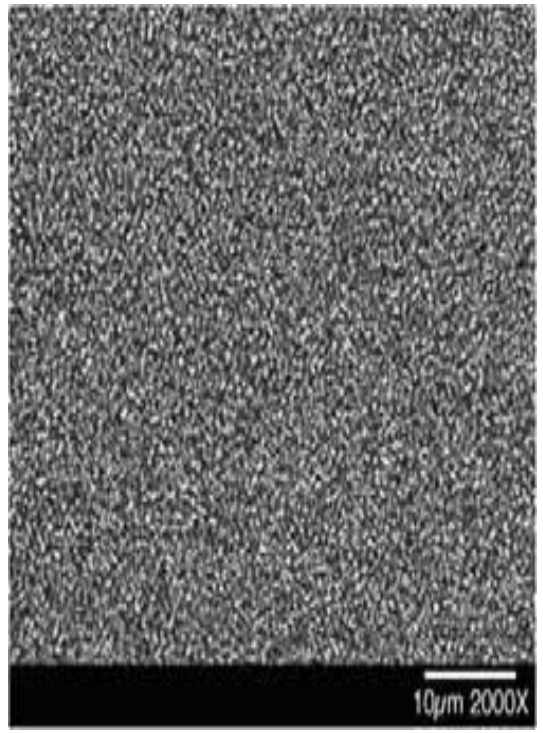
- Heating the product too high in temperature can cause melt-back or product collapse
- Condenser overload caused by too much vapor hitting the condenser.
    - Too much vapor creation
    - Too much surface area
    - Too small a condenser area
    - Insufficient refrigeration
- Vapor choking – the vapor is produced at a rate faster than it can get through the vapor port, the port between the product chamber and the condenser, creating an increase in chamber pressure.

- **Important Lyophilizer Terms**
- **Eutectic Point or Eutectic Temperature**
- Is the point at which the product only exists in the solid phase, representing the minimum melting temperature. Not all products have a eutectic point or there may be multiple eutectic points.
- **Critical Temperature**
- During lyophilization, the maximum temperature of the product before its quality degrades by melt-back or collapse

- **Crystalline**
- The material forms crystals when frozen.
-
- Has a eutectic point or multiple eutectic points
- Fast freezing creates small crystals which are hard to dry
- Annealing can help form bigger crystals



- **Amorphous**
- Multi-component mixtures which do not crystallize and do not have a eutectic point. They turn into a ‘glass.’
-
- Does not have a eutectic point
- For amorphous materials, freeze drying needs to be performed below the glass transition temperature



Collapse

- The point at which the product softens to the extent that it can no longer support its own structure. This can be a problem for many reasons:
-
- Loss of physical structure
- Incomplete drying
- Decreased solubility
- Lots of ablation (splat)

Reference

- https://www.millrocktech.com/lyosight/lyobrary/what-is-lyophilization/

Effort By Amaan Khan

contact: 0305-4716616